A tale of controllability: evolving network motifs in random Complex Networks



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We are all the outcomes of the interactions within an infinite network: the Universe.

Declaration

I hereby declare that except where specific reference is made to the work of others, the contents of this dissertation are original and have not been submitted in whole or in part for consideration for any other degree or qualification in this, or any other University. This dissertation is the result of my own work and includes nothing which is the outcome of work done in collaboration, except where specifically indicated in the text.

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Abstract

This thesis explores the common characteristics shared by complex systems, such as galaxies, natural food chains, and social communities, that are nurtured by the immeasurable amount of interactions between their atomic entities. While scientific frameworks have traditionally focused on analyzing smaller components of systems to understand the whole, this approach may result in overlooking important patterns and losing the big picture. To address this gap, our study makes use of the Complex Networks Theory as a framework able to examine the correlations between patterns of interconnections, or network motifs, and other components of varying scales present in the topology of the complex networks modeled from real-world systems. Our study suggests that network motifs could be a key variable for understanding the evolution and dynamics of complex systems. Based on our empirical evidence, we developed a new null-model that can generate random networks with motifs with a high accuracy and precision. This model is expected to have a significant impact on network motif analysis, providing researchers with a better understanding of how patterns emerge in real-world complex systems.

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List of publications

The rankings have been chosen in accordance with the classification of journals, as per the 2018 ranking published by UEFISCDI. Similarly, the classification of conferences is based on the CORE database, and the rank employed is the one that was active during the publication period. For instance, the 2018 rank (CORE2018) was used for publications made in 2019.

Publications in Web of Science - Journals Citation Index Expanded

[93] **Mursa, B.E.M.**, Andreica, A. (2023): Generating Random Complex Networks with Network Motifs using Evolutionary Algorithm-based Null Model.
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[98] **Mursa, B.E.M.**, Andreica, A., Dioşan, L. (2021): Network motifs: A key variable in the equation of dynamic flow between macro and micro layers in complex networks. Knowledge-Based Systems, 213:106648
doi: https://doi.org/10.1016/j.knosys.2020.10664 **(IF=8.66 on publication year 2021)**
Rank A, 8 points

[86] Mester, A., Pop, A., **Mursa, B.E.M.**, Greblă, H., Dioşan, L., Chira, C. (2021). Network analysis based on important node selection and community detection. Mathematics, 9(18):2294
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Rank A, 2 points

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[94] **Mursa, B.E.M.**, Andreica, A., and Dioşan, L. (2018): Parallel acceleration of subgraph enumeration in the process of network motif detection. In 2018 20th International Symposium on Symbolic and Numeric Algorithms for Scientific Computing (SYNASC), pages 191–198. Institute of Electrical and Electronics Engineers
doi: 10.1109/SYNASC.2018.0003 **(Index used = CORE2018)**
Rank C, 2 points

[97] **Mursa, B.E.M.**, Andreica, A., Dioşan, L. (2019): Study of connection between articulation points and network motifs in complex networks. In Proceedings of the 27th European Conference on Information Systems (ECIS), 127. AIS eLibrary
https://aisel.aisnet.org/ecis2019_rp/127 **(Index used = CORE2018)**
Rank A, 8 points

[95] **Mursa, B.E.M.**, Andreica, A., Dioşan, L. (2019): An empirical analysis of the correlation between the motifs frequency and the topological properties of complex networks. In Procedia Computer Science, 23rd International Conference on Knowledge-Based and Intelligent Information & Engineering Systems (KES), volume 159, pages 333–341. Elsevier BV
doi: 10.1016/j.procs.2019.09.188 **(Index used = CORE2018)**
Rank B, 4 points

[96] **Mursa, B.E.M.**, Andreica, A., Dioşan, L. (2019): Mining network motif discovery by learning techniques. In Proceedings of the Hybrid Artificial Intelligent Systems 16th International Conference (HAIS), pages 73–84, Cham. Springer International Publishing
doi: 10.1007/978-3-030-29859-3_7 **(Index used = CORE2018)**
Rank C, 2 points

Publications in journals and conference proceedings indexed by international databases (BDI)

[92] **Mursa, B.E.M.** (2023): Examining the social behavior of ant colonies using Complex Networks. In Proceedings of the Studia Universitatis Babes-Bolyai Informatica,

volume 67, pages 49–64.
Rank D, 1 point

Publications scores: 27 points.

Chapter 1

Introduction

What do a galaxy, a natural food chain, a social community, or other complex systems have in common? Although they are different, these structures share common characteristics by an evaluative dynamic nursed by the immeasurable amount of interactions existing between the entities they contain. Knowing that for each action, there is an equal and opposite reaction, the interactions are what give life to systems apparent static at first glance.

Being surrounded by an immense quantity of stimuli, our brain observed that filtering the information or paying awareness only to specific interactions can be more than enough to ensure the effectiveness of its daily activities [4, 82]. Moreover, by making use of abstraction, our brain can reuse information associated with patterns observed in specific systems in new and unknown systems later encountered [118].

Although unnatural to how our brain was used to perceive the world, our race evolved a scientific framework based on a common analytical approach of splitting the problem into smaller conundrums - divide et impera [28]. It is quite common now, even more since the popularity gain by statistics, to associate a global characteristic to a system by analyzing the most atomic components of its own. Although such an approach is simpler to follow and more comfortable to generalize proofs, it is not necessarily accurate as analyzing individually smaller components of a system one is in danger of losing the big picture.

Without any willingness to neglect the benefits brought by such analytical methodology, which significantly boosted the process of understanding the world, literature presents the need for more robust frameworks that are capable of understanding the apparent chaos as it is and not making order in it. Understanding the importance of patterns and how our brain perceives the world around us, an intriguing question arises: can the patterns be a missing variable in understanding the evolution of complex systems and their dynamics? By dynamic, we understand the non-linear mathematical behavior of a complex system over

a certain period [44]. Our work is willing to represent an empirical study of correlations between the properties of these patterns and other components of different scales [1]. We intend to use this empirical evidence as a foundation for a new null-model that can generate random networks with a specific concentration of motifs.

1.1 Motivation

We found network theory as a robust enough framework to serve as a tool for the research we intend to conduct. The mechanism approached by this field is to encode the entities and the interactions between them as topological structures represented as graphs, composed by nodes and edges. What makes these mathematical objects so powerful is that they encapsulate in their underlying graph the contextual information offered by the modeled real-world system. This means that each edge will have a meaning, and it is explainable by understanding a real-world dynamic [14]. Metaphorically, we can say that each complex network has a story.

The interdisciplinary applicability of network theory makes it such an elegant framework, emerging notions from physics, mathematics, computer science, biology, and many others. Statistics are used as a mechanism of quantifying properties of the topological spaces, as it can offer the analytical approach to study the essential components of the networks: the nodes. The study of these fundamental components has formulated a large number of microscale properties from which we can remind a few of them: nodes degree, clustering coefficient [134], assortativity degree [105] or power-law exponent [11]. Fundamental properties of Stability Theory of dynamical systems [17, 58] were used as the argument to sustain the focus of the study on microscale properties in complex networks. This argument states that a dynamic triggered in a complex system by a given perturbation at the macroscale will eventually converge to its smallest structural components, encoding in these microscale components the results of the macro dynamic [6, 47]. Hence, knowing that action at a global scale is persisted at the local scale was a robust argument for continuing a microscale-based research.

Applying this convergence principle in a reversed engineering manner, the studies were extended to a macroscale analysis. In this type of examination, the microscale properties are correlated with macroscale network dynamics. The literature recalls the real-world networks that are characterized by the famous theory of six degrees of separation [10], namely small-world networks. These networks tend to form clusters with short paths between every

[1] *scale (micro, meso, macro)* found in complex systems - a telescopic perspective of the analyzed structures

two nodes of the network, which makes them robust structures without any trade-off in the performance of information flow between its members [112]. Other remarkable networks studied in the literature are sparse networks, which are most of the real-world networks governed by a power-law degree distribution [116] or random networks - structures generated using probability theory to help the study of real-world networks [99]. All these properties are quantifying the relationship of the nodes with their neighbors, as it can be observed in the analytical principle of *divide et impera*.

Although the accepted analytical framework [59, 60, 81, 121] can be considered reliable and came with answers to many questions proposed by researchers, something is missing in the big picture. We consider that the convergence from macro to microlayers [2] and its impact in the in-between layer (meso) are topics not studied sufficient, which require a better emphasis in the current analytical framework of the complex networks. Motivated by the existence of applications that study the importance of mesoscale structures in domain-oriented contexts (biology, ecology or sociology) [1, 88, 113, 138] in respect to the other layers of the network, macro, respectively micro, we consider the mesoscale structures as bridges between the macro and microscale analysis, the missing key in solving the equation of the dynamical convergence from macro to microlayers. Such an approach could lead to a universal framework that could imply better precision of the dynamics of the complex networks surrounding us.

Our focus is on improving existing approaches of understanding complex networks by not following a simple analytical method but trying to observe these topological spaces in their entire complexity and beauty.

The state-of-the-art presents various mesoscale structures such as communities - groups of nodes which are more densely connected than the rest of the network [24], block models - decomposed functional classes of entities within a complex network [55], or modules - community-like components with specific contextual roles [53]. These structures are widely studied in applications addressing problems from fields such as Biology, Physics, or Social Sciences [5, 31, 41, 65, 104, 104, 130, 130], most of the times presented as stand-alone components and not correlated with properties of components found on different scales.

Wharrie et al. [137] studied how the presence of triangles is an essential factor in the evolution of communities - raising awareness on the existence of a relationship between the structures of a given network. Starting from their valuable results, our work is willing to follow a similar principle by presenting an empirical study proposing more detailed research

[2] *layer (micro, meso, macro)* - structural segmentation level of the networks

of the correlations between all layers of a complex network - macro, meso, and micro.

As a first step in the pursuit of this goal, in our proposal we are willing to have a research focused on the study of mesoscale properties named network motifs - patterns of interconnections which can occur in significantly higher numbers in complex networks than in randomized networks which hold the same degree of distribution as the first network. Considering their nature of existence, network motifs will always have behind them contextual reasoning brought by the dynamics of the network.

Learning from nature's wisdom and understanding strong network motifs contributing to the dynamics of the network, we consider them a perfect candidate as a first step in the direction of our cross-layer analysis. This thesis proposes an experimental analysis of the correlation between the properties of microscale components and the existence of network motifs, respectively, the existence of network motifs and the properties of macroscale structures. The results obtained will emphasize how the existence of these mesoscale structures can help to discretize types of networks in ones with a more complex source of information. This experiment is also willing to empirically validate the convergence of the dynamic flows between all layers of the complex networks, having network motifs as a bridge in the dynamic convergence from macro to microlayers.

By introducing the study of network motifs as possible components facilitating the convergence mechanism, we intend to strengthen existing conclusions about the micro, meso and macro-layers dynamic previously analyzed through the perspective of modular structures such as triangles [137].

Applied to fields where connections between structures or components play an essential role, this study can help further understand molecular, self-assembly, protein-based cellular adaptation, or social interactions.

Using the results of the experiments focused on describing the convergence mechanism, as this thesis's final resolution, we propose a network generator model to generate random networks that contain motifs. Such a model would be an original contribution to the models existing in the literature that currently do not explicitly consider the evolution of network motifs in the random networks they generate. Considering the uncommonness of real-world networks evolving network motifs, such a model would be a valuable tool for generating synthetic networks for experiments that need large network samples with motifs, improving the quality of the studies and their applications.

1.2 Thesis structure

The thesis is structured in 9 chapters as following:

- **Chapter I - Introduction** presents a general picture of the theory of complex networks and its state-of-the-art applications, with a particular focus on network motifs. The Motivation section outlines the main objective of the thesis which is to propose a new null-model able to generate synthetic networks that contain network motifs and the necessary steps to achieve it, highlighting the potential significance and impact in complex networks research, leading to a deeper understand of their intricate dynamics.

- **Chapter II - Theoretical insights** provides a comprehensive examination of the historical background of Complex Networks Theory (CNT), elucidating the scale viewpoint of network topology through Stability Theory and detailing the distinct properties of each layer at the micro, meso, and macro levels. This chapter serves as a robust theoretical foundation to aid in the comprehension of the subsequent chapters' studies and experiments.

- **Chapter III - Parallel acceleration of subgraph enumeration in the process of network motif detection** addresses the challenge of performance in identifying network motifs, known to be an NP problem. Despite the existence of several algorithms attempting to solve this problem, no perfect approach exists in terms of time performance. The chapter proposes a redesign of the ESU algorithm, which utilizes parallel programming with process-driven and thread-driven models.

- **Chapter IV - Articulation Points and Isolated Communities: Catalysts for Network Motif Evolution** reports on a research study aimed at investigating the potential relationship between articulation points and network motifs in complex networks. The analysis was conducted on a variety of real-world networks to determine the coexistence of these two properties and shed light on the topological features that may contribute to the development and evolution of network motifs.

- **Chapter V - Unraveling the relationship between micro-level topological properties and the propagation of motifs in network topologies** builds upon previous research presented in Chapter IV which established a strong correlation between articulation points and meso-level network motifs. The current study extends this by

investigating the relationship between micro-level topological properties and the appearance of motifs. By using network properties, the study aims to gain insight into the mechanisms underlying the appearance of motifs and how network dynamics propagate through its structures.

- **Chapter VI - Prediction of network motifs existence in Complex Networks** describes a novel approach to predict the existence of network motifs in complex networks using Artificial Intelligence (AI) models. This method uses the topological properties identified in Chapters IV and V as facilitators for the evolution of network motifs and trains AI models to determine if a given complex network contains motifs. The aim of this approach is to serve as a pre-processing step to save time in motif detection, which is known to be an NP-complete problem, by using the proposed AI model for prediction.

- **Chapter VII - The study of the empirical evidence of dynamic convergence existing between complex networks' layers** presents an analytical framework that unifies the findings of previous chapters by demonstrating the existence of dynamic convergence between layers of complex networks. The framework shows that the existence of network motifs is directly related to a variety of microscale and macroscale properties, providing evidence of a macro-meso-micro convergence. The ultimate goal of this work is to propose a null-model capable of generating synthetic networks with network motifs in a controllable manner, and this framework aims to facilitate the achievement of this goal.

- **Chapter VIII - Generating Random Complex Networks with network motifs using Evolutionary Algorithm-based null model** proposes a new null-model that generates random networks containing motifs based on a series of micro-scale properties. The chapter builds on previous research that established a correlation between network motifs and various micro- and macro-scale properties of network topology. The proposed null-model, and the end goal of this thesis, aims to facilitate further study of network motifs by providing a way to generate synthetic networks for experimental purposes.

- **Chapter IX - Conclusions** summarizes the findings and contributions of the previous chapters, highlighting their originality and potential benefits for the research community. It also identifies areas for future work to enhance the proposed null-model.

1.3 Original contributions

Throughout the course of this thesis, which aims to propose a null model capable of generating random networks with network motifs, several theoretical and experimental studies were conducted. These studies not only validated research questions and hypotheses but also led to a number of original contributions to the current state-of-the-art through a methodology focused on gathering empirical evidence via statistical significance:

- A detailed comparison between the state-of-the-art algorithms for extracting network motifs in terms of performance and scalability [94].
- Proposing a new parallel model that makes use of parallel and distributed programming to significantly improve the running time of an existing algorithm used to extract network motifs [94]. This parallel model will be used as a primary tool for motif detection in the experiments described in Chapters IV, V, VI, VII, and VIII.
- Proving empirically that articulation points will occur with a higher degree of probability in real-world networks than in randomized networks [97].
- Proving that articulation points through their isolated communities that they link are prone to evolve network motifs [97].
- Proving the altruistic behavior of the ants in the social networks, they are part of centrality measures and other network properties [92].
- Proving the performance of information flow in ant colonies, as optimal short paths between any two ants, and validating the small-world characteristic of the networks modeled from their social interactions [92].
- Revealing a structural overlap between the top nodes detected by centrality measures at network level respectively community level [86].
- Performing a comprehensive study over a large set of micro respectively macro scale properties that revealed a significant correlation between Local Clustering Coefficient and Assortativity Degree and the existence of network motifs, leading to the conclusion that network topologies evolving high values for these two properties will be prone to contain network motifs [95]. The results obtained served as a foundation for the experiments described in Chapters VI, VII, and VII.

- Proposing of classification model that can predict if a network contains or not network motifs based on its topological properties. This approach serves as a time-saving pre-processing step for the state-of-the-art solutions used to detect motifs [96].
- Formalizing an analytical framework that can prove that the convergence of a specific dynamic between the network's layers leads to the materialization of network motifs. This study was accompanied by an experiment that validated the possibility of controlling the evolution network motifs in random networks [98] and serves as empirical proof for the null model proposed in Chapter VIII.
- Proposing a null model architecture based on Evolutionary Algorithms that generate random networks with motifs based on a series of parameters which includes the two most impacting topological properties identified as factors of triggering the convergence between network's layers that stabilize in network motifs: Clustering Coefficient and Assortativity Degree [93].

Chapter 2

Theoretical insights

2.1 Graph Theory

The problem of the seven bridges in Königsberg, a popular problem around 1730, intrigued mathematicians due to its complexity in finding a solution. At that time, neither of the mathematical frameworks offered a problem set of tools to help the experts solve the puzzle. The problem required finding a path that starts from a point, crosses each of the seven bridges once and only once, and gets back to the same starting point. Geometry was not useful as there was no geometrical shape that could be used to represent the bridges, nor was algebra through its data structures. The problem needed a completely different approach or framework, one that enabled encoding systems into a data structure that is not necessarily a geometrical shape, but an irregular structure that is invariant under external forces such as stretching and bending, and that encodes the atoms of the systems and their links following these forces.

During the 18th century, the pioneers in the study of topologies emerged, and the term "topology" [125] was later coined by mathematician Johann Benedinct Listing in the 20th century. The Seven Bridges problem is considered one of the earliest examples of topological problems, and mathematician Leonhard Euler proposed modeling the problem in a structure never before used: a graph [14]. Euler's solution involved representing the cities and their bridges as a structure in which the cities were nodes and the bridges were links. The graph representation was not bound to a geometric representation, as nodes and edges could encode any information, making graphs a mathematical tool for formulating axioms and formulas similar to other geometrical shapes. This generic representation also meant that the nodes and edges did not have to be positioned exactly as on the map, as long as there was a link between their corresponding nodes.

As Euler proposed, graphs possess a range of properties that make them applicable in various scenarios. Among these properties, one of the most fundamental is that graphs consist of vertices (objects) and edges (relationships between objects encoded as vertices). Edges can be directed or undirected, depending on whether the relationship between objects is one-way or two-way, respectively. Additionally, vertices can encode a variety of quantitative relationships, such as the distance between two cities. When edges are assigned values, the resulting graph is referred to as a weighted graph.

The basic yet intuitive elements of graphs - vertices, and edges - allow for the creation of diverse structures, ranging from simple paths connecting indirectly connected nodes to more complex ones such as cliques and cycles (Figure 2.1). Cliques refer to groups of fully interconnected nodes, while cycles denote paths that begin and end at the same vertex (as in the Seven Bridges problem).

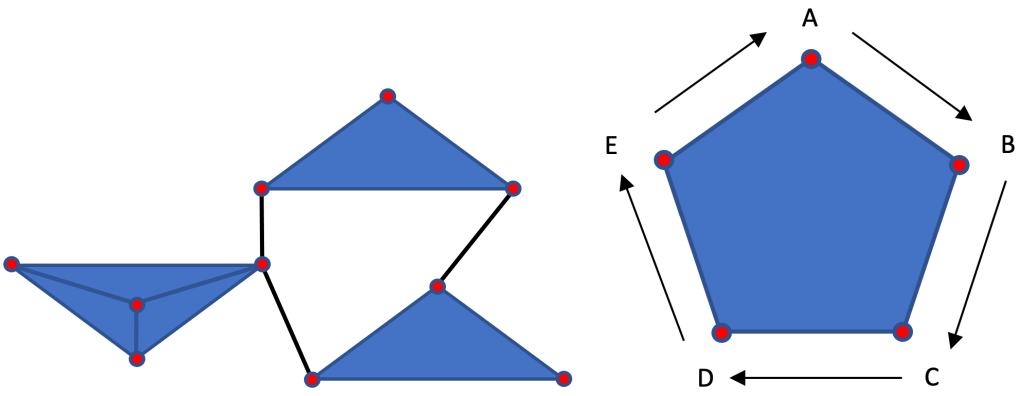


Fig. 2.1 Graph clique examples (left); Graph cycle example: A $\rightarrow$ B $\rightarrow$ C $\rightarrow$ D $\rightarrow$ E $\rightarrow$ A (right)

The fundamental structures and properties of graphs enable the resolution of a vast range of problems in various domains, including data modeling and analysis, optimization, and simulation. Graph Theory has paved the way for Complex Network Theory, which emerged as an interdisciplinary field combining mathematics and theoretical physics to study complex systems modeled as topological structures similar to graphs, where objects serve as nodes or vertices, and their interactions are represented by links or edges. A key distinction between graphs and complex networks lies in their connectivity patterns, with complex networks exhibiting non-random topologies that capture real-world behaviors, characterized by the presence of clusters or communities.

2.2 Complex Network Theory

In the 1960s, after a period of waning interest, Paul Erdős and Alfréd Rényi conducted groundbreaking research on random graphs to better understand their properties [35]. Their work is considered to be the precursor to Complex Network Theory, which gained momentum in the 1990s and 2000s due to the pioneering contributions of Duncan Watts and Steven Strogatz on small-world networks and Albert-László Barabási on scale-free networks. Complex Network Theory has since emerged as a crucial interdisciplinary field at the intersection of mathematics, physics, computer science, and other natural sciences, providing insight into the behavior and dynamics of complex systems.

Complex networks are topologies represented as graphs with non-trivial properties, where a node is an entity from a real-world system, and an edge between two nodes encodes an interconnection of any type between the entities [11].

Complex Network Theory provides a broad range of features that can be utilized to enhance the comprehension of complex systems, starting from node-level features like degree distribution, centrality, or assortativity, to group-level features like communities - a group of nodes that have more connections among themselves than with other nodes in the network, bridges - edges that link communities, connected only through that link, or hubs - a group of nodes that have significantly more connections than other nodes in the network.

In contrast to synthetic graphs, complex networks exhibit various properties and structures that can be attributed to the real-world dynamics of the system they represent. These networks are a static snapshot of the system at a particular moment and contain properties that result from multiple iterations of interactions influenced by internal or external stimuli. These interactions eventually converge and result in the formation of nodes, edges, and their corresponding metadata. In the following sections, we will discuss the convergence principles in Stability Theory and provide an overview of the important properties of structures present in complex networks at different levels, ranging from micro to meso and macro.

2.3 Dynamical convergence in Stability Theory of dynamical systems

One of the most addressed subjects in the field of complex systems is the study of their dynamics governed by the mechanisms of action-reaction present in our universe [103]. Such studies aim to understand better if there are any universal laws in the evolution of the complex system.

In dynamical systems, the perturbations give motion to the particles existing in its structure (Figure 2.2). Although the perturbations are spreading between the particles which become active in emanating interactions with its neighbors, the dynamics eventually converge into a stable state. Depending on the strength of the perturbation, the initial systems will end by having a diverse structure with particles having new states.

More fundamentally, we can say that a complex system S with a given dynamics forced by a perturbation p at the moment t will structurally evolve at the moment $t+1$ considering the change of the state existing in, if we suppose that a complex system is composed of particles state [108].

The main principle behind this theory explains how the dynamics of the system that starts at a macroscale spreads through the structure of the system until it reaches its most fundamental components, eventually reaching a stable state. Hence, it is learned about a mechanism of structural state convergence from an upper layer to a lower layer, namely macro to micro components.

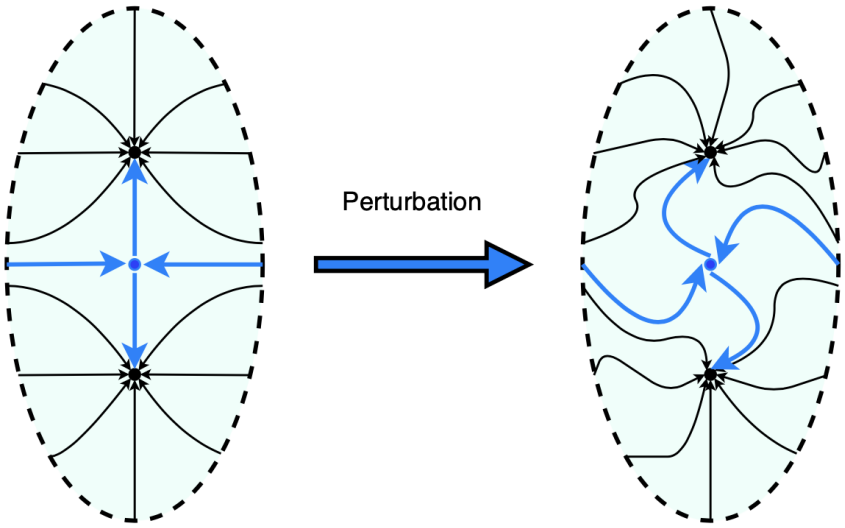


Fig. 2.2 Perturbations in dynamical systems - the dynamics eventually converge into a stable state once particles become active

On the other hand, divergence is the opposite dynamic motion starting from microscale components going to macroscale structures. Both dynamic motions are equally important and have applications in many fields (universe is governed by two forces which make it expand and contract in an almost equal manner) [133]. The focus of this thesis will fall on the study of the convergence mechanism.

Complex systems are a robust framework used to model the complexity that surrounds us. Complex network theory is one example of a tool that one can use to perform sophisticated analysis over the modeled complex systems [14]. A complex network is nothing more than a snapshot of a complex system, and it captures the dynamical state of the system in a given moment as a topological space.

Following this line of reasoning, the structural convergence stays as an argument of the focus gained to microscale analysis in complex networks as well, understanding that global actions will eventually react to microscale components. State-of-the-art in complex network analysis presents layered structural perspectives as micro, meso, or macroscales - referred to as micro, meso or macro layers while used to present hierarchical segmentation of the structure of a network [81, 121]. These scales were most of the time studied individually without a clear intention in unifying them in a universal framework that can answer the question about how perturbations spread from macro to microlayers in complex network topology layers.

2.4 The scale perspective of Complex Networks analysis

The study of complex networks is divided into three categories depending on the scale perspective addressed by the analysis: micro, meso, and macroscale. Macroscale analysis works in the study of the structures as a whole by considering its dynamics a sum of the actions and reactions between its substructural components and structures. This type of analysis can be considered one with a higher degree of complexity but with results providing multi-dimensional insights about the studied structure. Reminding about the fundamental components of complex systems, microscale analysis performs the study of the most atomic components from the structure of a system. These atomic components are studied as individual entities within the proximity area of their neighbors. The structures found between micro and macro layers fall in the mesoscale study, which is focused on the tendency of atomic entities to cluster into structures with specific characteristics.

This scale of perspective is also found in the study of complex networks. The nodes, being the atomic components of the network structures, are the focus of the microscale analysis. The literature presents various properties revealed by the study on the microlayer - to recall a few: node degree, density, clustering coefficient, k-core or relative edge distribution. The mesoscale analysis focuses on groups of nodes and interactions between them, which tend to have common characteristics and cluster in components such as communities [24] or network motifs [88]. The big picture perspective is analyzed by the macroscale,

Table 2.1 Studied topological properties

Property	Annotation	Property	Annotation
Articulation point	AP	k-core	k_{core}
Assortivity Degree	ρ	Local Clustering Coefficient	CC_L
Average degree	d_{avg}	Maximum degree	d_{max}
Average Triangles	tr_{avg}	Maximum triangles number	tr_{max}
Density	D	Minimum degree	d_{min}
Relative edge distribution entropy	H_{er}	Number of nodes	N
Number of edges	E	Power law exponent	γ
Gini coefficient	g	Triangles number	tr_{no}
Global Clustering Coefficient	CC_G		

category that characterizes complex networks as a whole using on their topology properties. The studies focused on this scale of perspective have revealed networks with distinguished topologies from which we can remind small-world [112], sparse networks [116] or random networks [99].

The upcoming subsections are willing to introduce or enhance readers' knowledge regarding the theoretical aspects of micro, meso, and macro scales existing in the state of the art of complex networks.

2.5 Microscale analysis

The analysis of microscale properties represents most studies on complex networks. These properties quantify measurements representing characteristics of the most fundamental components of the network, nodes, or edges. In graph and network theory, these properties vary in complexity from metrics that reveal a simple count to more complicated metrics that compute different distributions [12, 14, 75, 105, 111, 134].

Although following a simple analytical approach using these properties, it was possible to reveal important characteristics of the real-world systems, which were later formulated in theories that explain power-law degree distributions or preferential attachments [12, 131] between the nodes.

In the following subsections the emphasis will fall on detailing the properties established by the dynamical behavior of a network (Table 2.1). Considering that the topology of a network can be seen as a structural encoding of a real-world context, its correspondent properties can be used as one of the most objective measurements in performing an analysis of a real-world system. In our case these properties will be very useful in determining if

a given topology is more suitable to contain network motifs due to having certain values for its properties. The properties with a significant impact on our methodology will be gradually detailed by also considering how they might influence the existence of specific patterns encoded as network motifs.

2.5.1 Articulation Points

The importance of a node in the network's structure is a widely debated property [25, 80]. The ambiguity of this property arises from the diversity of factors that can be considered to conclude that a node is essential. The most common of them can be the number of connections that a given node has, which can say if a node is popular or not.

Similar to popular nodes, a topology can have the so-called articulation points (*AP*) [123], which are nodes that removed from network's topologies split the original network into two or more connected components (Figure 2.3).

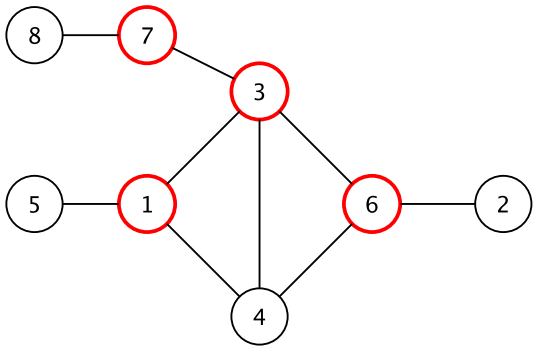


Fig. 2.3 Example of Articulation Points (in red)

We can see these nodes as communication bridges between two or more isolated communities. Understanding their capacity to link isolated communities, as a bridge that connects two inaccessible points in a space, one could learn the impact that an *AP* has in the controllability of the dynamics of these isolated communities and how they can inject a dynamic that can evolve specific behaviours encoded as motifs.

2.5.2 Assortativity degree

The study of complex networks revealed that nodes with similar properties favor the creation of new links with nodes having the same properties. An accessible real-world example is found in social communities in which popular members, where popularity means, in this

case, having a higher number of connections than the average, tend to surround themselves with other members that have the same level of popularity [107]. Complex Network applications study this characteristic of the nodes with assortativity degree ρ [105], a property that quantifies the tendency of the nodes to have neighbors with the same number of connections (degree) as theirs. In other words, ρ quantifies the inclination of high-degree nodes to connect with other high-degree nodes, respectively low-degree connect with other low-degree nodes.

A network that has a high ρ is a good candidate for obtaining motifs, as the preferential attachment between the nodes with the same degree can lead to specific patterns of interconnections. The formula below describes how ρ is computed:

$$\rho = \frac{\sum_i j_i k_i - [m^{-1} \sum_i \frac{1}{2}(j_i + k_i)^2]}{m^{-1} \sum i \frac{1}{2}(j_i^2 + k_i^2) - [m^{-1} \sum_i \frac{1}{2}(j_i + k_i)]^2}, \tag{2.1}$$

where m is the number of edges and j_i, k_i are the degrees of the vertices at the ends of the i_{th} edge ($i = [1, m]$).

2.5.3 Density

The volume of interconnections of nodes can be an excellent maturity level indicator of the structure of the networks. Such measurement is density (D)[111], property that measures how close a network is to reach the maximal number of edges. Density might not be a relevant property in the topological modeling for network motifs, but together with other properties it should reflect a more visible context of the maturity of the topology. The formula below describes the computation of D:

$$D = \frac{m}{n-1}, \tag{2.2}$$

where m is the number of edges, respectively n is the number of nodes.

2.5.4 Relative edge distribution entropy

In respect of the volume of interconnections of the nodes, it is essential to understand the uniformity of connections distribution between the nodes. A robust network structure should have such balanced interconnections distribution, a characteristic that would denounce the lack of critical areas prone to disconnect the network. In complex network studies, it is possible to learn and quantify the uniformity of the connections between nodes using relative edge distribution entropy (H_{er}) [75]. A network with an H_{er} close to 1 might be too

monotone to offer patterns of interconnections, hence we can presume that networks with a lower value of H_{er} are more probably to contain motifs. Relative edge distribution entry is defined as:

$$H_{er} = \frac{1}{\ln n} \sum_{u \in V} -\frac{d(u)}{2m} \ln \frac{d(u)}{2m}, \tag{2.3}$$

where n is the number of nodes, d is the degree of a given node and m is the number of edges.

2.5.5 Gini coefficient

Used in the economy to measure income inequality [48], the Gini coefficient (g) was introduced into the research of complex networks as an alternative to existing power law measurements. g indicates the degree distribution between nodes, having values between 0 and 1 - a 0 value being an equal distribution of the degree of the node and 1 the opposite of a balanced distribution. Real-world networks with $g = 0$ are rare due to contextualization and preferential attachment [11], which means a better chance for the network participants to evolve specific behaviours materialised as patterns of interconnections. g is defined by the following formula:

$$g = \frac{2\sum_{i=1}^{n} id_i}{n\sum_{i-1}^{n} d_i} - \frac{n+1}{n}, \tag{2.4}$$

where n is the number of nodes and d_i is the degree of node i.

2.5.6 Clustering Coefficient

The topological space created when a system is modeled as a complex network encodes properties that are studied can quantify real-world contextual behaviours of these systems. Such example are social networks on which their members tend to follow specific behaviors studied by fields like sociology. Societies, religions or communities are the results of a social network behaviours where theirs members tend to clusterise themselves into groups that share the same goals.

Topologically it is possible to quantity these behaviours by making use of different measurements like global clustering coefficient (CC_G) and local clustering coefficient (CC_L) [134]. Although both measurements describe a clustering behaviour of the network, the difference between them is in the level of localisation where they are focused. CC_L is com-

puted while considering how close are the neighbors of a node to create a clique together (Formula 2.5), hence a local behaviour, when CC_G (Formula 2.6) is studying the network as whole by computing a ratio between number of closed triangles and both open and closed triangles.

$$CC_L = \frac{T_i}{d(i)(d(i)-1)}, \tag{2.5}$$

where T_i is the number of triangles node i is part of, and $d(i)$ represents the degree of node i.

$$CC_G = \frac{\text{number of closed triangles}}{\text{number of all triangles (open and closed)}} \tag{2.6}$$

Considering the argument we introduced for other proprieties and considering isolation as an environment property prone to evolve specific patterns of interconnections, we can expect particular behaviors encoded as network motifs among the nodes part of the same clusters.

2.5.7 Power law exponent

Properties that indicate preferential attachment between nodes receive great interest as they are candidates that might explain the dynamics of different topologies modelled from real-world systems. A such example are scale-free networks [12] which are networks that hold a very specific degree distribution between nodes that is a correlation between network's nodes with a given degree d and a power $d^{-\gamma}$, where γ is the power law exponent. Explained in a more natural way, scale-free networks are networks with a Paretto distribution [101] (asymptotic distribution) of the node degrees, where large portion of degrees is held by a small fraction of the nodes. Following a similar analytic approach as for the other properties, we can say that rare nodes with a large fraction of connection can also favour the appearance the motifs on their proximity.

$$\gamma = 1 + n \left(\sum_{u \in V} \ln \frac{d(u)}{d_{min}} \right)^{-1}, \tag{2.7}$$

where V is the list of nodes and u is a node from the list, $d(u)$ the degree of node u and d_{min} is the minimum degree in the studied network.

2.6 Mesoscale analysis

In network theory, the mesoscale analysis is focused on the study of complex network samples of nodes with the interactions between them. The studies revealed that specific groups of nodes interact in distinctive ways that lead to remarkable properties. Mesoscale structures like communities or network motifs are used in applications from various fields to extrapolate theories about real-world complex network dynamics. Especially in social sciences, communities are represented in the literature as the fundamental and important structures in the evolution of a social system [104, 130]. Same goes for network motifs in fields such as biology [110, 117], ecology [120] or physics [65].

In this thesis, the focus will fall on network motifs - patterns of interconnections that can occur in significantly higher numbers in complex networks than in randomized networks that hold the same degree of distribution as the initial network [88].

The study applied to network motifs reveals that these structures have an essential role in the guidance of network dynamics; therefore, they have received particular attention for research focused on the structural evolution of networks.

These patterns, or motifs, are found in various real-world networks, such as in domains like astronomy, food chains, and microbiology. Figure 2.4 illustrates a few examples starting with large systems such as the universe where we can find clusters of astronomical bodies that share common characteristics and are linked together through the gravity of a star, unique relationships between the members of an ecosystem (hierarchical chains of predator-prey or flat chains of symbiosis), computer networks designed by following various topologies that are characterized by significantly more frequent types of routing between its components, or systems of proteins that evolve specific protein-protein interconnection.

From the perspective of Graph and Networks Theory, an example of a motif can be represented by a subgraph of a given size of k (number of nodes). In Figure 2.4 we can see possible motifs of size k=3 from directed or undirected graphs.

Literature presents various algorithms [52, 69, 70, 88, 106, 135] for network motif discovery, all of them following the same steps:

- enumerate all subgraphs of a given size of k.
- cluster the enumerated graphs by their properties and then compute their concentration concerning the entire network (clusters subgraph count vs. the total number of enumerating subgraphs).

As a definition, a network motif is a subgraph G' where its frequency in the initial net-

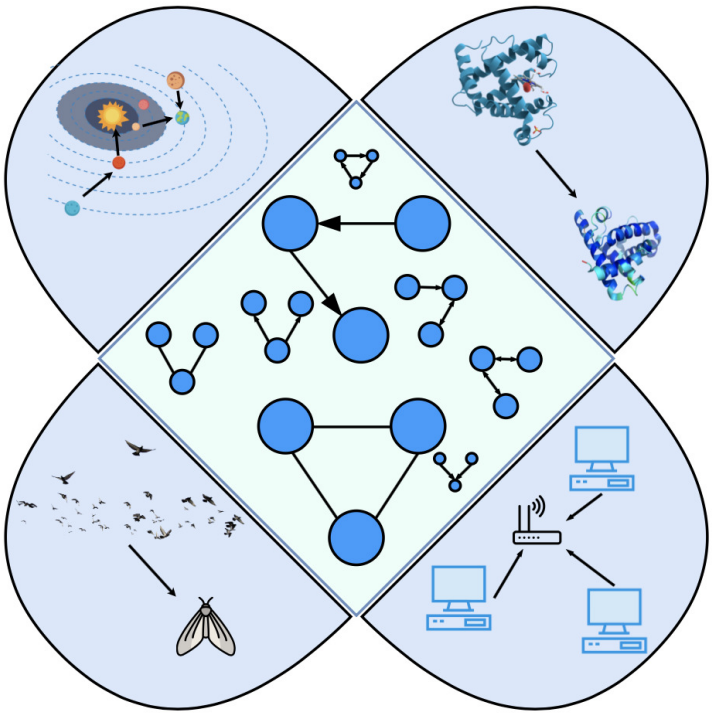


Fig. 2.4 Network motifs examples from astronomy, biology, microbiology and computer science. Compared with other structures, such patterns of interconnections are characterized by a unique frequency.

work G ($F_G(G')$) is considerably higher than its average frequency in n random networks $R(G)$, random networks that hold the same degree distribution as the initial network. This theoretical approach is confirmed if the result of the given $Z-score$ exceeds a given threshold:

$$Z(G') = \frac{(F_G(G') - \mu_R(G'))}{\sigma_R(G')}, \tag{2.8}$$

where $\mu_R(G')$ and $\sigma_R(G')$ are the mean and standard deviation of subgraph G' frequency in $R(G)$ [88]. The importance of a subgraph as a motif is directly proportional to the value of $Z(G')$.

Implementation-wise, network motifs discovery is an $NP-hard$ problem [9], so there is no perfect algorithm, only algorithms more efficient than others. Speed is not the only concern, but memory usage also. All these algorithms are CPU and memory-intensive processes, and because of that, there are raised limitations on the size of the processed network. Although the broad scope of the thesis is one studying the existence of motifs and not how

they are extracted, the $NP-hard$ implications became quickly essential to us, such as we had to process a large sample of networks of different networks in our experiments sizes to prove our hypothesis. Hence, we gave significant importance to the literature study for searching for the most efficient algorithms that could speed up the waiting time in our experiments.

Unfortunately, even if the literature presents a list of algorithms that are able to extract motifs from a given network or graph, all of them encounter a performance bottleneck once the motif size (wk) scales to larger and larger values.

Still, literature presents a solution that improves waiting time but involves a decrease in accuracy. It follows a sampling paradigm that implies extracting motifs only on a randomized partition of the original network and generalizing the entire network's results. Of course, this approach is becoming more statistically accurate as the partition size (sample) grows. Essentially, with this paradigm, it is considered a trade-off between accuracy and the waiting time.

Besides our main primary research track described in the sections dedicated to the experiments, we will start a secondary research track that is willing to investigate and propose enhancements to the solutions offered by the literature by disconsidering an eventual trade-off between performance and accuracy we consider as being not satisfactory.

2.7 Macroscale analysis

Although the understanding of the complex networks most of the time falls to an analytical approach focused mostly on microscale analysis, the complexity of understanding the system as a whole never ceased to be a focus. Connected or not, network topologies are encoding in their structure a contextual dynamic of the real-world system from which they modeled. By making use of a reversed engineering approach, the research applied to microscale components revealed that specific properties such as nodes clustering tendencies, short paths between nodes, or randomness in the interconnections set between the nodes could be used to classify networks into distinct classes with specific macroscale properties.

Small-world networks [134], sparse networks [40] or random networks [15] are widely studied in the literature in applications of real-world systems. In literature, we can find another type of network - spatial networks [18, 46], defined by geometrical topologies with nodes and edges that are spacial elements and positioned in a Euclidean space. This thesis will focus only on the first three network types, considering that spatiality would require a completely different analysis approach, which would distance itself from the thesis' initial

goal. Small-world, sparse and random networks are network types with topologies characterized purely by a classification of nodes interconnections (clustering tendencies or data flow efficiency), criteria which will facilitate the applications of our experimental methodology.

Using microscale properties, it was observed how sparse networks are structures with an edge distribution centered on preferential attachment between specific clusters of nodes. Random networks are used as null models in proving the hypothesis behind the existence of networks with specific characteristics; hence they need to be generated driven by a chaotical procedure that is not close to a contextual bias. Empirically, it was observed that random networks tend to have short paths distance, which makes them topologies that facilitate fast information spreading [64]. Small-world networks combine both sparse and random network characteristics at the same time as they tend to contain mesoscale structures as communities with short distances between the nodes that are part of the network.

Telesford et al. [122] propose ω as a coefficient that indicates the degree of membership of network topologies in each presented category, namely small-world, sparse, or random networks. This coefficient ranks a network with a score between -1 and 1, values closer to -1 pointing out a sparse topology, networks with values closer to 0 being considered small-world networks and networks with a rank closer to 1 having a random characteristic. ω is computed comparing the shortest path L and clustering coefficient C of a network G with the clustering coefficient of an equivalent lattice network (C_{latt}) and shortest path length from an equivalent random network (L_{rand}). The following formula defines this relationship:

$$\omega = \frac{L_{rand}}{L} - \frac{C}{C_{latt}} \tag{2.9}$$

Understanding the similarities between the topologies of the discussed network types, ω can be seen as a method to label network topologies that is not based on a binary-like logic but more likely on a gradient of truth methodology that is more flexible in the analysis.

We will make use ω to score the network studies in the experimental section and correlate its value with the discussed mesoscale property, namely network motifs.

Chapter 3

Parallel acceleration of subgraph enumeration in the process of network motif detection

Network motifs bring a great interest to many fields, because they are a perfect candidate to speed up the applied research in the understanding of complex networks dynamics. One of the biggest problems raised in the process of finding these motifs is the performance. Starting from splitting the initial network into subgraphs to the subgraphs clusterization, each algorithm used addresses an *NP* problem. Hence there are many algorithms that are trying to solve network motifs discovery, however there is no perfect approach talking from a time performance perspective, but only solutions that are faster than others.

In this chapter we describe a redesign of one of the most competitive algorithms for subgraphs enumeration found in the literature called ESU [136]. In the proposed approach a given network can be processed by using parallel programming, either with a process-driven model, or a hybrid one (process-driven and thread-driven). Finally, the existing and proposed models are compared using benchmark graphs, revealing competitive results.

The results were published in paper **[94]** and represent an original contribution to the existing state-of-the-art algorithms used to extract network motifs by significantly improving the running waiting time and not affecting the accuracy of the results.

Moreover, we consider the results proposed in this chapter as being complementary with all other experiments we led as it represented an indispensable tool we used to extract motifs from the networks we studied in the main research track.

3.1 Problem definition

Although the literature presents a list of algorithms that will follow the same two steps described in Section 2.6 which represents an *NP* problem individually. Each of these algorithms will encounter a performance bottleneck once the motif size scales to larger and larger values.

Due to its *NP* nature, in order to compute $F_G(G')$, no algorithm is better than others in a remarkable way. The existing solutions will encounter the same bottlenecks:

- memory performance problems - as the data structures used by the algorithms will increase exponentially in size once more subgraphs will be generated;
- time performance problems - the subgraphs extraction will be slow while processed using a single process;

Once graphs have a higher number of edges and nodes, these two key aspects will be more prominent. The algorithms that guarantee the maximum accuracy of the results obtained reach a level of exhaustiveness very fast.

Following an approach that implies a trade-off between accuracy and performance, some algorithms improve time performance based on a probabilistic method. Because it used just a partial sampling of the graph to do that, their results' accuracy will be influenced by the sampling size; hence their approach is not entirely reliable.

Considering the aspects described above, we can define the problem as being one of scalability. The current solutions encounter a bottleneck in memory and time performance or general accuracy through sampling due to the limitation in their sequential algorithmic design.

3.2 State of the art

Many NP-hard problems raised by the Computer Science community are problems about graph operations, and network motif discovery is one of them. In the literature we can find a list of algorithms which tackle the problem of motif discovery. Based on the paradigm used, each algorithm could be classified under one of these methods: counting methods, sampling methods and pattern grown methods.

Counting methods retrieve the entire list of motifs from a given network, without any statistical approximation, such that the results are completely accurate (mfinder [88], ESU [135], Grochow and Kellys' Algorithm [52], Kavosh Algorithm [69]).

A method that counts all possible subgraphs of a given size can be exhausting. Knowing that, there was proposed another paradigm which will sacrifice accuracy, to performance, by extracting a determined number of samples of subgraphs from a network and random networks with the same properties. Using these samples it is possible to compute an approximated $Z - score$, which is used to determine an approximate concentration of those subgraphs. This approximation will be more accurate as the number of samples is increased (Kashtan algorithm [70], Rand-ESU Algorithm [135], MODA Sampling [106]).

Pattern grown methods are models which use a reverse-engineering approach, that starting from an initial network, will grow it using a set of rules. After a given number of iterations the initial network will have a different structure which could reveal different motifs [16].

Although the paradigms are different, each algorithm follows almost the same steps, which are first to find and count the number of subgraph of size k (number of nodes) occurring in the network and then compute concentrations of subgraphs in respect with the entire network. We can say that a group of subgraphs with the same properties is important if it is widely spread in the network and has many occurrences.

Even if counting methods are exhaustive, they reach the ultimate goal of having results which are not approximated and give an exact accurate representation of subnetworks census.

From the existing list of counting algorithms proposed to find motifs, this thesis will focus on an algorithm called ESU proposed by Wernicke S. [136], which is one of the best candidates to be redesigned as a parallel model. This solution is a network motif discovery algorithm which works for networks that have a mandatory property, that all nodes need be indexed with numeric values.

During this algorithm run, a structure called ESU-tree (see Figure 3.1) is created, with a remarkable property which it will be later used to create parallel models. ESU-tree will contain $k-1$ levels in depth, k representing the number of nodes of the subgraphs we want to obtain.

The algorithm (see Algorithm 1) starts with a set of tuples positioned on the first level of the tree (see Figure 3.1), each tuple containing on the left-hand side a node of the network and in the right-hand side set its adjacency list with the property, that the list will contain only nodes with an index greater than the node index found in their corresponding left-hand side set.

These initial tuples are iterated recursively by adding one index from the right-hand side set at a time to the left-hand set, and computing a new adjacency list for all nodes found in

Algorithm 1 EnumerateSubgraphs(G, k) (ESU)

Input: Graph G =(V,E) and an integer $1 \leq k \leq |V|$
Output: All size-k subgraphs in G
for each vertex $v \in V$ **do**
    $V_{Extension} \leftarrow \{u \in N(v) | u > v\}$
    call ExtendSubgraph($\{v\}, V_{Extension}, v$)
end for
ExtendSubgraph($V_{Subgraph}, V_{Extension}, v$)
if $|V_{Subgraph}| = k$ **then**
    return $G[V_{Subgraph}]$
else
    while $V_{Extension} \neq \emptyset$ **do**
        Remove an arbitrary vertex w from $V_{Extension}$
        $V'_{Extension} \leftarrow V_{Extension} \cup \{u \in \{V_{Subgraph} \setminus \{w\}\} | u > v\}$
        return ExtendSubgraph($V_{Subgraph} \cup \{w\}, V'_{Extension}, v$)
    end while
end if

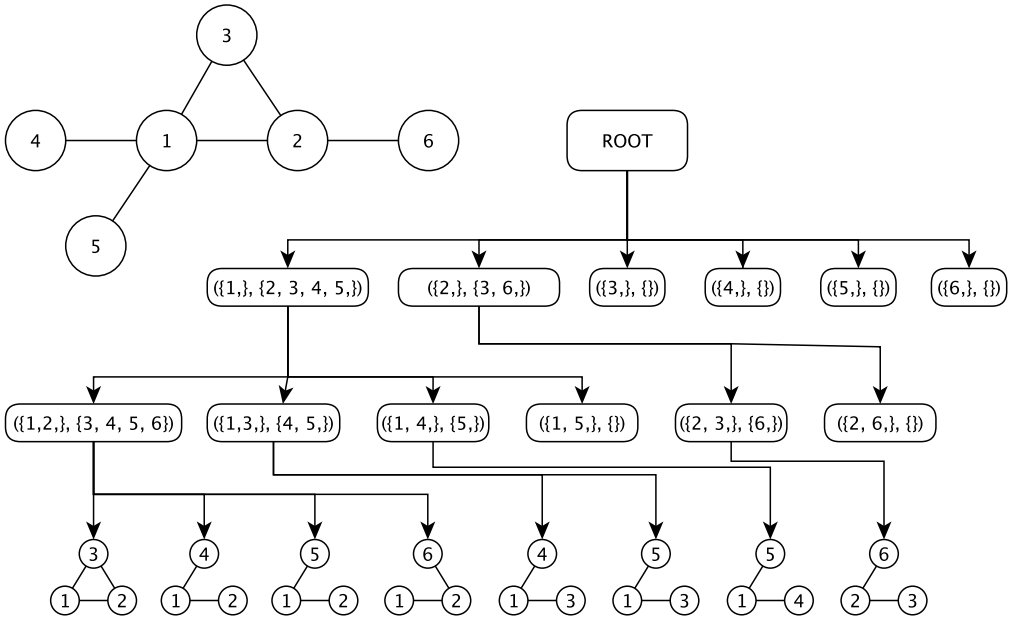


Fig. 3.1 ESU tree structure example

the new left-hand side set with the property presented earlier. Recursion stops when ESU-tree reach depth equal to value $k-1$. As a final step, all final tuples are used to expand subgraphs, by considering the nodes from left-hand side list and one node from the right-hand side list at a time.

Because on each level nodes from the left-hand side will have associated a list of nodes which have indices greater than their indices, ESU prevents subgraphs over-counting since it is not possible to expand the same subgraph from different ESU-tree branches.

After all possible tuples are expanded and subgraphs obtained, motifs will be obtained while computing the concentrations of the resulted subgraphs, using McKay's Canonical Labeling Algorithm [83], but this thesis focuses only on the process of subgraphs enumeration.

3.3 Proposed approach - Accelerating ESU

Compared to existing counting algorithms in the literature, ESU (Enumerate Subgraphs) [136] is one of the fastest solutions, for motifs sizes not greater than 8 [132]. Unfortunately, when implemented as a serial solution, ESU encounters hardware resources limitations. Once motif size becomes larger and larger, it follows the trend to be a solution that is memory exhaustive and slow. During the process of ESU-tree expansion two key aspects are brought into discussion, one is the memory performance as the ESU-tree data structure will increase exponentially once a new level is expanded, and second the time performance of ESU-tree expansion which will be slow while processed using a single process. Once graphs or the networks will have a higher level of edges and nodes concentration, these two key aspects will be more prominent.

Historically, parallelization was used as a mechanism of solving hardware resource limitations as the ones presented above. Making use of power offered by clusters of computers, algorithms that needed lots of memory or computational power, were redesigned into models that were distributed on clusters.

The same approach can be applied also in our case, because ESU has potential to be redesigned into a pure parallel programming model [32] due to the data structure it creates (See Section 3.1). Map-Reduce [30] is a processing technique used in distributed computation, which is a great candidate as a pure parallel model for redesigning ESU. This technique consists in two main steps:

- *map* - takes an input and generates a list of key/value pairs
- *reduce* - takes the key/value pairs generated by the mapper and creates an output using a given functionality

Literature presents a such parallel model that used Map-Reduce to redesign ESU [128]. Starting from the serial version of the algorithm, the authors of [128] used an approach of node-based scaling to process the ESU-tree expansion in parallel. Either if the model brought a significant improvement compared to the classical implementation, we think that once it is implemented and ran into a real-world ecosystem, this node-based scaling does

not benefit entirely from a parallel scaling capability. Although the parallelization factor would be a maximum of n (n being the number of nodes from an ESU-tree level), the node oriented parallelization will be always limited to a single level of the ESU-tree. Also it will be required a new Map-Reduce job for each level, which could lead to a dangerous overhead. Considering these facts, a branch-oriented scaling would bring a more powerful impact on the performance.

Even if parallelization orientation might have an important impact on the performance, both branch-based and node-based approaches, encounter the same weak point, which is the parallelization capability that is not entirely reached by none of these paradigms. A solution would be to combine both of these scaling paradigms, in order to create a more complete solution. A so called hybrid model will first follow the branch-based scaling approach to split the ESU-tree into an equal number of branches for each parallel process, then each parallel processed branch will be expanded node-based using threads following ESU algorithm (See function ExtendSubgraph from Algorithm 1).

No matter which paradigm is used, a parallelization approach is one of the best candidates to speed up the process of network motifs discovery. Comparing all these proposals using real implementations, it will be possible to find weak points, which later will be used to design a model closer to a more bulletproof and trustworthy solution, talking from a time-oriented performance perspective.

3.3.1 Subgraphs enumeration - Node-based parallelization

The model presented in this Subsection was firstly described in the paper [128] as a way to parallelize ESU algorithm. It makes use of Map-Reduce as technique to parallelize the initial model, containing a sequence of Map-Reduce jobs:

- ESU Job - this job will be run iteratively for each ESU-tree level until it reaches a k level (k representing motif size)
- Labeler Job - uses McKay's Canonical Labeling Algorithm [83] to label each generated subgraph
- Combiner Job - combine generated subgraphs using their labels into clusters, to calculates the frequency of each subgraph type

As ESU Job description says, this model will generate a Map-Reduce job for each ESU-tree level (See Figure 3.2).

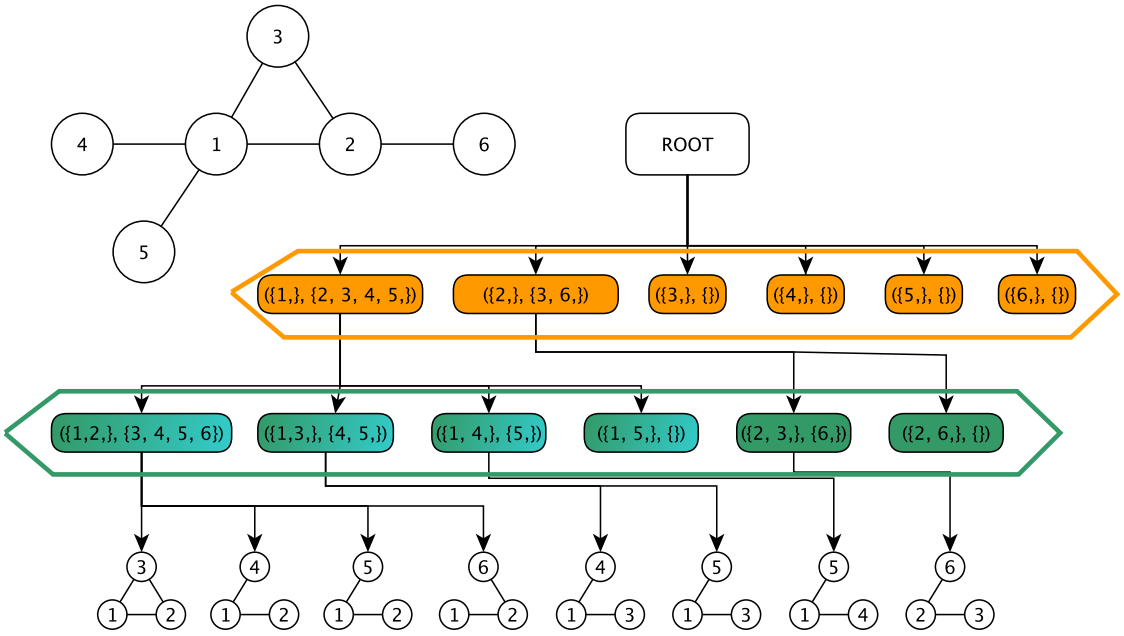


Fig. 3.2 Node-based scaling model applied to ESU-tree using Map-Reduce

This model processes in parallel nodes of an ESU-tree level . Although this approach use one of the finest granularity in terms of the parallelization of the ESU-tree, the scaling capacity is bound anyway to the ESU-tree levels. In theory, once a node from a given level is finished to be processed, its expansion can be applied to the next level and so on. Thus a Map-Reduce job is started for each level of the tree, a node expansion cannot be moved to the next level until the processing of all nodes from the current level is finished.

Implemented as a real-world application, this model will encounter a problematic processing step brought by the new Map-Reduce processes which need to be started after each level of the ESU-tree. Unfortunately, this overhead eliminates almost completely the benefits of the granularity factor brought by the parallelization of the nodes-based processing for an ESU-tree level. This assumption will be proved into a future section which contains a set of tests for different graphs and motifs sizes.

3.3.2 Subgraphs enumeration - Branch-based parallelization

Knowing that branches will expand an unique subgraph which will not be found in any other branches, we can use a similar approach as the model from Subsection 3.3.1, but using ESU-tree branches as level granularity, having as a start point the first level of ESU-tree (see Figure 3.4).

A great candidate that can be used to parallelize ESU-tree data structure as it was presented in Figure 3.4 is Map-Reduce, which is a distributed processing technique. Using it, the list of tuples created in the first level of ESU-tree (ESU root components), will be used as input to the Map-type processes, which will transform them into key/value pairs sent to Reducer-type processes that will apply ExtendSubgraph function (See Algorithm. 1) for each one of those pairs (See Figure 3.3).

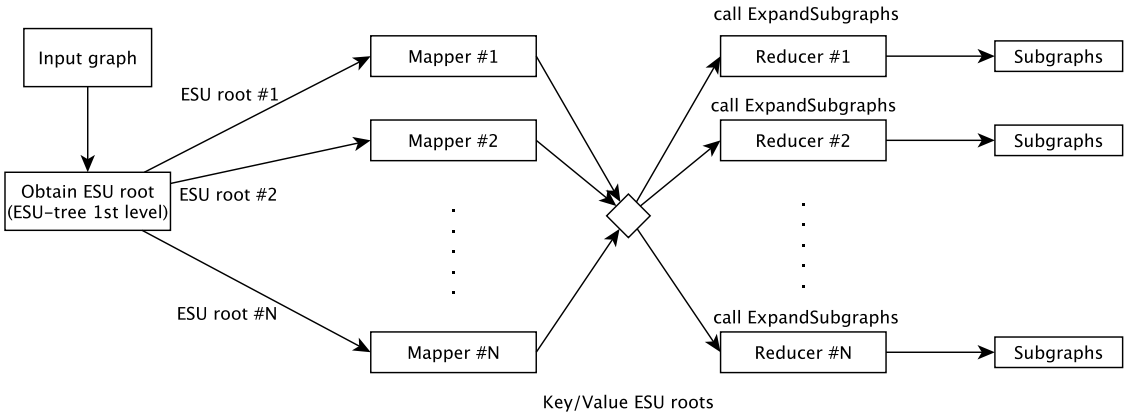


Fig. 3.3 Branch-based scaling model applied to ESU-tree using Map-Reduce

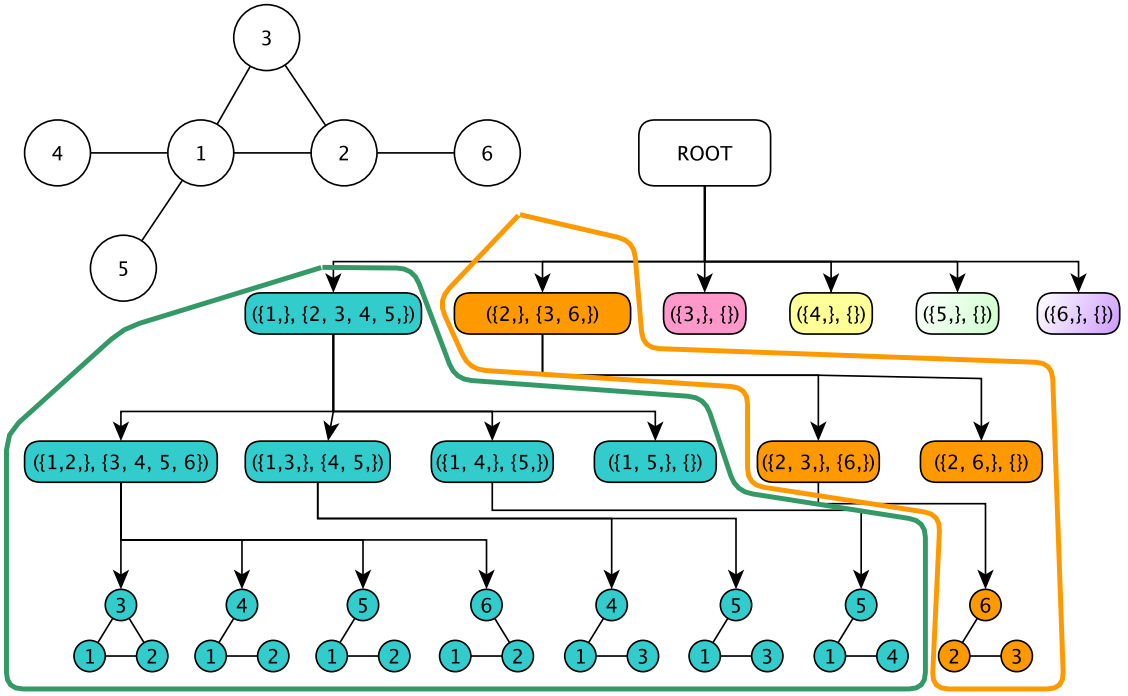


Fig. 3.4 Branch-based scaling paradigm applied to ESU-tree

The scalability factor in this model is proportional with the number of reducers that can process ESU root components in the same time. Taking a simple example, using the ESU-tree presented in Figure 3.4, it is recommended to have 1 to 2 reducers, as there are only 2 ESU root components which are real candidates for subgraphs enumeration (a tuple is considered a real candidate, if it has at least one element in the right-hand side tuple).

The simplicity of this technique comes with a great advantage in matter of scalability - a model implementing Map-Reduce can be easily scaled to thousands of processes. Also a noticeable mention is that only a single Map-Reduce job will be started with Reducers associated to each branch of the ESU-tree (see Figure 3.4), an approach which was missing on model from Section 3.3.1.

One of the biggest downside of this solution is caused by a recurrent problem. Even if we used ESU-tree branches to process them in parallel in order to obtain performance, this parallelization is not balanced, as a branch can have more nodes than others. This is a visible trend in the first ESU root components, caused by the ESU algorithm property which says that in the right-hand side tuple will be only nodes with indexes greater then the indexes of nodes from the left-hand side tuple.

This downside is partially solved by the node-based scaling parallelization which was discussed in the previous Subsection 3.3.1.

3.3.3 Subgraphs enumeration - Hybrid parallelization

Both node-based and branch-based parallelization models have benefits and trade-offs. In Subsections 3.3.2 and 3.3.1, we observed that the branch-based model proposes a scalable solution which is able to process the ESU-tree in parallel using the list of branches as input to a Map-Reduce job, but it comes with a inconvenience caused by the unbalanced volume of processing done by a parallel process at a given moment. Instead, node-based model seems to have a better approach for balancing, by isolating each parallel process at ESU-root level. Its inconvenience is caused by the overhead brought by each process of Map-Reduce which needs to be run for every ESU-tree level. Moreover, the scalability is not reached to a maximum potential by this fact.

Taking into consideration the benefits of both models, this section presents a new proposed model which uses a hybrid parallelization approach. We call this model hybrid as it was designed as a mix of node and branch based models presented in the Sections 3.3.1 and 3.3.2.

We can observe that level by level once an ESU-tree branch is expanded (Figure 3.6 and

3.7), it becomes by it self a tree with the same structure as the initial ESU-tree, hence we can call it a sub ESU-tree.

Starting with the branch-based parallelization, this sub ESU-tree would be associated to a Reducer which will expand it until all sub-graphs were obtained. The problem that comes with this approach was explained in Section 3.3.2 and it is related to the processing of the sub ESU-tree levels. Even if the branches are processed in parallel, to obtain a better performance of the parallelization, each branch expansion can be processed in parallel as well, thus using a node-based parallelization (Section 3.3.1).

This model will use processes (reducers) to process the ESU-branches and threads to expand each branch as a sub ESU-tree (Figure 3.6 and 3.7).

Following and extending the concept discussed in Subsections 3.3.2 and 3.3.1, Figure 3.6 gives us a visual perspective of how each branch will be also processed in parallel.

Figure 3.7 presents an isolated branch of the ESU-tree, in order to display a better explanation of model's parallelization paradigm, seen from the perspective of processes types.

We can see that the hybrid model tries to tackle the unbalancing issue raised during branch-based modeling, by making use of more parallel computation approaches that can be applied to diminish the mentioned problem.

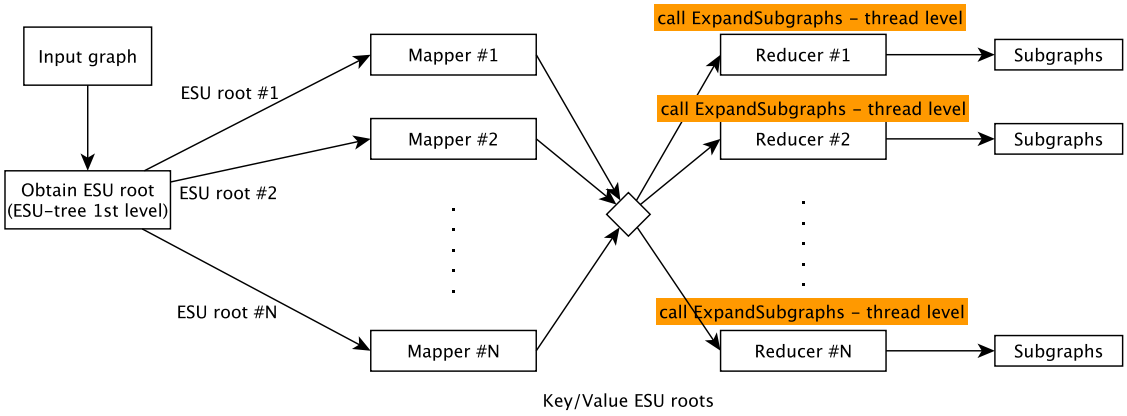


Fig. 3.5 Hybrid scaling model design

3.4 Experimental analysis

In order to have a validation for the theoretical assumptions formulated as comparison between the models described Subsections in 3.3.2, 3.3.1 and 3.3.3, this section's focus is to

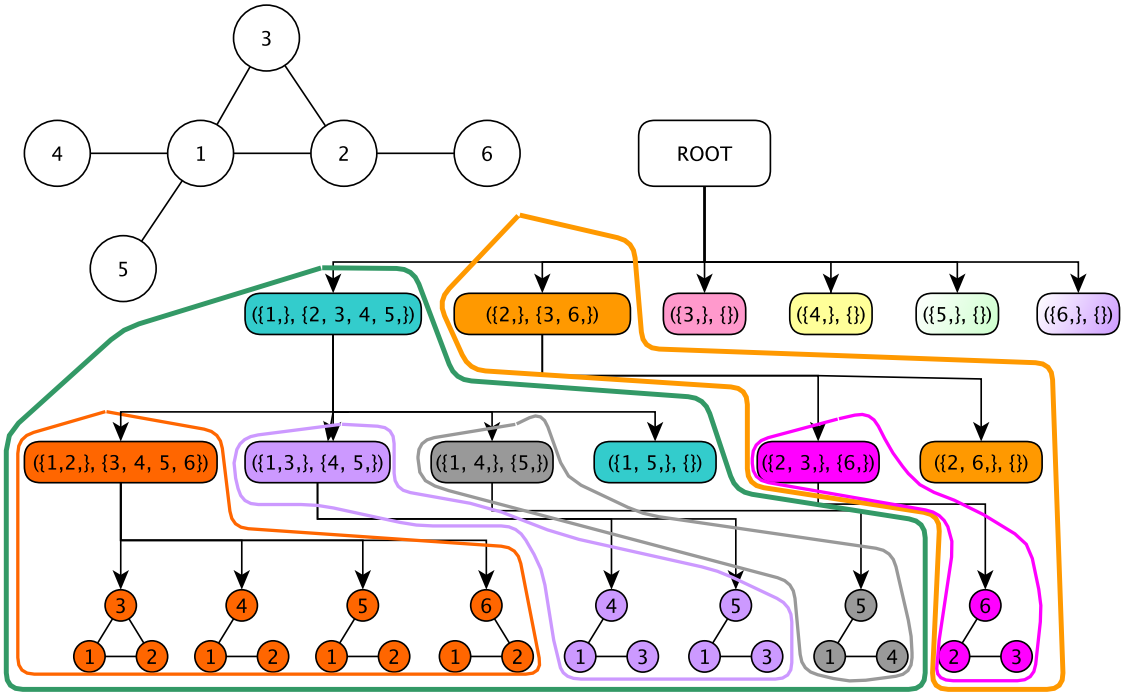


Fig. 3.6 Hybrid scaling model applied to ESU-tree

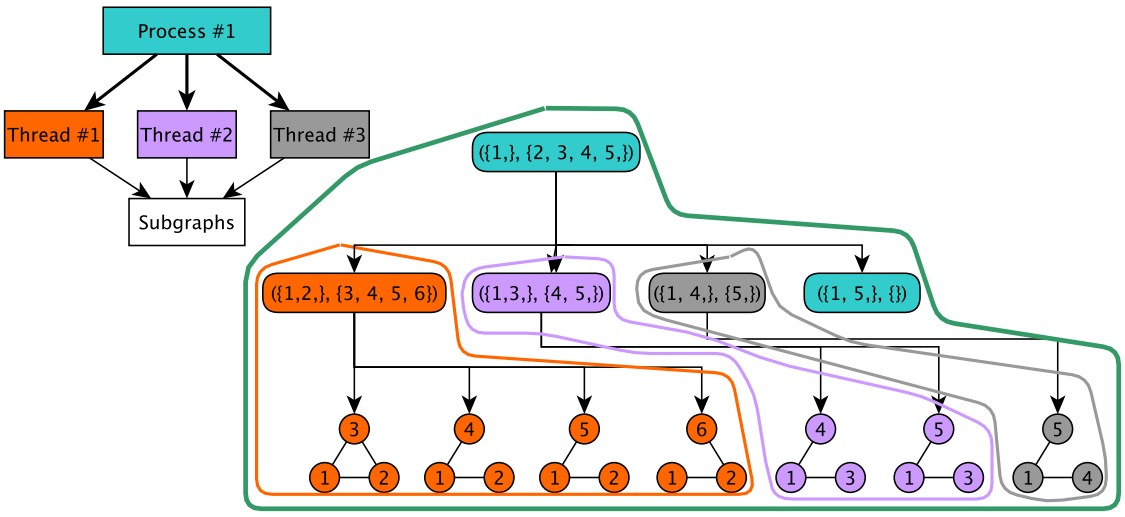


Fig. 3.7 Hybrid scaling model - parallel processing perspective

present the results of multiple test trials ran on a dataset consisting of 9 different real-world networks.

3.4.1 Methodology

The main criteria that will be used as comparison will be time performance represented in seconds, considering the model type and the network configuration. Being a research done on a time performance optimization problem, test trials should reflect a quantification of the models performance related to execution times.

This research compares three different parallel models which want to improve an existing algorithm of network motifs discovery. All three models make use of parallelization, the only aspect that differentiates them being the paradigm of parallelization which they apply.

There are two conjectures that test trials would like to confirm. One is the assumption that even if the branch-based scaling model deals with an unbalanced amount of processing for parallel processes, it is still more powerful than the node-based scaling paradigm implemented by V. Verma [128] (*H1*). Second assumption is formulated for the comparison between the hybrid model and the other two models, which should be less powerful than the first one. Due to its nature of modeling, the hybrid scaling model merged the best features of node-based and branch-based ones (*H2*).

During the test trials we encounter a list of independent variables like: network configuration (number of nodes, number of edges, network coefficients), the configuration of the cluster used to run the tests (number of cores, RAM size, networking). Based on this, we expect to have variations on the models' execution time which represent a dependent variable in this scenario.

As test trials are done for each combination of network and model type, time performance results represented as seconds are gathered in a table 3.1. For each of these combinations, multiple test trials were run in order to be sure that we have accurate results. Later we will compute a mean and a 95% percentile between these values, which will be used to do the final comparisons between models.

3.4.2 Data

The dataset used to run test trials, contains a set of 9 real-world networks (see Table 3.8 for the more specific properties of these networks) with low to medium sizes. These networks have been open-sourced by Stanford University [78] and represent different industry databases which have interactions that can be represented as networks.

These networks have various configurations (number of nodes, number of edges, Clustering Coefficient) which indicates that the dataset used for test trials is not homogeneous. One of the best indicators for the motifs discovery is Clustering Coefficient, which is a

measure of the degree to which nodes in a graph tend to cluster together.

Network Name	ID	Type	Nodes	Edges	CC_l
Gnu,Aug 4 2002	gnu-04 [78]	directed	10876	39994	0.0062
Gnu,Aug 5 2002	gnu-05 [78]	directed	8846	31839	0.0072
Gnu,Aug 6 2002	gnu-06 [78]	directed	8717	31525	0.0067
Gnu,Aug 8 2002	gnu-08 [78]	directed	6301	20777	0.0109
Gnu,Aug 9 2002	gnu-09 [78]	directed	8114	26013	0.0095
Gnu,Aug 24 2002	gnu-24 [78]	directed	26518	65369	0.0055
Gnu,Aug 25 2002	gnu-25 [78]	directed	22687	54705	0.0053
Gnu,Aug 30 2002	gnu-30 [78]	directed	36682	88328	0.0063
email-Eu-core	email-core [78]	directed	1005	25571	0.3994

Fig. 3.8 Networks used in test trials

3.4.3 Numerical results and discussion

The methodology applied to obtain the following set of results, was to run each model described in the previous sections against all networks from Table 3.8 in order to obtain a performance measured in seconds. Each run was repeated for 3 times, for a range of motif sizes (k) between 2 and 5.

Each Map-Reduce model was implemented in Go programming language [33] and it was run in a Hadoop ecosystem [127] created in Amazon Web Services (AWS) with 3 nodes (16 vCore, 32 GiB).

All parallel models used a parallelization factor of 15 Reducers ran in the same time. Although the theoretical maximal parallelization factor can be larger for Node-based model compared to Branch-based model, applied into a real world application, the Node-based model needs to start Map-Reduce jobs for each ESU-tree level and each Map-Reduce should wait the previous ESU-tree level to be processed in order to start its execution. The expansion per Map-Reduce job will be always limited to only one iteration, until all nodes are expanded - operation which can be too atomic and simple (for one iteration) to have a real benefit from being parallelized.

Considering this explanation, it was considered a fixed parallelization factor as the real problem to be highlighted during the test trials was the existence of the overhead brought by spawning new Map-Reduce jobs for each level. Keeping the node scaling constant and raising the number of ESU-tree levels (by raising the size of motifs) it is desirable to indicate that the overhead added by spawning of new Map-Reduce jobs is clearly in the detriment of the Node-based scaling.

Of course parallelization factor could be increased for Node-based approach in order to slowly remove the problem of the overhead (although not completely), but because in a real-world application we are always tight to limited computational resources, tests results want to prove that with the same computation power it can obtained a better performance using Branch-model and Hybrid-model.

During test trials each model was run on each one of the 9 networks described in Table 3.8. A test trial was considered complete only after the model was run for 3 times for each motif size configuration (2, 3, 4 and 5). If a test trial exceeded 300, respectively 600 seconds for motifs with size 4, respectively 5, by default we considered the performance being the corresponding limit, 300 seconds for motif size 4 and 600 seconds for motif size 5. These defaults were chosen as thresholds from a time-saving perspective of test trials and they want to point out that the test run is exhaustive if the threshold was reached.

Following test trials' results, *H1* is confirmed by checking the significant time difference between branch-based model and node-based model (See Figure 3.18 and Table3.1). Another statistical confirmation which strengthens *H1* can be obtained by doing an analysis of the Confidence Interval (95%) for each combination of model type and motif size k. This analysis reveals that the Confidence Intervals for node-based and branch-based models are disjoint for $k > 2$, with more time efficient values in favor of the results of branch-based model, hence we can say that the branch-based model is better than the node-based model once motifs have a significant size (See Table 3.2). Initial assumption was formulated using reasons mentioned in Section 3.3, which were namely about the scalability potential of branch-based model, which will process in parallel all ESU-tree branches and the overhead added by the mechanism of creating a new Map-Reduce job each time the node-based model processes a level. All these together, were proved by running these test trials on graphs with different configurations.

On the other hand, *H2* was not confirmed entirely. In all test results it is clear that hybrid approach is a more powerful computational solution than the node-based model, but not more powerful than the branch-based model. Moreover, on graphs like gnu-08 [3.12] branch-based model performs better. A possible reason behind this might be the fact that the configuration of graph does not have any benefit from the splitting the ESU-tree branches into small components that are processed by threads, as Hybrid model design proposes (See Figure 3.7). For this example, a good clue is the network's number of edges, which is the smallest from the entire list of networks from Table 3.8, and a higher CC_l which indicates that ESU root components will not have a large set of nodes in their structure, hence there is no need for more computational power to process them.

Another aspect that could have an impact on the Hybrid solution is the environment in which it was run. Hadoop [127] is a high level ecosystem which brings a lot of benefits for productivity, but not so much to the high performance computing. It is not very clear how it manages the threads started in the Hybrid model, hence there is no guarantee that the model reaches its full potential.

Graph	Model type	K=2 Mean	K=3 Mean	K=4 Mean	K=5 Mean
gnu-04	Branch-based	55.00	57.67	63.67	169.00
	Node-based	56.33	103.33	300.00	300.00
	Hybrid	55.67	56.00	63.33	167.33
gnu-05	Branch-based	54.67	55.33	63.00	210.67
	Node-based	55.67	106.67	221.33	600.00
	Hybrid	55.33	55.33	63.33	213.00
gnu-06	Branch-based	54.33	55.67	61.67	186.67
	Node-based	60.00	106.67	208.00	600.00
	Hybrid	56.00	55.67	61.67	190.33
gnu-08	Branch-based	55.67	56.33	65.33	312.67
	Node-based	55.67	108.33	239.67	600.00
	Hybrid	54.67	56.00	62.33	366.67
gnu-09	Branch-based	55.67	55.67	65.33	346.00
	Node-based	55.67	108.00	249.67	600.00
	Hybrid	55.67	56.33	64.00	329.67
gnu-24	Branch-based	56.00	57.67	115.67	600.00
	Node-based	56.00	107.00	300.00	600.00
	Hybrid	56.33	57.67	114.00	600.00
gnu-25	Branch-based	57.33	57.00	95.33	120.67
	Node-based	55.67	107.00	176.00	600.00
	Hybrid	55.67	58.00	62.67	110.00
gnu-30	Branch-based	55.67	57.00	65.67	173.33
	Node-based	55.67	109.33	209.33	600.00
	Hybrid	57.33	58.00	64.67	172.00
email-core	Branch-based	35.33	34.67	35.33	35.00
	Node-based	36.67	74.67	114.00	151.67
	Hybrid	36.33	36.33	36.33	37.00

Table 3.1 Test trials results of the proposed models

Model type	K=2		K=3		K=4		K=5	
	Mean	95% CI	Mean	95% CI	Mean	95% CI	Mean	95% CI
Branch-based	53.3	48.9 to 57.7	54.11	49.3 to 58.9	70.11	55.2 to 85	239.33	132 to 346
Node-based	54.15	49.8 to 58.6	103.44	95.9 to 110	224.22	186 to 262	516.85	407 to 627
Hybrid	53.67	49.4 to 58	54.37	49.9 to 58.9	65.81	52.7 to 78.9	242.89	133 to 353

Table 3.2 Proposed models comparison based on Confidence Intervals (95%)

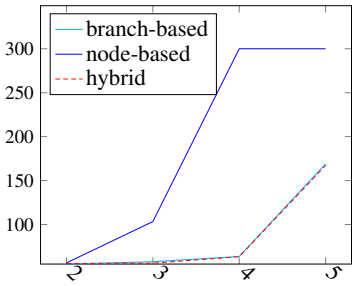


Fig. 3.9 gnu-04

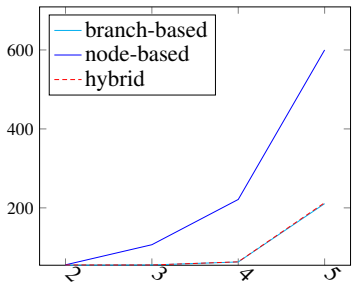


Fig. 3.10 gnu-05

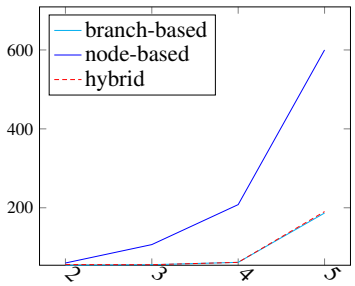


Fig. 3.11 gnu-06

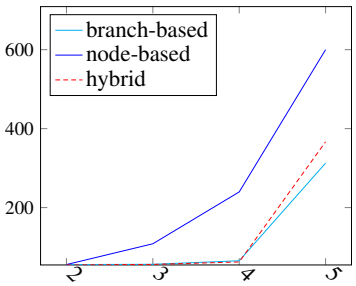


Fig. 3.12 gnu-08

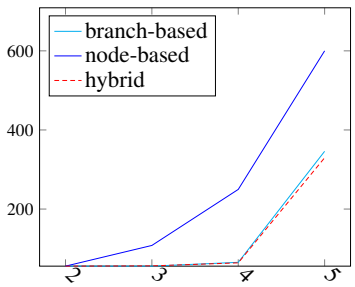


Fig. 3.13 gnu-09

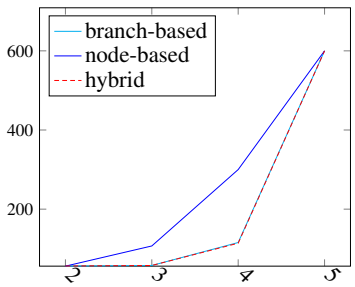


Fig. 3.14 gnu-24

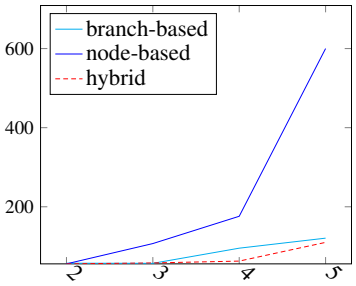


Fig. 3.15 gnu-25

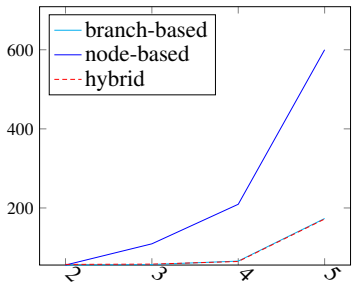


Fig. 3.16 gnu-30

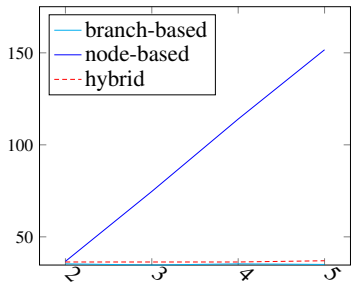


Fig. 3.17 email-core

Fig. 3.18 Plots of test trials for each network and proposed model (O_x - Motifs size, O_y - Seconds)

3.5 Conclusion

The purpose of this work was primarily to come up with a contribution to the network motifs discovery field, and secondly to improve the waiting time existing in the experiments led in the primary research track of this thesis. Making use of high performance computing and parallel programming, we proposed two new models which represent a parallel redesign of an existing algorithm from literature called ESU [136]. Models are implemented using different scaling paradigms, branch-based respectively hybrid, which are widely explained in previous sections. Using an existing parallel model, which redesigned ESU as well but following a different paradigm (node-based)[128], in Section 3.4 were formulated two con-

jectures:

1. Branch-based scaling model should perform better than the node-based scaling model;

2. Hybrid model should perform better than the branch-based scaling and node-based scaling models;

Then using a set of networks with different configuration, a rich set of results was obtained and analyzed in Section 3.4.3. Using the set of results, *H1* 1 was confirmed. *H2* was partially confirmed, as the Hybrid model performed significantly better than the node-based model [128], but not that different than the branch-based model. A reason behind this behavior might be driven by the graph configurations which were used in the trials, which did no require a processing at ESU-tree branch level. Another reason might be the environment where the model was run, which is a high-level ecosystem where threads and processes are not controllable by external factors. This is why it is not known whatever the threads' model reached their full potential.

This work represents an original contribution to the network motifs discovery field using two new parallel models which were validated and compared to a similar solution from literature [128]. Comparisons proved that both models are more powerful than this existing solution. Moreover, the Hybrid model has potential to become more powerful once it will be implemented on a more low-level ecosystem.

Chapter 4

Articulation Points and Isolated Communities: Catalysts for Network Motif Evolution

Complex networks are powerful mechanisms one can use to model real-world networks as topological spaces. The beauty of these structures is provided by the infinite degree of analysis one is allowed to do using them. Biologically it is almost impossible for the human mind to comprehend the behaviour of these systems, but when modelled as complex networks different properties of the network topology can reveal precious information. Starting from the two key properties of the participants in a complex network and the relations between them, one can derive further properties that reflect specific behaviours for entities or groups of entities.

Examples of these further remarkable properties include entities which create unique bridges between two or more communities (known as articulation points) or the appearance of patterns of interconnections between entities (known as Network Motifs).

In this chapter it is presented a study on the co-existence of these two properties, articulation points and network motifs, and how their appearance is correlated, by using results obtained in analysing a variety of real-world networks.

The results of this experiment were published in paper **[97]** and represent a first step in our research project that is willing to prove that meso-structures such as network motifs are key components facilitating the dynamic propagation between network layers.

4.1 Understanding the Problem Landscape: Defining the Current State of the Art

Complex networks are data structures that can be represented as graphs having non-topological features that occur only in modelled networks or real-world systems, such as the World Wide Web, chemical systems, financial, social or biological networks [119]. The modelling of a real-world system as a complex network can reveal a set of properties about the system structure (number of entities, number of connections between entities, density of connections etc.). These properties raise the interest of the studies as they are affected directly or indirectly the dynamics of the system.

An example of such a property that raises this interest is where the removal of an entity leads to the disconnection of the network [123]. In an abstract way, we can view these entities as points of interest or articulation points. In real-world applications we can find these specific entities almost all the time. Depending on the context presented by the real-world application these entities can be seen either as influencers (such as popular persons in social networks) or vulnerability points (a router which links all computers from a network).

Often in social real-world systems, influencers can also be seen as opinion leaders with a gravitational force over their community of followers, a force which can attract new entities creating more and more connections, depending on the influencer's ideologies or behaviour (an entity with more connections will have a stronger gravitational force than an entity with fewer connections).

Knowing this, in this chapter it is presented a study and a hypothesis that it seeks to prove it, which states that articulation points can spread behaviours to the entities with which they have connections. An analogy that can serve as an example is the role of an influencer in a social network, which can easily induce ideologies to its followers. With a large number of followers, an influencer's ideology can get more weight and trust in the community. The trend of persisting an influencer's ideology can also be continued by its followers and then their own followers and so on.

We considers these behaviours injected by articulation points to their neighbors as patterns of connections which can be detected in the network structure. The study of these patterns of connections or network motifs [88] reveals their significant weight in guiding complex network dynamics.

Recently, more and more studies [50][45] are focused on proving that the current mechanisms of generating realistic networks are based on theories which are not empirically realistic. An example of such a theory, which can be considered the foundation of the lat-

est 20 years of studies in the field, is Scale-free network Theory [11]. Unfortunately, recent studies [29] revealed that these kinds of networks are quite rare in real-world systems, hence the community needs to come with new mechanisms of generating more realistic networks.

This research project aims to serve as a platform for new proposals that will make use of the most powerful concept of a network, the connection between its nodes. Our proposal contends that using simple or non contextualized information, such as node degree, loses a lot of topological and spacial network context, which we now know is very important in real-world networks.

The goal of this study is to indicate that complex structures like motifs, correlated with other key concepts of networks, can be used as inputs to new and revolutionary models which will generate networks which more closely resemble a real-world behaviour. This work intends to be part of a series of experiments that will like to serve as a theoretical and statistical foundation for our main research goal which is to propose a generator model for networks, with specific concentrations of motifs.

The proposed study is heading to an unexplored subject of the Complex Network Theory, as from our knowledge there is no research which considers a study of the correlations between articulation points. Considering this void in the domain's literature, a study like the one presented in this chapter is crucial to our long-term goal of proposing the network generator model.

4.2 Theoretical insights

In order to better understand the methodology of the analysis approach in the next section a few key theoretical concepts will be introduced in this section:

4.2.1 ESU-tree

To prove that the remote modules that an *AP* link are prone to evolve network motifs, we introduce the step of subgraph enumeration presented on algorithm 1 as a method to split the network and isolate each module that *AP* connects for the analysis described in the further sections.

This algorithm generates a structured called an ESU-tree (Figure. 3.1) which has a tree-like structure with $k-1$ levels, k representing the size of subgraphs that it was asked to enumerate.

ESU (Algorithm. 1) starts by computing the root of the tree, as a list of tuples for each graph node with the following structure:

- left-hand side list: starting node (root node r)
- right-hand side list: adjacency list of the node from left-hand side list, with the property that the list contains only neighbors with an index greater than that of the node from the left-hand side list

This list of tuples generated in the ESU-tree root is expanded recursively one by one, by adding one node at a time from the right-hand side list to the left-hand side list and then recomputing a new adjacency list for all nodes found in the new left-hand side list. When recursion generates a tree with $k-1$ levels, the algorithm expands the tuples into subgraphs for each combination of nodes from the right-hand side list and one node from the left-hand side list.

In the ESU-tree we use the term branch (ESU_{branch_r}) for the structure expanded from an root node r until it reaches a complete state with a depth of k-1.

A remarkable property of this ESU-tree structure is opportunity to enumerate unique subgraphs for each branch, enabled by the condition of having on the right-hand list only elements with an index greater than those of the elements from the left-hand side list [70].

This property represents an important factor that guarantees the correctness of the analysis approach, presented later in our methodology which uses as its main data structure the ESU-tree and its branches.

4.2.2 ESU-tree branch proximity area

Starting from the ESU-tree structure, we propose a new concept named ESU-tree branch proximity area ($ESU_{PA}(ESU_{branch_r}, L)$) as the connected component built for a ESU-tree branch ESU_{branch_r} with a depth L.

We can define the ESU_{PA} of a given ESU_{branch_r} as a connected component C found in a graph G, where $V(C) \subseteq V(G)$ and $E(C) \subseteq E(G)$, and where V and E are the set of vertices v, respectively, of the set of edges e of the graph G and a

$$max(d(ESU_{branch_r}, v)) \leq L$$

where r, $v \in V(C)$ and r is root node of the ESU_{brach_r}.

Figure. 4.1 offers a visual representation for the proximity area generated for the ESU-tree branch that starts with node 1 from the left-hand side network ($ESU_{PA}(ESU_{branch_1}, 2)$). The connected component of $L = 2$ has root node 1 connected directly to its neighbours (orange) and then indirectly to its neighbours of neighbours (purple).

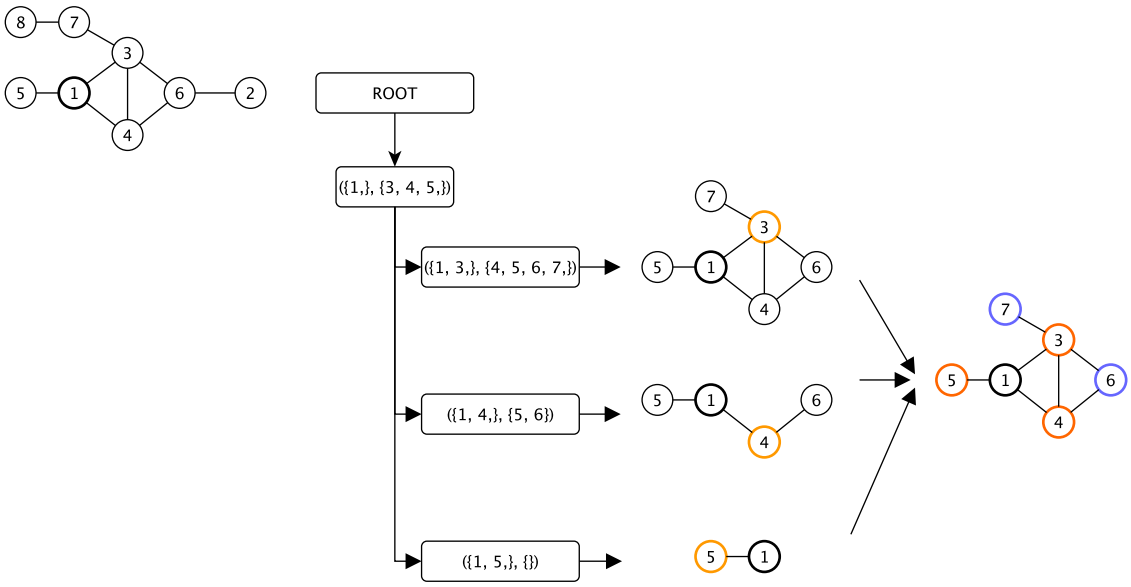


Fig. 4.1 Level 2 proximity for ESU_{branch_1} - with black level 0 or root node, with orange level 1st nodes or direct neighbours, with purple 2nd level or neighbours of neighbours

Although proximity areas cannot be seen as completely isolated connected components, due to the fact that they might overlap with other proximity areas generated for another ESU_{branch_r}, the overlap is minimized by the ESU-tree structure which will generate adjacency lists that will contain only neighbours with an index greater than a nodes' index from the left-hand side list (Section 4.2.1) - the same property that guarantees the enumeration of unique subgraphs on each ESU_{branch_r}.

When considering a complete overlap between two ESU_{PA}, f and g, we refer to sub-ESU_{PA} the one ESU_{PA} which is completely included in the other one (E.g: f is sub-ESU_{PA} if $f \subset g$).

4.2.3 ESU-tree branch Articulation Points expansion

Another measure that we introduce is the Articulation Points Expansion ($ESU_{AP_{exp}}$) of a given ESU-tree branch ESU_{branch_r}, which represents the difference between the number of

APs (AP_{no}) from an ESU_{branch_r} depth $k-1$ and the number of *APs* from depth 1 of the same ESU_{branch_r}:

$$ESU_{AP_{exp}}(b) = AP_{no}(b_{k-1}) - AP_{no}(b_1), \tag{4.1}$$

where b is an ESU_{branch_r}.

The scope of this measurement is to help better understand the importance of *APs* found in the proximity area of an ESU_{branch_r}, such that a proximity area which gains more *APs* while expanded is prone to produce more subgraph motifs.

Figure 4.2 visually explains how $ESU_{AP_{exp}}$ is computed for two of its branches.

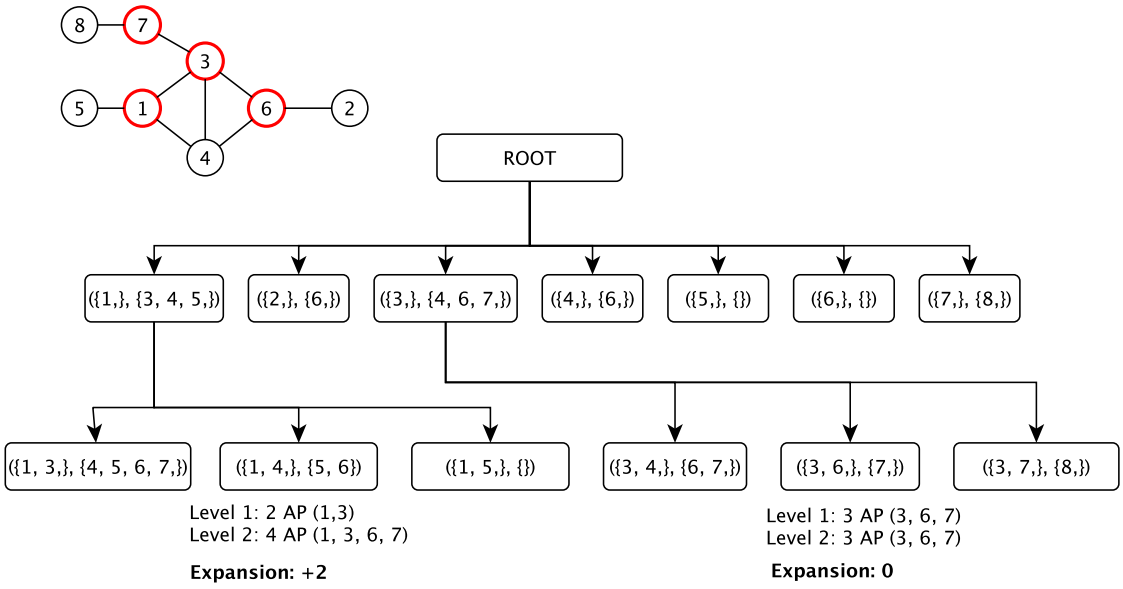


Fig. 4.2 ESU_{PA} computation for ESU-tree branches

4.3 Research questions

The previous section was intended to introduce the reader to a list of key concepts that will be used in the research analysis approach.

Our analysis is mainly focused on finding correlations between the number of motifs and the number of *APs*. The analysis uses as its primary data structure the ESU-tree (Section 4.2.1) over which we formulate the following notions:

- *LAP* - ESU_{branch_r} that starts with an *AP* only in left-hand side list (Figure. 4.3 - purple color)
- *RAP* - ESU_{branch_r} that starts with no *AP*s in left-hand side, but at least one *AP* in the right-hand side list (Figure. 4.3 - turquoise color)
- *BAP* - ESU_{branch_r} that starts with *AP*s in both the right-hand side and the left-hand side list (Figure. 4.3 - orange color)
- *G* - ESU_{branch_r} that does not have any *AP*s in the tuple lists (Figure. 4.3 - grey color)

This tree structure is capable of enumerating unique subgraphs from each of its branches, a property guaranteed by a condition found on its expansion logic and discussed in Section 4.2.1. Knowing this, we can make use of the concept of ESU-tree branch proximity area described in Fig. 4.1 to isolate in to certain degree (but not completely) the connected components which correspond to an ESU_{branch_r} and then compute our metrics.

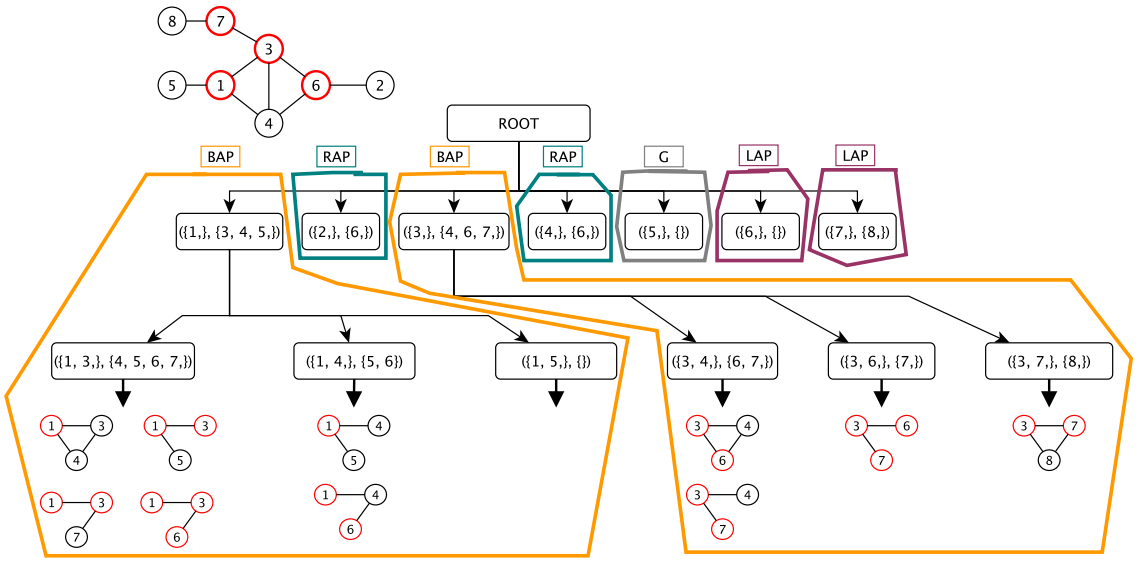


Fig. 4.3 Correlating ESU-tree branches of the graph depicted in Figure. 3.1 with the existence of Articulation Points (in red) in the branch tuples

The metrics we intend to compute and correlate are the concentration of subgraphs considered as motifs, the *AP*s concentration and the $ESU_{AP_{exp}}$ for each type of branch. Using these metrics we can formulate answers to question like *"does an LAP branch produce more motifs than an RAP branch?* or *"is an BAP branch is the most probable configuration of ESU-tree branch to enumerate subgraph motifs?"*

Starting from the metrics we want to correlate, we proposes two ideas formulated as research questions:

- *RQ1* - do the *AP*s occur with a higher degree of probability in real-world networks than in randomized networks ?
- *RQ2* - in an ESU_{PA}, is the occurrence of *AP*s correlated to the occurrence of network motifs ?

4.4 Experimental Analysis

We begin investigating the empirical evidence of the correlation between the occurrence of network motifs and the existence of articulation points by using the metrics obtained from the structural analysis of the network data set presented in Subsection 4.4.1. To aid a graphical representation of the results, we will present metrics for motifs of size k=3 extracted using the algorithm presented in Subsection 4.2.1 while considering n=2 randomly generated networks using Erdos-Renyi model [35]. Considering the size of the motifs and the undirectional nature of the networks, the list of possible motifs is restricted to two types of subgraphs, 2-edged and 3-edged.

4.4.1 Data

The data set used in test trials is made up of 30 networks from real-world systems obtained from KONECT [74]. These networks come from various fields: social science, infrastructure, informational science or biology.

The size of the networks can be an important factor in the evolution of the network's topology. Our focus is to study networks grouped by their real-world context and not by the topological differences between their properties, hence the studied networks that come from the same field will have approximately the same number of nodes or edges and similar topological properties.

By considering only the provenience or the context of the real-world system from which the networks were modelled, some expected behaviours can be intuitively or scientifically explained. For example, a social networks context favor higher values for the property CC_L [77] than in an infrastructure networks case, which could be viewed as the relative ease with which one creates new interactions between nodes in social network compared to between nodes in infrastructure network (e.g: to build a road between two cities is a more complicated process than to build a connection between people).

Due to this real-world contextual importance, the characteristics of each network context will be considered during the analysis comparing the network motif appearance metrics of each network type presented in the next subsections.

4.4.2 Articulation Points and Motifs occurrence in real-word vs. randomized networks

As stated in Section 2.6, network motifs represent patterns of interconnections which appear with a higher degree of probability in real-world networks than in randomized networks.

Although current literature does not yet recognise a similar property for *AP*s, our analysis starts by proving that not only do subgraph motifs appear in a higher concentration for real-world networks, but also *AP*s in higher concentrations as well.

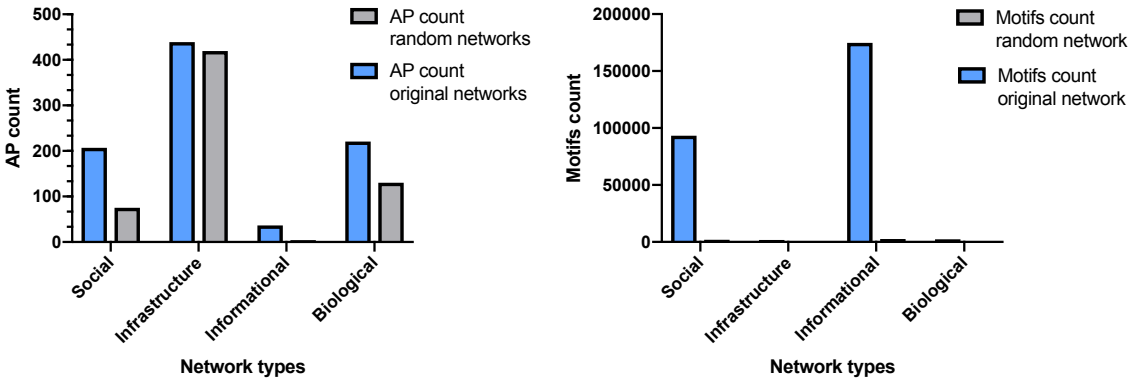


Fig. 4.4 Articulation Points and Motifs appearance in considered networks (original vs random)

Figure. 4.4 displays a comparison for the both number of motifs and the number of *AP*s found in original networks versus the generated random networks using Erdos-Renyi model [35]. Considering the results shown, for motifs there is no doubt that they appear in a higher degree of concentration in the original networks compared to the randomized networks. Using the same principle of empirical evidence shown in Figure. 4.4, we can also conclude that *AP*s tend to appear in real-world networks with a higher probability than in randomized networks and can serve as an answer to RQ1. The argument is the same for network motifs. They both represent structures modeled by a sequence of events which cannot be simulated only by considering network structural properties, but also require external factors that cannot be encoded in a networks' topology.

The outlier of this conclusion is the infrastructure type networks which have a close equality between the compared metrics. One reason for this behavior could be revealed by the structure of infrastructural networks. In our data set, these type of networks have a ratio between number edges and nodes of ~1.23 compared to a general ratio of ~12.22 (roughly a neighbour for each node compared to 12 neighbours for each node). This means that a node in an infrastructure network could be easily considered an *AP* even in a randomly generated network, due to the lack of links between nodes, which is indeed a natural behaviour in this type of network (in everyday terms, one does not expect to find direct links between cities too often, more likely one would find a chain of links between cities).

The analysis presented in this section could also serve as an indicator to the existence of a correlation between *AP*s and network motifs, something discussed further in the following sections.

4.4.3 Motifs appearance by ESU_{branch_r} type considering ESU_{branch_r} type frequency

Considering an ESU_{PA} with $L = 2$ for an ESU_{branch_r} of a given type (*RAP*, *BAP*, *LAP* or *G*, check Section 4.3), we sought to compute the frequency of the motifs in their list of enumerated subgraphs. The purpose of this metric was to find out if the motifs tend to appear in a specific ESU_{branch_r} - ie, structures that may contain one or more *AP*s, whose existence can be correlated with the appearance of the motifs as explained in Section 4.3.

In Figure. 4.5 we can see how motifs tend to appear generally and particularly in each studied network type, in branches that contain at least an *AP* (*BAP*, *RAP*, *LAP*). Observing that all of the ESU-tree branches that contain more than 90% motifs have an ESU_{PA} with at least an *PA* in their structure and a high concentration of motifs, it encourages one to consider a connection between these two phenomenons (RQ2).

Considering the ratio between the frequency of a given ESU_{branch_r} the type and the total number of ESU-tree branches, we found a slight correlation between the number of motifs and some frequencies of ESU_{branch_r} types. However, due to the inconsistent behaviour between network types we could say that empirically it is not clear from this study that a higher frequency of ESU-tree branches with *AP*s rather than without *AP*s will lead to an unbiased metric result.

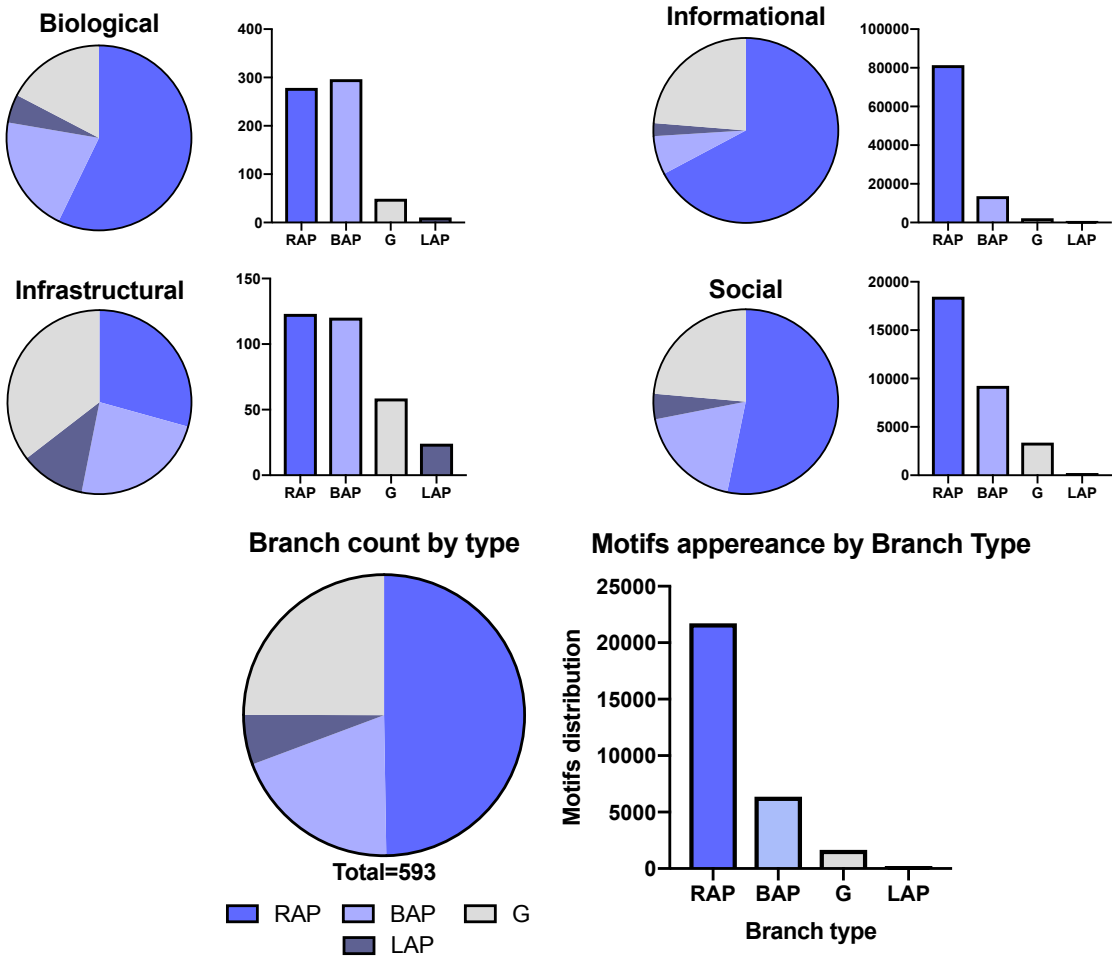


Fig. 4.5 Motif appearance by Branch Type considering Branch Type distribution

4.4.4 Motifs frequency and *AP* count by ESU_{branch_r} type

In order to reinforce the statement that serves as an answer to RQ2, Figure. 4.6 shows the head-to-head metrics of the distribution of the motifs by ESU-tree branch type and the number of *AP*s in these branches (the distinct count of *AP*s in an expanded branch). This visual representation of the obtained results also confirms a trend between *AP*s and motifs, this time justifiable in quantity terms. Branches *BAP* and *RAP* both contain a significantly higher number of *AP*s which is well correlated with the frequency of motifs, higher than for other branches as well.

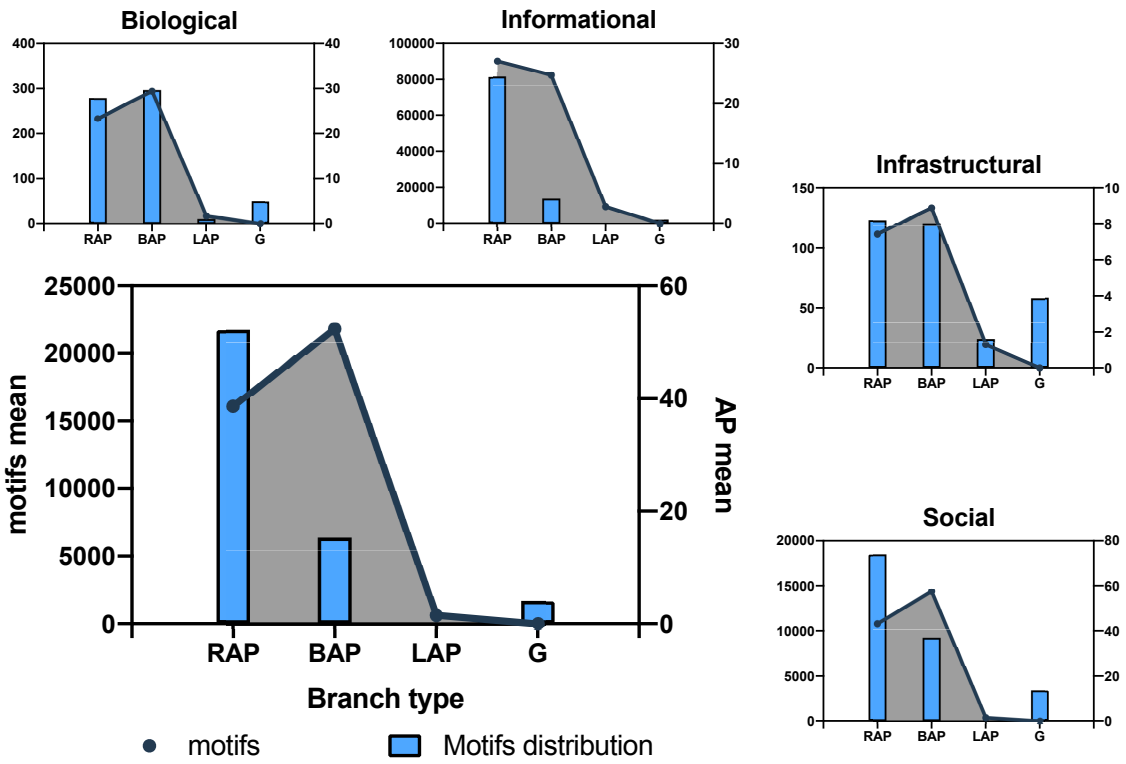


Fig. 4.6 Motifs frequency vs *AP*s count by Branch Type

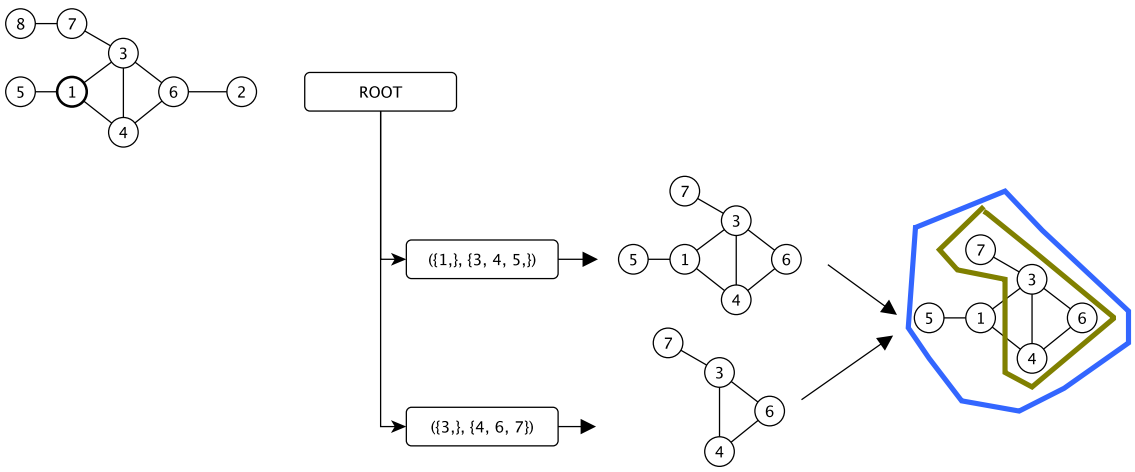


Fig. 4.7 An example of sub-ESU_{PA}

In theoretical terms, this can be explained by the existence of multiple sub-ESU_{PA}'s in the ESU_{PA} of a *RAP* or *BAP*, each one corresponding to an *AP* found in the initial ESU_{PA}. Using the study for RQ2, each of these ESU_{PA} contained in a given ESU_{PA} indicate an

authority over their area and can spread motifs over it. Hence a union of ESU_{PA} could only amplify the occurrence of motifs, either if they overlap in a certain way or another. In Figure. 4.7 we can see, as an example, how *AP* 1's ESU_{PA} (corresponding to a *BAP* branch, Figure. 4.3), overlaps the ESU_{PA} of *AP* 3.

4.4.5 Motifs frequency and $ESU_{AP_{exp}}$ by ESU_{branch_r} type

Similar results to RQ2 are obtained when one correlates the $ESU_{AP_{exp}}$ and the distribution of the motif by branch type (Figure. 4.8). *BAP* and *RAP* branches hold a similar distribution as in Figure. 4.6, demonstrating that branches that expand tend to gather more and more *AP*s and will also enumerate with a higher probability a larger quantity of motif).

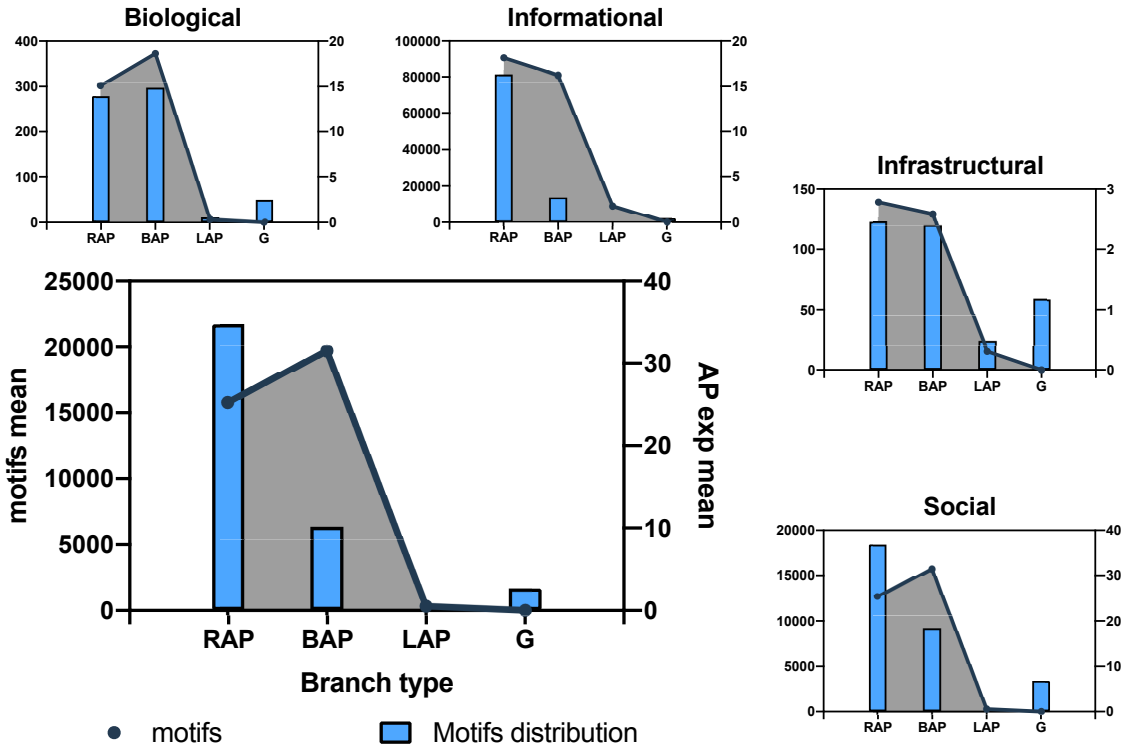


Fig. 4.8 Motifs frequency vs $ESU_{AP_{exp}}$ by Branch Type

An argument that might explain these results is one that is revealed by the ESU_{PA}'s structure, which from a probabilistic perspective has a lower chance of having a high number of *AP*s on L=1, than on a $L > 1$. While expanding an ESU_{PA} from a L=n to L=n+1, the number of *AP*s from the newly expanded ESU_{PA} can only increase or stay the same. Hence a high number of *AP*s from a given branch is very probable to correlate with a high $ESU_{AP_{exp}}$

of the same branch. This assumption can be sustained by the similar proportions between charts from Figure. 4.6 and Figure. 4.8.

4.5 Conclusions

The scope of this experiment was to present empirical proof for a possible connection between the existence of articulation points (AP) in complex networks and the appearance of network motifs around their proximity areas (ESU_{PA}).

We proposed an analytic approach that used as a data structure an ESU-tree. The results obtained from the analysis of 30 real-world complex networks sought to answer the following research questions:

1. do APs occur with a higher degree of probability in real-world networks than in randomized networks, similar to motifs? (RQ1)

2. in an ESU_{PA}, is occurrence of APs correlated to the occurrence of network motifs? (RQ2)

Using the results presented in Section 4.4, we provided answers to RQ 1 and 2 using metrics, graphical representations and discussions based on the analysis approach presented in Section 4.3.

Considering the empirical evidence presented in this study, we can say that in real-world networks, articulation points and network motif appearance are correlated.

Driven by the results of this experiment, the following chapter will introduce a study that tries to correlate even more topological properties with the existence of the network, proving the strong bond between the micro, respectively, the meso-layer structures named network motifs.

4.6 Study extension - Examining the social behavior of ant colonies using Complex Networks

The question of what defines a node as being important lacks a widely accepted definition in the Complex Network Theory community and requires a philosophical explanation that can be always placed in a grey area. However, various centrality measurements have been proposed as a framework to compute node importance based on the number and types of associated links. For instance, degree centrality is a simple measure that considers a node's

importance based on the number of associated links, while more complex measures such as closeness centrality and betweenness centrality consider the importance of a node based on its position in the network and its role in information flow. These centrality measures have been widely studied in the literature [19, 115]. In the context of this chapter's study, an articulation point (AP) can also be regarded as an important node. This is because an AP has the ability to control isolated communities by serving as the sole bridge of connection between them.

Despite not being directly related to the study of network motifs, our research uncovered an intriguing secondary research direction, investigating the influence of important nodes, as measured by state-of-the-art centrality metrics in the network dynamics, with a focus on communities. Our study aimed to establish a relationship between different node centrality measures and nodes dominant in hubs identified through community detection in various networks.

4.6.1 Social behavior in nature

In nature, a variety of species exhibit a pronounced social behavior, whereby members of the same species tend to interact in order to increase their chances of survival. Such behavior is not restricted to mammals alone, but also found in insect colonies, schools of fish, and, to a lesser extent, reptiles. Certain species of lizards have been observed to display social behavior and organize themselves into social hierarchies [23].

In species with smaller members, individual survival rates tend to be lower, leading to the formation of complex communities characterized by hierarchies and defined roles. Ant colonies are a prime example of such communities, often comprising millions of members [7, 84]. These colonies exhibit remarkable synchronization in tasks such as food gathering, cleaning, and protection. Ant behavior has been widely studied in numerous experiments [51, 87, 129], although logistical challenges often pose a significant obstacle to researchers.

As the complexity of interactions between members within a system is difficult for human observation and measurement in real-time, researchers have sought frameworks to model the underlying dynamic system. Complex Network Theory (CNT) has emerged as an increasingly popular framework in recent years [35], as it enables the modeling of complex systems as topological spaces, specifically as graphs, where the members are represented as nodes and their interactions as edges.

By studying a modeled network, it is possible to gain insight into the behavior of the network's members using a range of properties. Some properties provide information at

the level of individual nodes, such as node degree, assortativity degree, and centrality measurements [19, 105, 115]. Other properties are oriented towards characterizing the network as a whole, such as clustering coefficient [77], communities [38, 49], and network motifs [6, 69, 70].

Various studies have demonstrated the advantages of utilizing CNT to model ant colonies as intricate networks, enabling researchers to investigate ant social behavior and dynamic processes such as food collection and communication. However, most of these studies have focused on the colony as a whole rather than individual ants. Given the existence of hierarchy in ant colonies, the lack of research on individual behavior is notable. Therefore, we aim to explore whether individual ants prioritize their hierarchical position and strive to climb higher or exhibit purely altruistic behavior by acting for the benefit of the community without expecting anything in return.

A set of research inquiries will be developed to steer an experimental analysis of six ant colonies monitored for a duration of 41 days [85]. The objective is to employ properties at the node and network levels derived from CNT to measure and confirm the central query: do ants act altruistically or selfishly? The objective is to distinguish between selfish and altruistic behavior in ants by employing centrality measurements, which are commonly used to evaluate the significance of a node in complex networks. We will also utilize network properties that optimize the exchange of information within the colony, such as maintaining a low average shortest path, to identify altruistic behavior. Ultimately, a qualitative analysis will be conducted to compare the two methods of quantifying altruistic and selfish behavior.

The following sections aim to provide an overview and clarification on the challenges that arise when studying small-sized creatures, as well as how CNT can help overcome these challenges based on the existing state-of-the-art. Additionally, we will detail each of the metrics employed in the experiment, explaining their relevance and how they will be applied. We will then introduce the dataset and the research questions that will guide the experiment, with the results being validated against the network properties obtained. Finally, we will conclude with a discussion that will confirm or reject the initial assumptions regarding the research questions posed.

4.6.2 Problem definition

From a logistical standpoint, studying ant colonies is challenging due to the small size of the ants, making them difficult to observe by the naked eye. Additionally, the high number of members that share a similar appearance can complicate the study of their behavior, as

they move quickly and often exhibit chaotic movement patterns. Manually observing the behavior of ants in such a scenario would be time-consuming and exhaustive, requiring video recording and subsequent frame-by-frame tracking by one or more researchers [89, 139].

Following the logistical difficulties of studying ant colonies, there is a secondary issue that arises during the process of analyzing the interactions between ants. This issue is caused by the need for a framework to facilitate the entire process, starting from defining interactions and culminating with the impact a group of interactions has on the dynamic of the colony. One solution to this issue is to model ant colonies as systems that can be studied using mathematical and statistical approaches. This allows for the numerical quantification of ant behavior, leading to a feasible and valid way of answering research questions. However, even with a colony modeled as a system, it can be difficult to understand the apparent chaotic behavior between ant interactions. To address this challenge, a set of tools is required to extract properties from the system that may lead to paths not initially intended by the researchers. Complex Network Theory is one such framework that provides a wide range of tools to extract properties about the system modeled as a complex network. By modeling the colony's system as a network, where ants are nodes and their interactions are edges, we can explicitly analyze the interactions a given ant or group of ants have, defining statements about their social or individual behavior [66, 124].

In the subsequent section, we will provide an overview of the fundamental properties existing in complex networks that are pertinent to the examination of social interactions among ants belonging to the same colony, accentuating the practical implementation of the theoretical aspects in the context of a real-world ant colony.

4.6.3 Theoretical insights of the state of the art

Any complex network is characterized by a collection of nodes and a set of edges that connect them. Although these two fundamental components are simple, they give rise to multidimensional complex topologies with unique properties that can be explored, underscoring the advantages of representing real-world systems as complex networks. One of the most widely researched concepts, particularly in social networks, is the definition of critical, important, or popular nodes. However, this is more of a philosophical question that has been debated extensively in the literature [76]. Nevertheless, the literature proposes a group of metrics known as centrality measures that aim to offer various ways of characterizing important nodes.

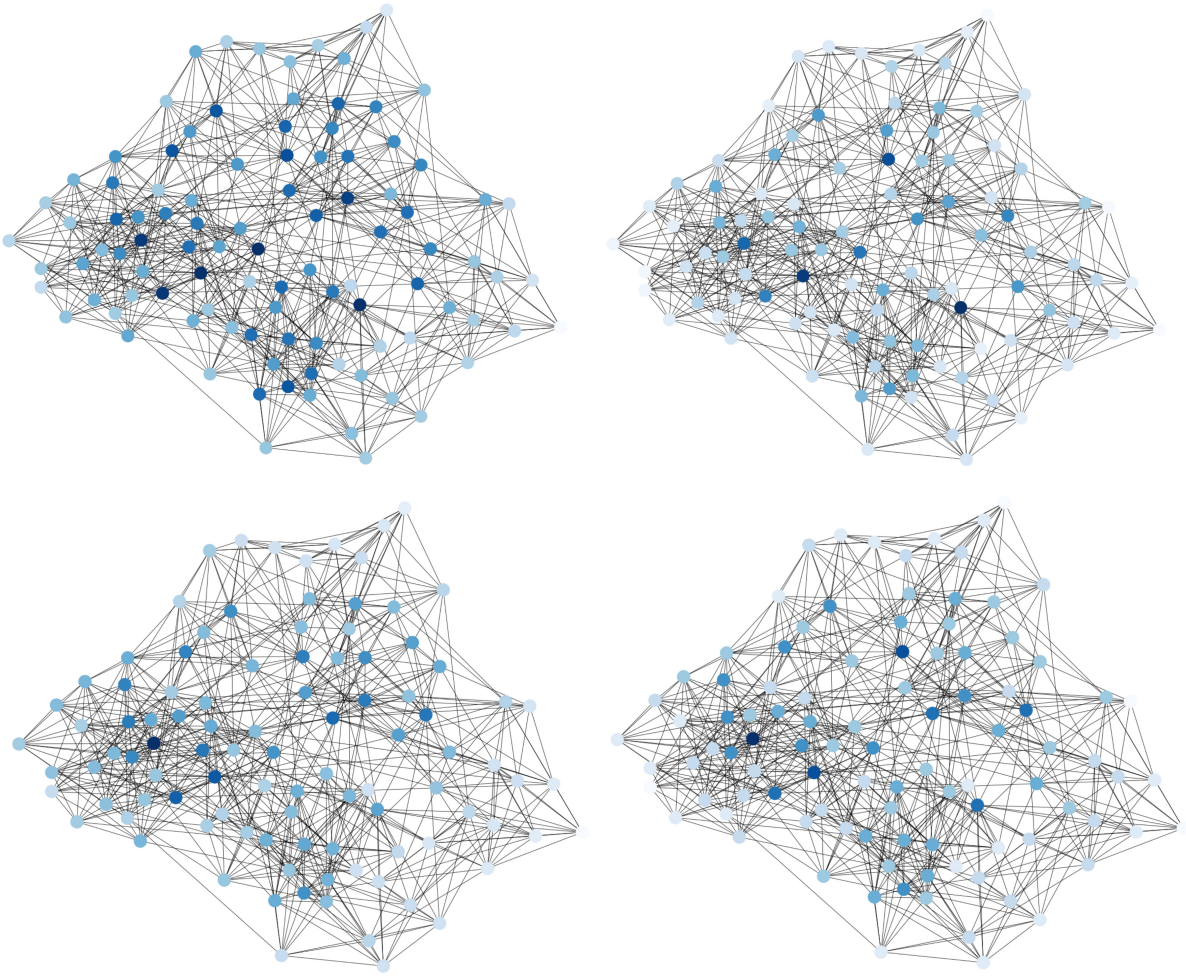


Fig. 4.9 Centrality measures in a Newman Watss Strogatz graph with 100 nodes. Light blue - low value, Dark blue - high value (Top-Left: Closeness centrality, Top-Right: Betweenness centrality, Bottom-Left: Eigenvector centrality, Bottom-Right: Degree centrality)

In this section, we will provide detailed descriptions of the graphs and their respective nodes' centrality measures, as shown in Figure 4.9. The figures illustrate that different nodes are identified as "important" by each centrality measure, highlighting that each metric has a distinct approach to determining a node's significance.

Degree centrality (D_c) is one of the fundamental measures of centrality in complex networks. It quantifies the importance of a node based on the number of edges it has with other nodes in the network. Nodes with a high number of edges have higher degree centrality and are considered more important. The mathematical formula for degree centrality is as follows:

$$D_c(x) = \frac{d_x}{n-1} \tag{4.2}$$

where $D_c(x)$ is the degree centrality of node x, d_x is the degree of node x, and n is the total number of nodes in the network [115].

In addition to degree centrality, another centrality measure that takes into account the number of links and goes further in assessing a node's importance is *eigenvector centrality* (E_c) [115]. This property evaluates a node's influence in the network based on the degree centrality of its neighboring nodes. From a real-world perspective, a node with a high eigenvector centrality is connected to other nodes that are also important, meaning that being connected to popular nodes increases one's own popularity. The eigenvector centrality formula follows a recursive approach that calculates a value for a node by using the values computed for its neighboring nodes, as follows:

$$E_c(x) = \frac{1}{\lambda} \sum_{u \in N(v)} E_c(u) \tag{4.3}$$

where $E_c(x)$ is the eigenvector centrality of node x, $N(x)$ is the set of nodes that are connected to node x, and λ is a constant called the leading eigenvalue of the adjacency matrix of the network [76].

A distinct perspective on defining node importance or popularity is based on the flow of information in the network, whereby nodes that enable the flow of information in the network tend to be more important or popular to other nodes. In this regard, the next group of centrality measures uses the path between the nodes and their lengths to evaluate the significance of the nodes. *Betweenness centrality* (B_c) assesses the importance of a node by its ability to control the flow of information through the number of shortest paths between any two nodes that pass through it. Having more shortest paths passing through it means that the node is a hub facilitating the information flow in the most rapid manner through the shortest paths it is part of. The formula below depicts the quantification of this centrality measure:

$$B_c(x) = \sum_{s \neq x \neq t \neq x} \frac{\sigma_{st}(x)}{\sigma_{st}} \tag{4.4}$$

where x is the node for which we compute the metric, σ_{st} represents the overall count of shortest paths from any starting node s to any destination node t and $\sigma_{st}(x)$ denotes the subset of these paths that go through intermediate node x (excluding cases where x is an ending point) [115].

Closeness centrality (C_c) is another measure that quantifies the importance of a node in a network. Unlike betweenness centrality, C_c considers how quickly a node can be reached by all other nodes in the network. A node with a high C_c is considered to be "close" to all other nodes in terms of its shortest paths, making it an important hub that facilitates the flow of information in the network:

$$C_c(x) = \frac{1}{\sum_{u \neq x} d(u,x)} \tag{4.5}$$

where $C_c(x)$ is the closeness centrality of node x, $d(u,v)$ is the shortest path distance between nodes u and v, and the summation is taken over all nodes $u \neq v$ in the network [115].

In recent studies, a new approach to defining the significance of nodes has emerged, focusing on their role in maintaining network integrity using *Articulation points* (*AP*) (discussed in Subsection 2.5.1). In social networks, an *AP* could be a social media influencer or politician, while in an ant colony, the queen can be an *AP*. This concept is gaining popularity as it provides insights into the structure of networks and can inform strategies for improving network efficiency and stability [123].

In the experiment described in the following section, we aim to investigate the social behavior of ants in their colonies by analyzing their network properties. Specifically, we will explore whether nodes with high values for centrality measures, such as degree centrality, eigenvector centrality, betweenness centrality, and closeness centrality, but also existence of articulation points, are associated with the emergence of individual behavior among ants, as they try to ascend the colony's hierarchy.

4.6.4 In-depth study of centrality measures in network communities

Prior to examining ant colonies, we deemed it advantageous to undertake a preliminary investigation encompassing a broad-based study that entailed an exhaustive analysis of node centrality metrics described in the preceding subsection, and community detection algorithms and presents several experiments on real-world networks to validate the overlap between the top nodes detected by centrality measures and community detection, emphazing the roles of the important nodes in social networks [49]. The results were published in paper **[86]** and indicate that centrality measures and community detection methods yield common important nodes, highlights the intrinsic connection between an individual's significance and their role within a community [67]. The main contributions of the study are the com-

prehensive literature review and empirical validation of the relationship between centrality metrics and community detection-based hub-dominant nodes, and these results will be used as an empirical evidence to the experiment described in the next subsection.

4.6.5 Experiment

The following section will overview the experiment proposed to analyze the social behaviour of ants, making use of a comprehesive dataset for which we validate a series of research questions that will be evaluated using a defined methodology. The results of the extracted properties (detailed in Section 4.6.3) obtained in the experiment will be analyzed and a series of conclusions will be drawn to conclude the formulated research questions.

4.6.6 Dataset

To conduct a robust and meaningful experiment, it was deemed necessary to utilize a diverse dataset that is both horizontally and vertically scaled. To this end, we employed a complex dataset of complex networks that consist of six ant colonies that were completely isolated, as proposed by Mersch D. et al. in their research paper [85]. The complex networks were modeled by observing each ant colony over a timespan of 41 days, utilizing a video tracking system that was based on fiducial identification labels. Each ant's position was tracked twice per video frame, resulting in a vast amount of data - 2,433,250,580 ant positions and 9,363,100 social interactions. Social interactions were defined as instances where one ant's front end was within the trapezoidal shape representing another ant. From this data, a total of 246 networks were modeled, utilizing both the ant positions and the tracked interactions.

Table 4.1 provides a comprehensive summary of the modeled networks based on their nodes and edges - ants are represented as nodes, and the interactions between them are represented as undirected edges. The weight of each edge is determined by the number of interactions between the same ants. Since the ant colonies were observed for a period of 41 days, changes in the number of individuals within each colony could occur, resulting in variations in the network's characteristics. These changes were captured by the mean, standard deviation, and 95% percentile values, enabling the dynamics of the colonies to be analyzed.

he metrics obtained from the modeled networks exhibit a significant degree of diversity with regards to the number of members in each colony, which is reflected in the number of edges present in each network. This phenomenon can be attributed to the fact that larger colonies tend to have more interactions among members, resulting in a higher number of

colony id	metric	mean	95% percentile	Stdev
1	nodes	89.21	[82.35, 96.06]	± 21.15
	edges	2803.97	[2429.41, 3178.54]	± 1155.48
2	nodes	100.13	[91.47, 108.79]	± 26.71
	edges	3541.03	[2909.48, 4172.57]	± 1948.25
3	nodes	130.05	[122.01, 138.09]	± 24.80
	edges	6036.79	[5274.19, 6799.40]	± 2352.55
4	nodes	68.77	[60.73, 76.80]	± 24.79
	edges	2067.03	[1640.79, 2493.26]	± 1314.88
5	nodes	113.31	[102.26, 124.35]	± 34.07
	edges	4905.05	[4034.66, 5775.44]	± 2685.03
6	nodes	131.85	[123.07, 140.63]	± 27.09
	edges	6338.82	[5451.44, 7226.20]	± 2737.44

Table 4.1 General properties of the studied networks [85]

edges.

4.6.7 Research questions

The experiment is designed to address two research questions, a primary one (**RQ1**) and a secondary one (**RQ2**):

RQ1 - What is the nature of social interactions among ants, specifically whether they demonstrate altruistic or selfish behavior?

RQ2 - Does the dynamics of ant colonies optimize the flow of information through interactions among the ants?

4.6.8 Methodology

In accordance with the formulated research question, the methodology of the experiment will involve the extraction of centrality measures and articulation points from each network to investigate the occurrence of nodes with abnormally high values for these metrics, indicating their inclination to strategically position themselves to maximize their valuable connections. In addition to the previously mentioned topological indicators, we will also extract two additional measures, namely local clustering coeffient (CC_l) and assortativity degree (ρ), to gain a deeper understanding of the global social behaviour of ants and to investigate if any specific behaviour emerges. CC_l measures the tendency of nodes to clusterize into

communities (Subsection 2.5.6), while assortativity degree measures the tendency of nodes to form links with nodes having a similar degree as themselves (2.5.2). These measures will aid in addressing the research questions posed in this experiment.

By quantifying and validating **RQ1** through this approach, we can employ the concept of popularity, which is defined in various ways as discussed in Section 4.6.3.

To investigate whether the networks and colonies aim to optimize the performance of the information flow among their members, we will use two topological indicators: density and average shortest path. Density (d) is a measure of how close a network is to being fully connected, where all ants interact with one another. Networks with high density tend to optimize the information flow by ensuring that all members are easily reachable through a high density of edges. The average shortest path is a measure of the average number of steps needed to travel between any two ants in the network. We will compute and use these indicators to demonstrate the colonies' dynamics and to answer the research question. The calculation of the average shortest path (AVG_{sp}) can aid in determining if ants strive to optimize the formation of valuable links within the context of an efficient information flow. A low average shortest path, which denotes the average number of links needed to create an optimal path between two members, suggests that ants can easily reach each other. By analyzing this metric, we can provide a formal response to **RQ2**.

4.6.9 Results

In accordance with the methodology outlined in the preceding section, we derived all the centrality measures outlined in Section 4.6.3 as well as other topological properties, including CC_l, d, AVG_{sp}, and ρ (Table 4.2). This was done to enable the quantification of the propositions that could potentially serve as answers to the formulated research questions.

Ants are social animals with highly organized behavior and specialized roles throughout their lifetime [85]. Their division into groups is strictly task-oriented, which may result in a large number of interconnections between ants within the same group. This behavior is well-reflected in the high values of CC_l observed in all the studied colonies.

It is well known that each of the examined ant colonies has a queen, whose exclusive responsibility is to lay eggs. Given its high hierarchical position, it is reasonable to classify the queen as an articulation point (AP). However, the analysis showed that the queen is not often an articulation point in any of the modeled networks. This indicates that despite the queen's critical role, it is part of a well-connected community, and her death would not necessarily lead to the colony's disintegration.

At a macro level, the social behavior of ants does not exhibit a particular tendency to create links with new ants, as indicated by the value of ρ which is equal to 0. This suggests that the creation of links is not driven by individual preferences, but rather by the collective behavior of the group as a whole.

This hypothesis is strongly supported by all of the computed centrality measures, which do not identify any notable group of nodes that exhibit exceptional values for their centrality measures. By computing the 95% percentile interval and mean values, we can observe the homogeneity of centrality measure values across all nodes, with insignificant standard deviation.

Using these two observations we can conclude the altruistic behaviour of the ants (**RQ1**), meaning their interactions are ones established purely on their role in the community they are part of and the only thing that is important to them is to succesfully complete their task.

The current findings strongly indicate the well-organized nature of ant colonies. The AVG_{sp} metric is another measure demonstrating that colonies are optimized not only in terms of creating specific groups for efficient task completion, but also for the rapid flow of information among members. According to this metric, it takes approximately 1.30 edges for any ant to reach another ant using the shortest path in the modeled network. This value is remarkably low for any network, highlighting the communication efficiency present in ant colonies and confirming the answer to **RQ2**.

Another notable finding is that the ant colonies exhibit both a low AVG_{sp} and a high CC_l, which are characteristic features of small-world networks. These networks are commonly observed in various real-world systems, such as transportation networks, and have been the subject of many studies aimed at understanding their effective information flow and how to replicate it in other contexts. Given the well-organized structure of ant colonies, it is not surprising to find that the networks derived from observing them exhibit small-world characteristics.

4.6.10 Conclusions

Ants have been a subject of fascination for the scientific community for a long time due to their ability to develop highly intricate social structures organically, which enables them to efficiently accomplish tasks such as foraging, cleaning, and defense. However, research on ants has largely focused on their collective behavior rather than individual behavior. While it is widely acknowledged that ants exhibit altruistic behavior at the group level, it remains to be seen whether this behavior is universal across all members of the colony or if some

ID	Metric	Mean	95% percent.	Stdev	ID	Mean	95% percent.	Stdev
1	AVG_{sp}	1.31	[1.30, 1.33]	0.05	2	1.36	[1.34, 1.38]	0.07
	CC_l	0.79	[0.78, 0.80]	0.03		0.77	[0.76, 0.78]	0.03
	ρ	0.03	[0.02, 0.04]	0.03		0.09	[0.07, 0.11]	0.06
	d	0.69	[0.67, 0.70]	0.05		0.64	[0.62, 0.66]	0.06
	AP	0.08	[0.00, 0.16]	0.27		0.03	[0.03, 0.08]	0.16
	C_c	0.77	[0.74, 0.80]	0.08		0.75	[0.72, 0.77]	0.08
	B_c	0.0035	[0.003, 0.005]	0.0027		0.0041	[0.003, 0.006]	0.0038
	D_c	0.69	[0.64, 0.73]	0.15		0.64	[0.59, 0.70]	0.16
	E_c	0.11	[0.10, 0.12]	0.02		0.10	[0.09, 0.11]	0.03
3	AVG_{sp}	1.31	[1.30, 1.32]	0.04	4	1.20	[1.19, 1.21]	0.04
	CC_l	0.79	[0.78, 0.80]	0.03		0.86	[0.85, 0.87]	0.03
	ρ	0.01	[0.00, 0.02]	0.02		0.03	[0.00, 0.05]	0.04
	d	0.69	[0.68, 0.71]	0.04		0.80	[0.79, 0.82]	0.04
	AP	0.15	[0.04, 0.27]	0.37		0.00	[0.00, 0.00]	0.00
	C_c	0.77	[0.75, 0.80]	0.08		0.85	[0.82, 0.88]	0.08
	B_c	0.0025	[0.002, 0.003]	0.0015		0.0032	[0.002, 0.004]	0.0020
	D_c	0.69	[0.65, 0.73]	0.15		0.80	[0.76, 0.85]	0.14
	E_c	0.09	[0.08, 0.09]	0.02		0.13	[0.11, 0.14]	0.02
5	AVG_{sp}	1.29	[1.28, 1.30]	0.04	6	1.31	[1.29, 1.32]	0.05
	CC_l	0.80	[0.79, 0.81]	0.03		0.79	[0.78, 0.80]	0.03
	ρ	0.01	[0.00, 0.02]	0.05		0.03	[0.02, 0.03]	0.02
	d	0.71	[0.70, 0.72]	0.04		0.69	[0.68, 0.71]	0.05
	AP	0.00	[0.00, 0.00]	0.00		0.03	[0.00, 0.08]	0.16
	C_c	0.78	[0.76, 0.81]	0.08		0.78	[0.75, 0.80]	0.08
	B_c	0.0029	[0.002, 0.004]	0.0021		0.0025	[0.002, 0.003]	0.0015
	D_c	0.71	[0.67, 0.75]	0.15		0.69	[0.65, 0.74]	0.15
	E_c	0.10	[0.09, 0.11]	0.02		0.09	[0.08, 0.09]	0.02

Table 4.2 Network properties extracted for each of the six ant colonies

individuals display a more self-centered approach aimed at maximizing their own benefit.

Continuing this line of inquiry and utilizing a set of intricate network models based on observations of six distinct ant colonies over a span of 41 days, our research aimed to address two fundamental questions. These questions were formulated to resolve the previous uncertainties:

RQ1 - What is the nature of social interactions among ants, specifically whether they demonstrate altruistic or selfish behavior?

RQ2 - Does the dynamics of ant colonies optimize the flow of information through interactions among the ants?

During the study experiment, we obtained various topological properties and centrality measures of the examined networks. The analysis revealed that there is a considerable consistency among the centrality values of the individual nodes, with insignificant standard deviation. Furthermore, based on the property ρ, we observed that the ants do not have any macro-level inclination towards preferential attachment, but rather establish connections through their task-based activities. These two observations led us to confirm the altruistic behavior of ants as an answer to **RQ1**.

In addition to ants' ability to naturally and organically evolve complex groups that optimize task performance, the observations indicate that the information flow within their networks is highly efficient, confirming **RQ2**. The average shortest path between any two ants in the network is close to one edge, indicating that every ant is almost directly connected to every other ant. Furthermore, all studied networks show a high CC_l and exhibit characteristics of small-world networks, which optimize information flow performance and evolve strong and complex structures, as commonly observed in other real-world systems such as transportation networks.

The experiment corroborates the widely accepted behavior of ants as altruistic individuals that prioritize the collective good over individual interests, while also demonstrating their capacity to naturally develop sophisticated systems that optimize task performance for the group.

This experiment employs Complex Network Theory to model the interactions among ants and investigate their social behavior. The study examines whether the behavior of six ant colonies is community-oriented or individual-oriented. To measure the existence of social or individual-oriented behavior, the research utilizes several node properties that define nodes' importance. The study's original contribution to the literature is the use of Complex Network Theory and centrality measures to quantify the altruistic behavior of the ants. The results were published in paper **[92]**.

Chapter 5

Unraveling the relationship between micro-level topological properties and the propagation of motifs in network topologies

Based on the evidence found in our previous studies presented in Chapter 4, which revealed a significant correlation between the existence of articulation points and meso-level components such as network motifs, in this chapter we will extend this study by presenting analytical research between a consistent set of micro-level topological properties from Graph and Complex Networks Theory and the appearance of the motifs. The purpose of this study is to use network properties to provide a better understanding of how and why network motifs appear, learning how the dynamic of a network is propagated through its structures.

The results presented in this chapter were published in paper **[95]**. They represent an original contribution in understanding what micro-level properties are indicators to the existence of network motifs. With these results, we can unify the co-existence of various micro and meso-level structures, namely network motifs, and continue studying the co-existence of network motifs with macro-level properties that will bring us to the final step of presenting a complete picture of the convergence phenomena happening in network topologies.

5.1 Problem definition

Recognising the importance of motifs and understanding what factors cause their appearance, this experiment continues to present an in-depth analysis of the presence of motifs in complex networks in respect of the topological properties of the networks. Presuming that motifs appear in contextual circumstances and the topology of the network can be seen as a structural encoding of the real-world context, there must be a correlation between specific properties and motifs. Our approach is to examine this presumption by using a consistent set of networks from various fields to which a statistical analysis is applied that will confirm a list of stated hypothesis.

Most fundamental theories regarding complex networks are based on a micro-level analysis that makes use of the structural property of complex systems which states that the behaviour of the system converges to its smallest topological components - nodes, in case of complex networks. Although such level of analysis following a course grain methodology may offer a simplistic set of techniques to explain the behaviour of a system, this micro-level analysis can miss aspects of the general truth found in a wider meso- or macro-level dimensionality [71].

The reason for this research is the lack of similar studies present in the literature. From our knowledge there is no research which considers a study of the correlations between the appearance of network motifs and other topological properties. Considering this void in the domain's literature and the need to answer to questions such as 'can network motifs appearance be probabilistic detected by looking at network topological properties?', we think that the subject addressed in this chapter is one of great interest.

Following our desire to have a better understanding of complex network behaviour, our goal is to understand how meso-level components such as network motifs are related to properties retrieved from a micro-level analysis. Although there are no explicit studies between the correlation of network motifs and other topological properties, the literature presents various studies which range from micro-level to macro-level analysis [56] [57] [68]. An field of study which demonstrates the benefits of explaining the behaviour of networks from a higher dimensionality is named telescopic analysis which is a technique that tries to correlate general micro-level properties to macro-level properties [109][81].

Inspired by these works and considering the lack of answers in the micro to meso-level analysis between the core topological properties of a network and network motifs, the following sections present details of the steps followed by our research to prove a set of formulated hypothesis.

5.2 Experimental analysis

5.2.1 Data insights

In order to obtain relevant analysis for the hypothesis testing, we used a consistent set of approximately 200 undirected networks with homogeneous topological properties, coming from various fields such as social science, biology, chemistry or infrastructure [74][114].

We begin by understanding the data distribution and the relationship between the properties we are using in our analysis. Presuming that a network motif property is a dependent property of all other topological properties, it is possible to model the analysis as a regression problem.

In the data preparation step, we processed and extracted statistics for each network individually. Therefore the data set used in analysis is a multidimensional set of values with approximately 200 rows and 9 columns. We have not considered a step of averaging values for groups of networks from the same field or other clustering criteria as it might bring unbiased results for our further analysis. As a general description, the studied networks do have between [10, 3323] nodes and [45, 23455] edges.

For the simplicity of the analysis, we considered only motifs of size N=3 (Fig. 5.1).

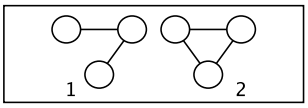


Fig. 5.1 Network motifs examples from a topological perspective.

Fig. 5.2 offers an almost complete graphical representation of the data insights. Plots found on the main diagonal are offering information about value distributions for each property.

Interpreted individually, each of these plots follows different types of distributions, from bimodal distributions for CC_L, CC_G, H_{er}, or ρ to regular polynomial distributions for γ and M_f. What can be learned from considering these distributions together is that the networks have sufficient differences from each other such that any biased behavior has a smaller chance of occurring during our analysis.

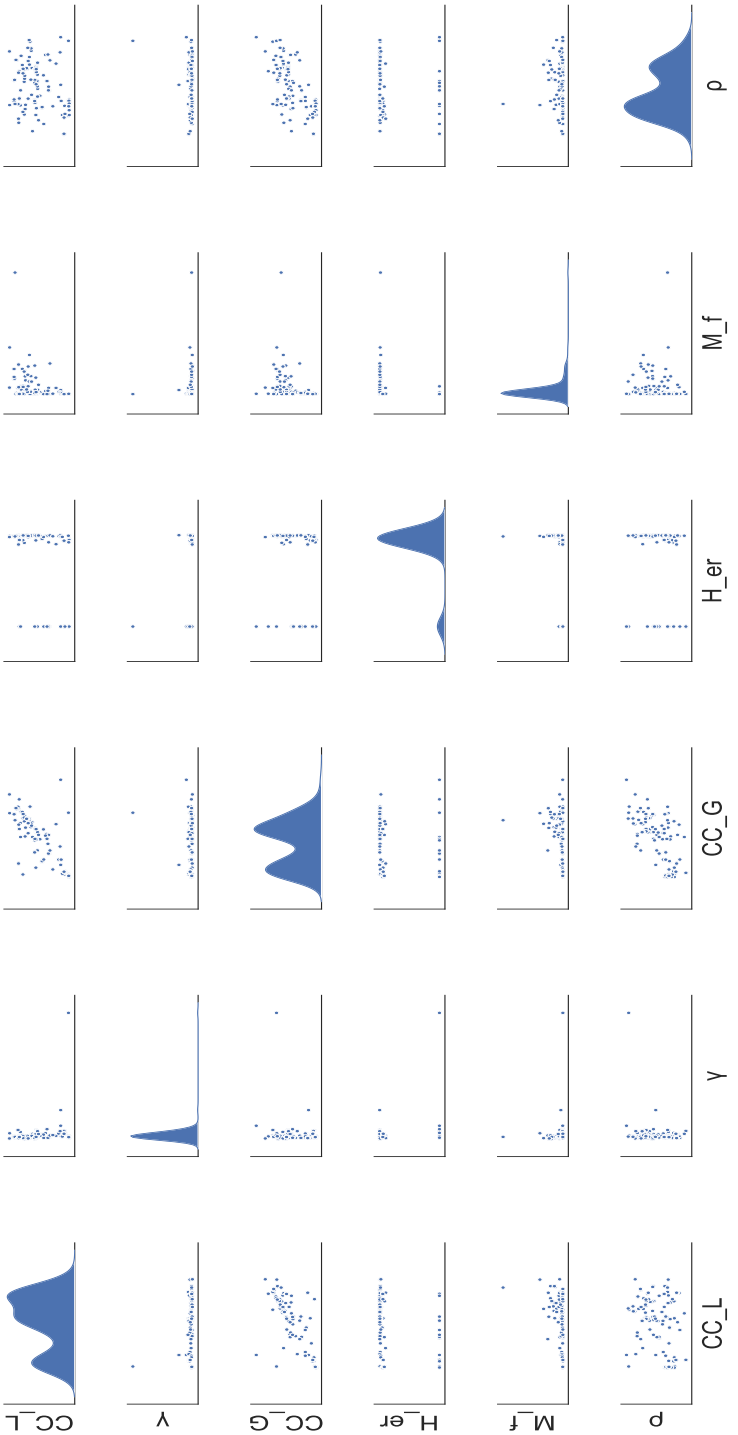


Fig. 5.2 Set of plots for data distribution and their relationship based on several features

All other plots found above and underneath the main diagonal represent a visual representation of the relationship between the combination of the metrics we are using in our research. Focusing only on the relationship of M_f (frequency of motifs sub-components found for our networks) and all other topological properties, we can interpret the plots visually. Although there is nothing clearly visible between M_f and some properties there is a shy linear trend (CC_L) or Gaussian representation (ρ) which will help us in formulating a hypothesis for our experimental analysis. Of course these statements might be too trivial to serve as a proof, but they will be reinforced while being analyzed again in the following subsection, this time using sharper statistical tools of data correlations.

5.2.2 Hypothesis formulation

We present a list of hypothesis that the study would like to confirm. These hypothesis were formulated using the theoretical and data insights described in Section 2.5 and Subsection 5.2.1 respectively, and they serve as the core component of the analysis structure, such that the analysis will have a follow logical narrative thread.

- *H1* - networks with a tendency of local clustering have a higher chance to provide specific types of interconnections, namely network motifs.
- *H2* - networks with a dynamics that is suitable to preferential attachments have a topology that is more likely to favor the appearance of network motifs.

5.2.3 Hypothesis testing

In Subsection 5.2.2 two hypothesis were formulated by using a purely theoretical approach about a possible relationship between topological behaviours and M_f. The next step in our analysis is to confirm $H1$ and $H2$ by using the measures obtained from the set of networks presented in Subsection 5.2.1. In the same section, by applying a rudimental statistical approach, there was observed a shy correlation between a few parameters and M_f, a correlation which had to be studied using sharper statistical tools.

The analysis will firstly clusters the topological parameters presented in Section 2.5, considering the hypothesis formulated, such that for $H1$ we will have to study both CC_L and CC_G, and secondly serves as an element of antithesis to highlight the possible confirmation of $H1$. To prove and confirm $H2$, the most relevant measures for its study could be the preferential attachment properties ρ and γ.

Correlation statistically measures the strength of association between two variables which numerically can vary from [-1; +1]. A perfect degree of association, either negative or positive, is represented by a value of ± 1. The so-called direction of the relation is indicated by the sign of the value, such that a + sign indicates positive relationships and a - the opposite. State of the art in correlation modeling it is represented by: Pearson correlation, Spearman correlation and Kendall rank correlation [26]. As each model behaves differently depending on the data distribution, our analysis will use each proposed model so that = in the interpretation of the results we will exclude an unbiased opinion.

We start by using the Pearson correlation, the most common correlation out of the three mentioned. From the analysis obtained using this solution (Fig. 5.3 and Table. 5.1) one can observe a positive trend in the correlation between CC_L and ρ and M_f with a value that can not be neglected. Both values represent key measures to prove $H1$, and, respectively $H2$. However because Pearson correlation can be sensitive to variables with a distribution that is not normal (which we can consider as a worst-case scenario, although it is not completely the case), we will use the other two methods to understand if the trend persists.

The Spearman correlation and the Kendal rank correlation are non parametric measures which are not sensitive to the variables' distribution. Therefore we can assume that the resultant correlations are not be influenced by abnormal distributions. Keeping focus on CC_L and ρ we can observe that their correlation with M_f still follows a positive trend and also are the highest two in terms of numerical comparison, for both measures Spearman, Kendall respectively (Fig. 5.3 and Table. 5.1).

As all three correlation coefficients indicate the same trend and powerful degrees of strength for the relationship between our variables of interest and M_f, we can formalize a confirmation for both $H1$ and $H2$. Having CC_L as the one of the most correlated variables with M_f in all measures, we can use this numerical interpretation to confirm $H1$. With the positive trend and its degree value being one of the highest in all correlation coefficients can be use it as a proof to say that networks with local clustering favor the appearance of network motifs. Another correlation that will strengthen the confirmation of $H1$ is the analysis of the values obtained for CC_G, which although they are positive they are significantly lower than the values obtained for CC_L. Hence we can say that a local clustering behaviour favours the appearance of the motifs due to the type of relations that are set between closer entities which tend to have local characteristic and behaviours not found in global behaviours.

The same structural explanation can be used for $H2$ while considering variable ρ, which in all the correlation coefficients functions is also ranked as one of the most correlated variables with M_f. Its high value of the degree of correlation is one of the best proofs for

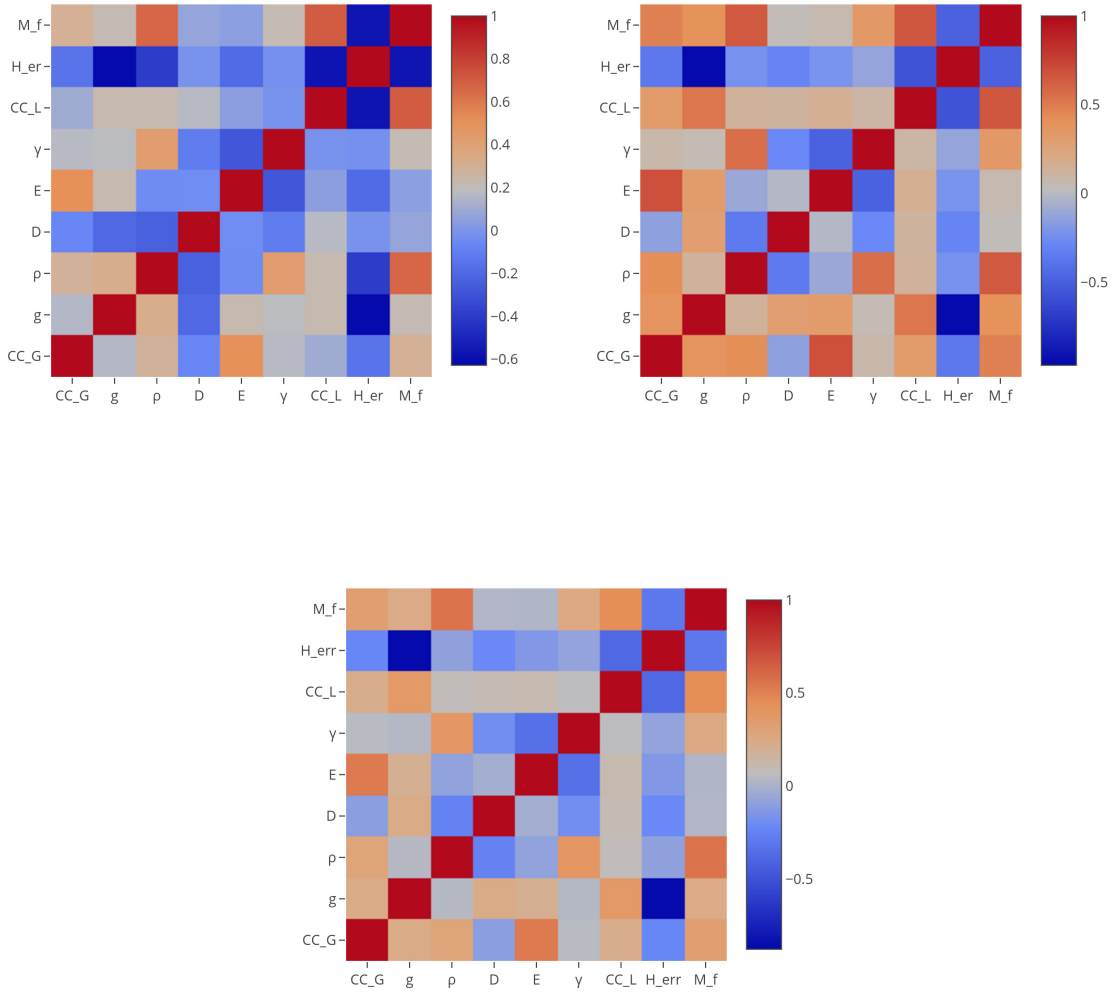


Fig. 5.3 Heat map of the statistical correlations between the studied properties and measure M_f (Left: Pearson, Right: Spearman, Center: Kendall)

saying that $H2$ is also a confirmed research question. ρ represents an important topological measurement for a preferential attachment such that these results can only confirm $H2$.

5.3 Conclusions

The purpose of this experiment was to present a comprehensive and objective analysis of the possible correlation between network motifs and a set of topological properties found in

Table 5.1 Numerical results of the statistical correlations between the studied properties and measure M_f

Algorithm	CC_G	g	ρ	D	E	γ	CC_L	H_{er}
Pearson	0.23	0.22	**0.67**	0.07	0.042	0.21	**0.7**	-0.58
Spearman	0.48	0.41	**0.65**	0.04	0.07	0.35	**0.66**	-0.49
Kendall	0.34	0.24	**0.56**	0.02	0.018	0.25	**0.45**	-0.31

complex networks.

Driven by the lack of similar studies from literature, we intended to come up with a set of hypothesis which, when confirmed by our analysis, could serve as a solid theoretical and numerical proof for other studies that might use the results obtained.

Considering our purpose, in Subsection 5.2.2 we formulated two hypothesis:

- *H1* - networks with a tendency of local clustering have a higher chance to provide specific types of interconnections, namely network motifs.

- *H2* - networks with a dynamics that is suitable to preferential attachments have a topology that is more probably to favor the appearance of network motifs.

Subsection 5.2 presents a strong evidence for the confirmation of both hypothesis. After we obtained a set of topological measurements and extracted network motifs from a consistent set of complex networks, we sought correlation between M_f and the set of topological measures while applying statistical correlation measures, namely Pearson correlation, Spearman correlation and Kendall rank correlation. In the test results, two of the topological measures, CC_L and ρ, were strongly correlated, with the existence of M_f having a positive trend in their relationship. Using the definition of CC_L and the results obtained, we confirmed $H1$, the same outcome was also present for ρ which helped to prove $H2$.

Considering the results from this experiment as solid proof that there is a strong bond between micro and meso-level structures, we will proceed in the following chapters with two research tracks:

- secondary - considering the strong evidence of the correlation between micro-level properties and network motifs, we can model this property as a classification model that can predict if a network contains network motifs or not using as independent variable the network micro-level properties highly correlated with the existence of

network motifs. Such a model would improve the waiting time implied in the procedure of the motifs extraction, as it could be used to predict if the network contains or not motifs even before running one of the algorithms capable of extracting motifs.

- primary - finalize the study convergence phenomena between all the layers. We will use this unification as empirical evidence to the theoretical foundation to our final goal of this research project: propose a network generator model that will generate topologies with a specific concentration of network motifs.

Chapter 6

Prediction of network motifs existence in Complex Networks

Properties of complex networks represent a powerful set of tools that can be used to study the complex behaviour of these systems of interconnections. They can vary from properties represented as simplistic metrics (number of edges and nodes) to properties that reflect complex information of the connection between entities part of the network (assortativity degree, density or clustering coefficient). Such a topological property that has valuable implications on the study of the networks dynamics are network motifs - patterns of interconnections found in real-world networks. Knowing that one of the biggest issue with network motifs discovery is its algorithmic NP-complete nature, in this chapter we present a method to detect if a network is prone or not to generate motifs by making use of its topological properties while training various classification models. This approach wants to serve as a time saving pre-processing step for the state-of-the-art solutions used to detect motifs in Complex networks.

The results and the model proposed were published in paper **[96]** and they represent an original contribution to the optimization of the processes used to extract network motifs. Similar to the improvements presented in Chapter 3, the results described in this chapter represent a complementary work to our main research track as it facilitates the waiting time implied in our experiments for motifs extraction.

6.1 Problem definition

From a topological perspective, networks motifs are frequent patterns of nodes and interconnections with a contextualised cause of their appearance, which context is given by the

real-world origin of the network. In order to confirm that a given configuration of nodes and interconnection is indeed a motif, the current state-of-the-art presents a series of algorithms that follow a common paradigm which unfortunately is ranked as a NP-complete problem [62]. While implemented in a real-world application, all proposed solutions reach time or memory bottlenecks as long as the size of the motifs grows.

Although given its NP-complete nature, the problem was not solved yet by a perfect algorithm, the literature presents various approaches to partially improve the time or memory bottlenecks. Part of the algorithms are using a statistical approach that process only samples of the original network, but having as a trade-off an approximate result. The most recent trend is to make use of computational power existing nowadays, while re-modelling existing algorithms as parallel solutions able to run on cluster of computers (Chapter 3).

Encouraged by the need of more solutions that could improve the time spent while analysing networks that are suitable to contain network motifs, in this chapter we will research and propose a method of deciding if a network with a given topological configuration is prone to contain network motifs or not. While making use of our previous studies presented in Chapters 4 and 5 that revealed important statistical correlation between network motifs and other network properties from which we can remind the ones with the strongest correlation: articulation points, local clustering coefficient or assortativity degree, our goal is to train a series of classification models that can be further used to predict the existence of motifs in a network. A such solution could drastically cut the time spent by researches while analysing networks, while using this solution as a pre-processing step in their process, approach which from our knowledge is not currently presented by literature.

6.2 State of the art of the classification methods

Our approach is one of using a set of independent variables to predict if a network contains network motifs or not (dependent variable). The independent variables will be presented later in Section 2.5.

Following the "no free lunch" theorem for supervised learning which states that there is no perfect algorithm for classification or regression problems, only models that perform better in specific circumstances [39], our goal is to test most promising state-of-the-art statistical models for our classification problem. These models will be evaluated by using the same test data set, such that in the end it would be possible a fair comparison between their results.

Our analysis will be performed while using 4 different statistical classification tech-

niques, different by their paradigm: Gradient boosting (XGBoost) as boosting algorithm [27], Decision Tree Classifier and Random Forest Classifier as tree-based models and last but not least SVM.

Knowing that each of these models is known to perform better under specific data distribution, in the next section it will be described the analysis approach it was applied starting from pre-processing steps of data normalisation and model training to the validation of the results obtained while performing a comparison of their performance.

6.3 Classification modelling

6.3.1 Data

While performing the classification modelling, we used a consistent data set of undirected networks of different sizes coming from various fields ([10, 3323] nodes and [45, 23455] edges). For each of these networks we extracted 17 topological properties that will be considered independent variables in the classification training that was performed - those properties considered to be the most important from a correlation factor perspective in our previous research described in depth in Chapters 4 and 5. For each network, the frequency of motifs of size N=3 was already extracted. Table 6.1 shows a numerical description of the networks used, by considering the mentioned properties.

Table 6.1 Average values for the studied topological properties

Property	Mean	Property	Mean
AP	340.85	k_{core}	14.39
ρ	0.19	CC_L	0.3
d_{avg}	16.8	d_{max}	117.86
tr_{avg}	372.2	tr_{max}	611.79
D	0.05	d_{min}	8.85
H_{er}	0.9	N	5047.73
E	19371.89	γ	3.72
g	0.22	tr_{no}	143000.86
CC_G	0.28		

The training set was labeled with two possible outcomes, 1 being for networks that contain at least one motif in their topology and 0 being for network missing motifs. A question that may derive from the applied labeling is why it was not considered the frequency of motifs also during this process. Our approach does not consider the frequency of motifs as

being relevant in this case, as by its definition (Section 2.6) a topology sub-component that is considered to be a motif it is already an important validation.

From a total of 386 networks, 184 have at least one motif in their structure, hence we can consider the data set as being a balanced one for the intended binary classification training. Although networks which contain motifs are rare in a real-world context, in line with literature studies that indicate a possible biased accuracy between unbalanced classes [126] we kept a balance between the classes found in the training set using manual intervention as a first iteration and use complex fitness functions to evolve unbiased classifiers [20] if the training set became unbalanced in the next iteration of the model.

6.3.2 Model training and results comparison

Knowing that properties range scale is different, before training the classifiers it was applied a pre-processing step to normalise the data using a scaler robust to outliers.

The training set, respectively testing data set was obtained by splitting the initial data using a 80/20 proportion.

The models were trained using *sklearn* Python package, and there were not required adjustments to the configurations of their default implementation, such that we run the models using the parameters recommended by *sklean* in their documentation.

Table 6.2 Validation test results

	XGBoost	Decision Tree Classifier	Random Forest Classifier	SVM
Accuracy	**0.95**	0.88	0.91	0.72
Precision	**0.92**	0.84	0.89	0.67
Recall	**0.97**	0.92	0.92	0.78
F1-score	**0.95**	0.88	0.91	0.72

The validation tests revealed that the boosting method is the most qualitative one in each testing categories (Table 6.2). By doing a deeper interpretation analysis of these values, our use case is to look for models that are that are able to predict as many true positives as possible, even if there is a small fraction of false positives in the prediction. The reasoning behind this expectation is that using in a pre-processing phase a classification model to say if the network is prone to contain motifs or not, it will not be bothering to know that there is a small chance to process a network that will not contain motifs as long as most of the predictions will indicate expected values. In this worst case scenario, the network will be processed without any expected motifs.

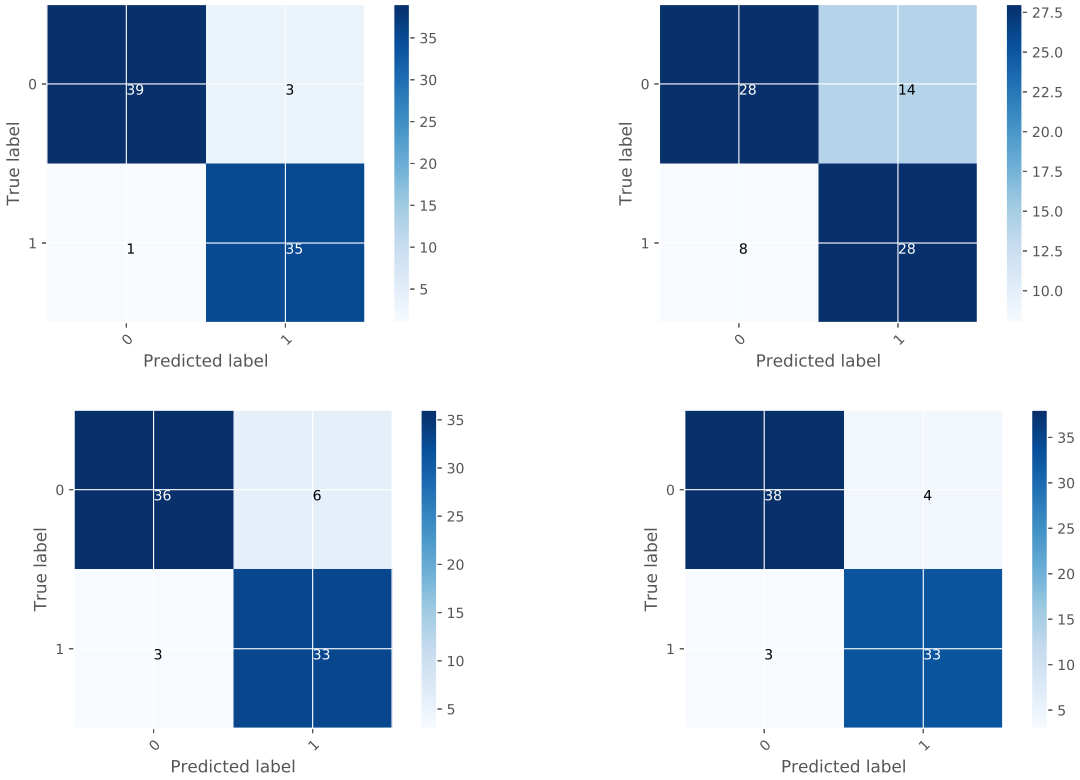


Fig. 6.1 Quality evaluation of the trained model using confusion matrices (Top-Left: XGBoost, Top-Right: SVM, Bottom-Left: Decision Tree Classifier, Bottom-Right: Random Forest Classifier). 0 label - network without motifs, 1 label - network with motifs

Using these results we can conclude that using our previous study on the correlation between the topological properties and appearance of the network motifs it is possible to train a statistical model to classify given networks as networks prone to contain motifs or not. Moreover this analysis was performed on various models such that it was possible to propose the boosting method as the best solution for this use case.

6.4 Conclusions

The discovery of network motifs in complex networks is a very intensive process being ranked as an NP-complete problem. Literature presents various solutions of optimising this process, either by making use of statistical tricks to reduce the number of computations but with a trade-off in the accuracy of the results, or by re-modelling existing algorithms into parallel models able to run on cluster of computers. From our knowledge there are no methods applied as a pre-processing step to optimise this process.

Driven by this void in literature, our work presents a method of detecting if a network is prone to contain network motifs or not by using its topological properties. Considering the results presented in Chapters 4 and 5 that revealed correlation of different strengths between these topological properties and the existence of the motifs, this being a good indicator to consider the properties as independent variables to predict the presence of the dependent variable, network motifs.

Modeling this as a binary classification problem, Section 6.3 presents the steps applied to train a set statistical models using a consistent data set of networks. Making use of various tests, it was observed that the statistical model using a boosting method excelling in all test categories. Moreover, this method presented a high recall value, which is one of the most relevant measure for our use case as it will not be bothering to have a small fraction of false positives as long as the prediction will return as many true positives as possible.

Acknowledging the presented results, we consider the proposed model one with a highly probable impact in the field of complex networks, helping both our experiments and the community to execute computation faster while performing a network motifs research. As an improvement, we could consider a perspective of transforming the model from a binary classifier into a regression model that is be able to predict the frequency of the motifs.

Chapter 7

The study of the empirical evidence of dynamic convergence existing between complex networks' layers

As often described in the narative of this thesis, Complex Networks theory represents a powerful tool to model real-world systems as graphs with non-trivial topological features. Static by their definition, complex networks are limited to be the reflection or the snapshot of the dynamical systems they encode in a given moment. Frankly, studies show that the network preserves the characteristics of the dynamic flow between various structures evolved at different scales: micro, meso, and macro, respectively.

Considering that micro components will end by encapsulating any stability state initiated by a perturbation of the macroscale, the most popular framework used to analyze complex networks associates in a reverse-engineering manner properties of the micro-scale components with global network characteristics. Although the principle of convergence is widely agreed upon, we consider the transition from micro to macro one not studied enough, and the highlighting of an in-between layer being necessary exemplifies the convergence mechanism better. Our work is willing to improve existing approaches of understanding the intricate beauty behind network dynamics by embracing the chaos they evolve and not order it using inflexible analytical mechanisms. We formulate a series of research questions that are willing to demonstrate that the existence of a mesoscale structure, namely network motifs, is connected with a variety of microscale, respectively, macroscale properties - hence strengthening the proof of macro-meso-micro convergence.

Our approach could provide a substantial improvement in the models used to understand molecular self-assembly, protein-based cellular adaptation, social interactions, and many

other fields, where connections between structures or components play an essential role. The research presented in this chapter was published in the paper **[98]** and it outlines a comprehensive an original analysis framework that ultimately aims the proposal of a null-model for the evolution of network motifs in networks. This analytical framework is based on the principle of causalization of topological properties, which foster the existence of motifs. The results of this analysis represent a significant contribution to the field of network science and provide a valuable resource for future research.

7.1 Exploring the current state of the art: An In-depth Analysis of Problem Definition

The state-of-the-art literature shows a tendency to study complex networks by making use of its most fundamental components, nodes. Driven by principles found in the Stability theory of dynamical systems that relates the structural state convergence from macro to microscale components (Subsection 2.3), the conventional analytical framework proposed by the literature is using the nodes in most experimental proofs of the studies done by the community [59, 60, 81, 121].

Starting from applications that are focused on microscale properties, nodes were also used to describe macroscale characteristics of the network. Making use of a reversed engineering approach of the structural state convergence from macro to micro, it was observed how measurements on microscale could help understand macroscale properties such as networks with significant characteristics (small-world networks, sparse networks or random networks). Although this type of analysis improved the way we understood the complexity of topological spaces such as complex networks, we think that the transition between micro and macro is a too significant leap with questions for what is happening in between.

In order to have a robust analytical framework, we consider essential the addition of the meso layer in the equation of network analysis. This layer can be seen metaphorically as a structural glue between micro and macro, the missing variable in the equation of state convergence from macro to micro in complex networks. Centered around the interactivity between layers, these framework capabilities of analysis are flexible, and they can be extended to more than the understanding of state convergence.

To prove that such a framework is valid and has real-world implications, in the next sections, an experiment over a large set of networks will be presented. The results aim to prove the existence of correlations between each layer of the network's properties that

can serve as empirical evidence to a new analytical framework that will be centered on the interconnectivity between the layers of the complex networks.

7.2 Methodology

As it was discussed in Subsection 7.1, the conventional analytical framework of complex networks is one mainly based on studying microscale properties. Although the literature presents various applications for meso and macroscales, all these network layers are widely studied individually without a precise determination in unifying them under a universal framework.

The literature presents a particular interest in this direction with applications that study the connection between meso and macro, or micro and meso-layers. Understanding the void in the literature, we propose unifying the currently distinctive micro-meso and meso-macro analysis under a single framework micro-meso-macro. Figure 7.1 offers a graphical representation of how the network layers can be seen as a multiplex network where each upper layer has a hierarchical impact on each inferior layer. In this sense, a perturbation on the macro level will impact the structures from the meso layer (network motifs, communities), which will impact the behavior on the micro-level of each atomic entity (nodes).

In order to prove that such a framework is reliable and robust for real-world applications and is aligned with the existing concepts from Networks Theory we will propose, in this subsection, a methodology for the experimental to ensure the correctness of the suggested concepts.

When we discuss the meso layer, communities and network motifs are structures that gained significantly more popularity in complex network applications [37, 49, 54]. Our focus for this study will fall on network motifs as structures with an occurrence being significant to contextual reasoning. We are looking for a correlation between microscale and macroscale properties and the existence of network motifs.

Desiring to provide an elegant and clear structure analysis, the experiment will be split into the following steps:

- study the correlation between microscale properties (Subsection 2.5) and the existence of network motifs.

- using ω score (Subsection 2.7), study the correlation between macroscale properties and the existence of network motifs.

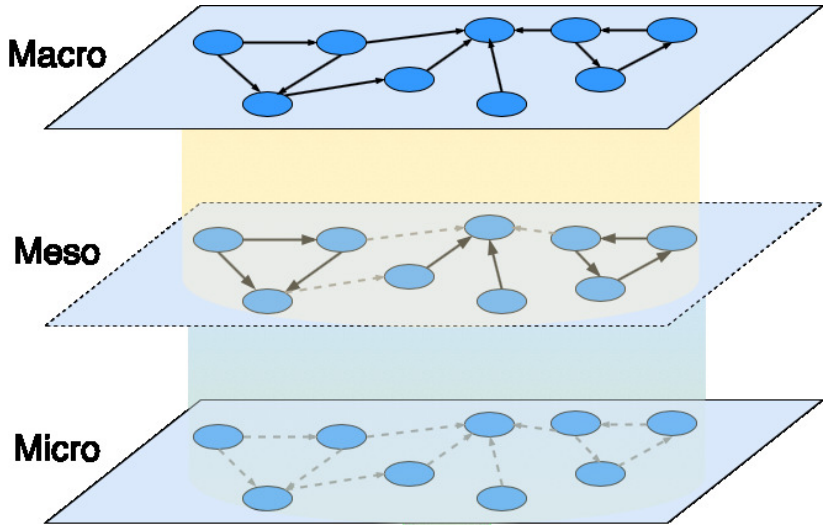


Fig. 7.1 Interconnectivity of the components of the network by scale. Similar to the perturbation mechanism discussed in Section 2.3, the macro level will impact the structures from the meso layer (network motifs, communities), which will affect the behavior on the micro-level stabilizing at the nodes level

- unify micro-meso and meso-macro correlations under the micro-meso-macro framework making use of state-of-the-art micro-macro studies.

In order to achieve valid empirical evidence for the analysis steps, we will use a data set of real-world networks modeled from complex systems coming from various fields.

7.3 Research questions

Our experimental analysis is looking forward to responding to a list of research questions starting from the methodology proposed in Subsection 7.2.

These research questions aim to serve as a guideline for both reader and researcher in understanding and following the goals of the study with respect to methodology steps. Considering these aspects, we formulate the following research questions:

- *RQ1* - Is the existence of network motifs correlated with the properties of microscale components?
- *RQ2* - Is the existence of network motifs correlated with the networks' macroscale ω?
- *RQ3* - Is the existence of network motifs correlated with the properties of microscale components that impact the computation of networks' macroscale property ω?

7.4 Data

While performing the analysis, we used a data set of 88 real-world networks that come from various fields: sociology, biology, infrastructure, ecology, or informational science (between [19, 3323] nodes and [78, 23455] edges).

To exclude a biased result, our data set contains networks with homogeneous topological properties and a medium to large size, size being measured in the number of nodes. For each of these networks, we extracted 17 microscale properties and computed ω coefficient as a macroscale way to determine the type of each network: sparse, small-world, or random. Having this topological classification, it is easier to follow how specific properties are more prominent in a given network type or others. In this sense, we can observe how small-world networks tend to have a significantly higher number of *AP* compared with sparse or random networks because their topologies are one rich of interconnected communities. High values for CC_L, CC_G or ρ are characteristic to sparse and small-world networks due to the tendency of their nodes to cluster together following a preferential attachment behavior.

For each network, the extraction methodology for motifs with size k=3 was applied (Subsection 2.6). From a structural perspective, motifs with k=3 are equivalent to the famous structures from social networks named closed triangles and open triangles (triadic closures). Nevertheless, motifs with k=3 and triadic closures should not be confused from a functional perspective, as triadic closures refer to a structural property that is limited to a k=3, when a motif is a structure characterized by a function given by its frequency and contextuality. Moreover, a motif can have any given size of $k < N$, N being the number of nodes in a network.

We did not apply any pre-processing procedures of the networks because we did not want to alter in any way the original structure of the networks.

Table 7.1 contains a numerical representation of the microscale properties grouped and averaged by network type. The networks were labeled with a type as follows:

- sparse: -1 $\leq \omega <$ -0.5
- small-world: -0.5 $\leq \omega \leq$ 0.5
- random: 0.5 $< \omega \leq$ 1

Table 7.1 Average values for the studied properties by network types

Property	Sparse	Small-world	Random	Property	Sparse	Small-world	Random
$AP_{0\leq AP<inf}$	1.05	35.8	2.43	$k_{core_{0\leq k_{core}<inf}}$	6.79	8.87	6.86
$\rho_{-1\leq\rho\leq 1}$	0.52	0.15	0.16	$CC_{L_{0\leq CC_L\leq 1}}$	0.48	0.4	0.01
$d_{avg_{0\leq d_{avg}<inf}}$	10.34	11.36	11.28	$d_{max_{0\leq d_{max}<inf}}$	13.26	29.5	27.79
$tr_{avg_{0\leq tr_{avg}<inf}}$	37.23	72.09	7.12	$tr_{max_{0\leq tr_{max}<inf}}$	50.58	185.3	44
$D_{0\leq D\leq 1}$	0.02	0.11	0.04	$d_{min_{0\leq d_{min}<inf}}$	6.95	5.55	5.36
$H_{er_{0\leq H_{er}\leq 1}}$	1.0	0.98	0.99	$N_{0\leq N<inf}$	832.37	408.87	1018.21
$E_{0\leq E<inf}$	3597.05	1543.76	5920.43	$\gamma_{0\leq\gamma<inf}$	9.12	3.40	2.82
$g_{0\leq g\leq 1}$	0.07	0.17	0.14	$tr_{no_{0\leq tr_{no}<inf}}$	13761.16	9590.50	3174.21
$CC_{G_{0\leq CC_G\leq 1}}$	0.46	0.35	0.09				

7.5 Micro-meso scale correlation

Considering the results obtained in Chapters 4, 5 and 6 that tackled the problem of correlation between properties of microscale components and the existence of network motifs while bearing in mind the questions how and why the perturbation from macro layer converges to mesoscale structures and then to microscale components. In another train of thought, we consider that it requires an iterative analysis to generalize the results into a framework able to explain the entire mechanism of state convergence. The study of the correlation between microscale properties of a network and network motifs can be seen as the first step following this approach.

To come with an answer to RQ1, this subsection serves as a reviewing platform for these recent researches in this area, ended by a series of conclusions of overall studies.

Starting from a philosophical question, widely discussed in Chapter 4, the focus was to study the co-existence between *AP* and network motifs. In real-world systems, *AP* can be seen as bridges that connect independent communities. This gives them an apparent power in controlling and injecting certain stimuli in the isolated communities they connect. Encoded in the network topology, the dynamic generated by these stimuli can be encoded as pattern-like structures, namely network motifs. Hence seen either as influences or as critical points, the hypothesis was that there must be a significant correlation between the existence of *AP* and the existence of network motifs.

The study revealed that networks that contained *AP* were prone to contain motifs found in the structure of the complex networks (Figure 7.2). Moreover, the number of *AP* tends to be correlated with the numbers of motifs found in the proximity areas of *AP*, such that the majority of network motifs are found in the proximity areas linked by the *AP*, and only a small fraction of network motifs is found in sub-networks that are not linked by any *AP* (Chapter 4). With this finding, it was possible to confirm the initial hypothesis that stated the role of *AP* as controllability factors in the network topology - favoring the evolution of motifs in the independent communities they link.

Understanding that the study of *AP* does not have a strong state-of-the-art yet, in Chapter 4 we focused on correlating more popular microscale properties and the existence of network motifs. Although there were studied a number of 17 properties, the focus had fallen on two properties: CC_L and ρ. Knowing that network motifs appearance is characterized by contextual reasoning, these two properties seemed to be susceptible to encourage the evolution of motifs.

As a first hypothesis, the study proposed that networks with a tendency of CC_L could

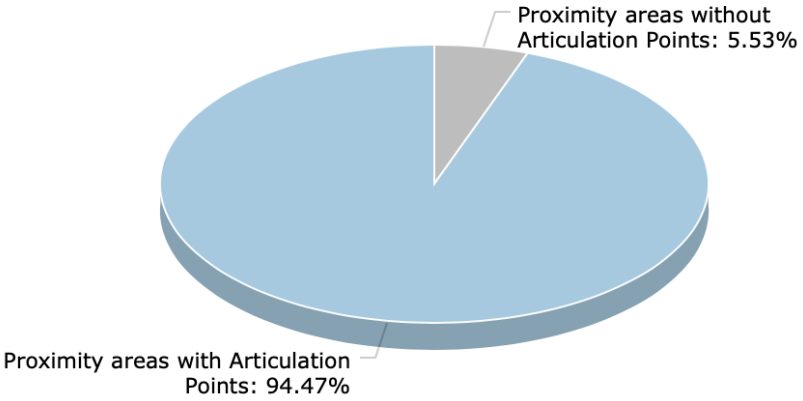


Fig. 7.2 Motif frequency considering the existing of *AP* in the studied proximity areas, indicating that areas with *AP* are prone to evolve network motifs.

have a higher chance to provide specific types of interconnections, namely network motif. Following the same principle of local clustering characteristics, the second hypothesis stated that networks with dynamics that are suitable for preferential attachments have a topology that is more probably to favor the appearance of network motifs. Both hypotheses were proven statistically using a series of correlation tests: Pearson correlation, Spearman correlation, and Kendall rank correlation. Unanimously, all three correlation measures proved a significant positive correlation between the existence of network motifs and CC_L for the first hypothesis, respectively ρ for the second one. Figures 7.3 and 7.4 display the distributions of the studied networks by ρ respectively CC_L) - for a better visual representation, 2 classes (low and high) were defined as follows:

- low: ρ or $CC_L < 0.4$
- high: ρ or $CC_L \geq 0.4$

Value 0.4 was considered as being a good enough threshold to divide networks by considering properties ρ or CC_L. A value greater than 0.4 already indicates a clear emerging trend in most networks dynamic.

Bearing in mind the strong empirical evidence found and applied in these studies regarding the correlation between micro and mesoscale layers presented in Chapters 4 and 5, we consider the presented research to be more than satisfactory to serve as an answer to RQ1.

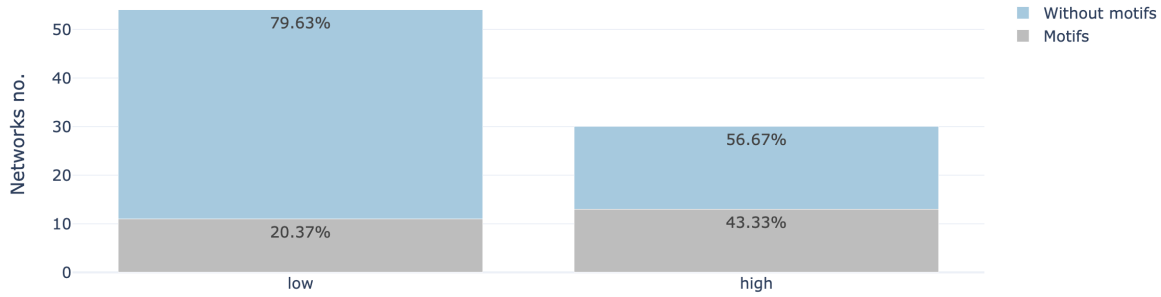


Fig. 7.3 Networks distributions by ρ indicating that networks with a preferential attachment characteristic are prone to evolve motifs.

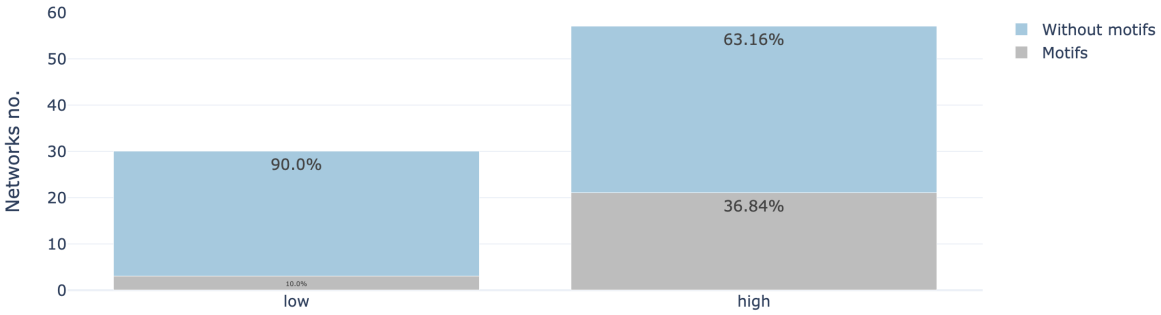


Fig. 7.4 Networks distributions by CC_L suggesting that networks with a tendency of local clustering are prone to evolve motifs

7.6 Meso-macro scale correlation

Following the same principle of correlating mesoscale structures, namely network motifs, with microscale properties, in this subsection, we present a corresponding analysis between the appearance of network motifs and ω - a coefficient able to score the topology of a network as being sparse, small-world or random. After computing to each network a corresponding type by making use of ω (Subsection 7.4), we want to understand which of these types is more susceptible to contain network motifs.

From a total of 88 networks, 24 networks contain at least a network motif, and 64 do not have any. Considering ω intervals presented in Subsection 7.4, based on which a network will be labeled with a given label, Table 7.2 displays a numerical distribution of the networks types with information about network motifs existence.

Table 7.2 Network count with/without motifs by type

Type	With motifs	Without motifs	With motifs ratio	Total
Sparse	15	4	0.79	19
Small world	8	47	0.15	55
Random	1	13	0.06	14
Totals	24	64	0.27	88

It is empirically observable how networks with a topology closer to a sparse characteristic are prone to contain more motifs than networks with a more random topology. To have stronger evidence for this initial lightweight assumption, we propose a more stratified and detailed attribution of network types by considering intermediate network types with an in-between two main categories characteristic, sparse and small-world, respectively small-world and random. In this manner, we will have three primary intervals associated with the network types: sparse, small-world, and random, and 2 "transition" intervals that intend to quantify the transition behavior between the primary intervals. The goal of this interval definition is to be used to count the existence of motifs and associate it to one of these intervals, transposing the frequency of motifs with the continuous omega interval that is segmented in labeled intervals.

Considering these aspects, we introduce new types as follows:

- sparse: $-1 \leq \omega \leq -0.66$
- sparse-small-world: $-0.66 < \omega \leq -0.33$
- small-world: $-0.33 < \omega \leq 0.33$

- small-world-random: $0.33 < \omega \leq 0.66$
- random: $0.66 < \omega \leq 1$

Table 7.3 Network count with/without motifs by reinforced types

Type	With motifs	Without motifs	With motifs ratio	Total
Sparse	11	2	0.85	13
Sparse to small-world	7	11	0.39	18
Small-world	5	33	0.13	38
Small-world to random	1	13	0.07	14
Random	0	5	0	5
Totals	24	64	0.27	88

In this new stratified configuration, it is more visible how the networks with motifs are not found anymore in the random network group, but more likely in groups of networks characterized by high CC_L and ρ such as small-world, and even more sparse network.

Figure 7.5 displays a segmentation of the networks by their correspondent reinforced type (formulated earlier) considering if they contain motifs or not. This distribution offers a clearer view over the formulation of an answer for RQ2, as it is observable how networks with CC_L and ρ are more probably to evolve network motifs than networks with a random characteristic. A strong argument to enforce this statement is given by the theoretical insight of network motifs presented in Subsection 2.6 which states that a contextual meaning characterizes network motifs and that they will appear only in real-world networks and not in random networks even if they hold the same degree of distribution as the initial networks. Hence networks driven by an evolution based on preferential attachment in microscale components will be more probable to evolve network motifs in these isolated structures, described in Chapters 4, 5 and 6.

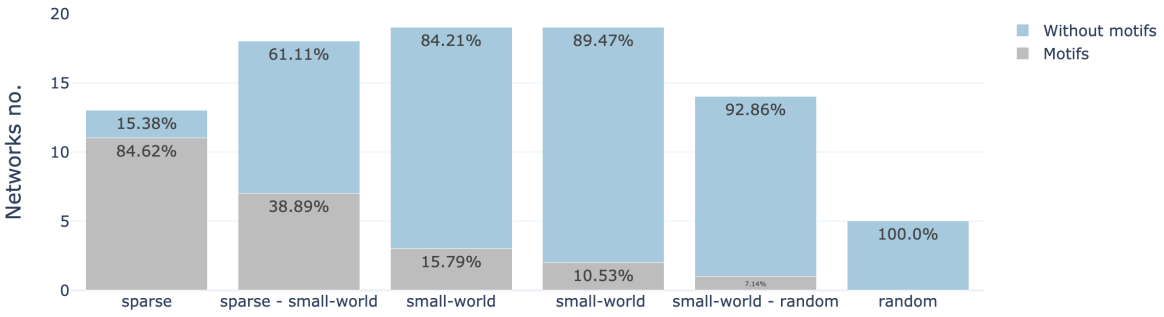


Fig. 7.5 Network with motifs by reinforced types indicating that networks with prominent clustering characteristic facilitate the evolution of network motifs

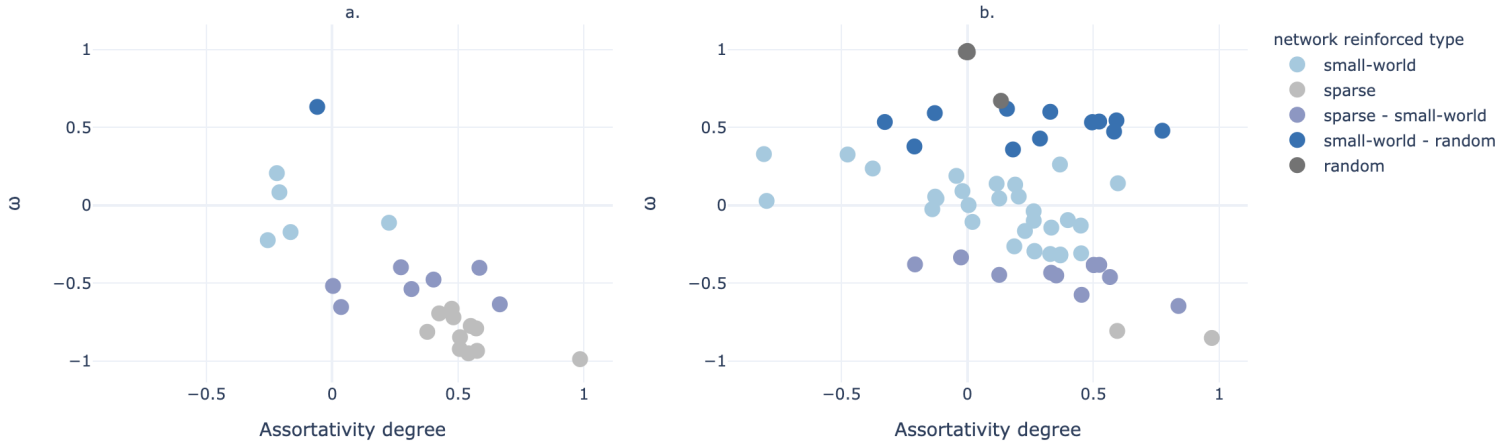


Fig. 7.6 Networks distribution by ω score and assortativity degree with emphasis on reinforced types (a - networks with motif, b - networks without motifs). Networks with motifs are more frequent in reinforced types with high CC_L and ρ values (sparse, sparse-small-world, small-world). In a head-to-head comparison between subplots 7.6.a. (networks with motifs) and 7.6.b. (networks without motifs), the trends indicate that most of the networks with motifs are found in the lower-right frame of the plot (area with high CC_L and ρ values), where networks without motifs follow an upper trend heading to the upper-left frame (area with low CC_L or ρ values)

Considering the empirical evidence found between the study network types and the existence of network motifs, we can say that there is indeed a strong correlation between certain types of networks and the existence of network motifs. Due to topological properties, networks such as sparse or small-world are prone to contain motifs in their structure (RQ2). Figure 7.6 emphasizes this argument even better, as it displays a distribution of the networks based on ω (which includes CC_L) and ρ. It is observable how networks with motifs are more frequent in reinforced types with high CC_L and ρ values (sparse, sparse-small-world, small-world). This specific trend is observable in a head-to-head comparison between subplots 7.6.a. (networks with motifs) and 7.6.b. (networks without motifs), where most of the networks with motifs are found in the lower-right frame of the plot (area with high CC_L and ρ values), where networks without motifs follow an upper trend going to the upper-left frame (area with low CC_L or ρ values).

Following this trend, we can say that we found that specific networks are correlated with the appearance of network motifs, which are also correlated with microscale properties characteristic to the macroscale properties of the network ($RQ3$). With this evidence, it is possible to conclude that there is a connection that can not be neglected between these three layers, which has a convergence dynamic that is sustained by the meso-layer, which can be seen as a bridge between micro and macro components.

7.7 Conclusions

The scope of the experiment presented in this chapter is to represent a solid experimental foundation of a new way of understanding the interconnectivity between micro, meso, and macro layers of complex networks while demonstrating that the existence of a mesoscale structure, namely network motifs, is connected with a variety of microscale, respectively, macroscale properties.

Starting from the experiments discussed in Chapters 4, 5 and 6, where a detailed statistical analysis was performed between properties of microscale components and mesoscale property network motifs, we intended to extend the experiment by studying the relation between network motifs and macroscale properties while using a homogeneous data set of networks.

In the sections describing the experiment, we discussed the findings of the exercise led accordingly with the methodology presented in Section 7.2. We observed that the existence of network motifs is correlated with microscale properties indicating a preferential attachment behavior that induces a clustering dynamic of the microscale components. Moreover,

networks characterized by such microscale properties (sparse or small-world networks) are more prone to evolve network motifs compared to networks that lack features of nodes clustering or nodes preferential attachment.

By considering the empirical evidence found in the correlation between the properties of microscale components and network motifs (Section 7.5), respectively network motifs and macroscale properties (Section 7.6), we concluded that the studied mesoscale property could be seen as a bridge facilitating the dynamical convergence from macro to microlayers.

As a next step, we intended to unify the analysis performed between microscale properties, network motifs, and macroscale properties under a framework able to analyze complex networks by making use of the interconnectivity between micro-meso-macro layers we discovered. By introducing network motifs as a variable in the function that returns a more specific macroscale property to a given network, we propose a theoretical framework that can make use of both microscale properties and mesoscale - such as network motifs - to compute macroscale properties with better accuracy. This approach could lead to a better discretization of the networks, from the traditional classification based on microscale properties only to a more precise categorization using network motifs (e.g., a small-world network containing specific configurations of network motifs which define the network as being a small-world network but with a unique characteristic given by the function of the motif).

Our current research focused on networks with or without motifs but did not take into account their size, type, or frequency. In order to bring evidence to our study, we consider this future research to be one of great interest. An additional analysis that could give more clarity to the state of dynamical convergence, which we intend to study, is the introduction of other mesoscale structures such as communities.

The modeling of such a study under a system of functions that can classify networks by mesoscale properties would represent the ultimate goal of the better discretization of network types we are aiming to achieve.

Understanding the void in the literature in this direction, we considr that such a proposed investigation can open new and undiscovered horizons in the analysis of complex networks that will increase the accuracy with which we comprehend the dynamic of complex systems and will serve as an analytical framework to propose the null-model capable to generate networks that contain motifs.

Chapter 8

Generating Random Complex Networks with network motifs using Evolutionary Algorithm-based null model

As discussed in the previous chapter, the existence of network motifs is mainly attributed to the apparent chaotical behavior of the atoms in the systems that the networks are modeling, behavior that emerges in patterns that will be encoded in the network. Motifs were not formalized as structures due to their rarity but only explained through their contextual appearance in the networks. The series of experiments described in the previous chapters are part of the goal of explaining the dynamics of network motifs and they have demonstrated a correlation between the presence of network motifs and various micro- and macro-scale properties of the network topology. This chapter introduces a novel null model architecture that employs Evolutionary Algorithms (EA) to generate random networks containing motifs. The proposed architecture is characterized by a set of parameters, including the Clustering Coefficient and Assortativity Degree, which have been identified as key factors in triggering the convergence between network layers leading to the emergence of network motifs. Such a model has the potential of significantly improve the study of network motifs by generating any number of synthetic networks that can serve as experimental material for the community of researchers.

The following sections present the results of our research submitted in **[93]**, which conclude with the proposal of a null-model that controls the evolution of synthetic networks with network motifs using Evolutionary algorithms. This proposal is the crowning achievement of our extended research efforts described throughout this thesis.

8.1 Problem definition

Complex Networks are powerful models for representing real-world systems, but obtaining and processing data to build these models can be both logistically and programmatically challenging. To overcome these challenges, the research community has proposed null models as a solution [29, 79]. These null models generate synthetic networks with specific topological properties and encode them as mathematical models that can be easily generated from input parameters.

However, the complexity of this approach lies in need for a deep understanding of the dynamics of the network being modeled. This leads to an "egg-chicken" problem, where proposing a null model based on the backbone system is difficult without first studying many real-world networks. In turn, obtaining these real-world networks is difficult without a null model to generate them.

This ongoing challenge explains why null models for Complex Networks are rare and difficult to propose. To generate accurate and meaningful null models, researchers must strike a delicate balance between understanding the underlying dynamics of the network and the practical challenges of obtaining and processing the data [22]. Ultimately, this requires a multi-disciplinary approach that incorporates mathematical modeling and domain experts' expertise in the field of study.

8.2 State of the art

The process of generating synthetic networks is prevalent in the field of Complex Networks from a practical point of view, as it represents a viable process of obtaining an indefinite number of networks that can be used in the researchers' experiments, getting as close as possible to analyzing real-world networks that are harder to be obtained.

As the ultimate need is to generate networks as similar as possible to real-world networks, proposing null models capable of generating networks that are not purely random but hold specific topological properties becomes a reverse engineering problem: understanding the dynamics of the networks you want to generate and use to simulate the same dynamic with its law as a mathematical model.

The literature presents various null models grouped based on their approach to generating the synthetic network. In this section, we will concentrate on the most representative models of their group, emphasizing the approaches used as state-of-the art and how our proposal will differ with respect to them.

8.2.1 Classical model generators

This category of models is one that can be considered to generate networks based on the most straightforward set of rules. Most of the time the generated networks do not have a direct real-world equivalent, as the applied controllability is one that intends to generate topologies with structures that replicate geometrical shapes.

One such example is *Lollipop graph* [34], which as the name indicates is a graph with a topology embodying the shape of a lollipop (Figure 8.1).

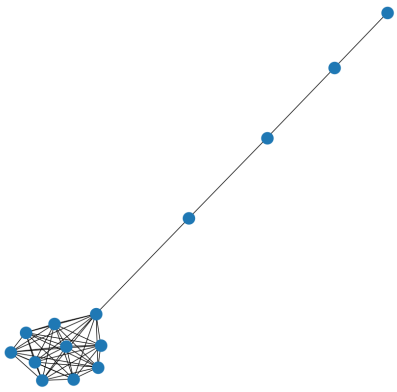


Fig. 8.1 Generated lollipop graph with $K_n = 10$ and $H_s = 4$.

As the figure indicates, this type of complex network contains a fully connected sub-network (or a so-called clique) that is attached to a node connected to a hub of other nodes, each being linked through edges, creating a "line". The clique is considered to be the actual lollipop candy, and the linear hub is the stick holding the candy. From an algorithmically perspective, the null model capable of generating a lollipop graph receives as input the clique size (K_n) and the hub size (H_s) in a number of nodes and:

- generating a clique by spawning K_n nodes and adding edges between each node until they become a fully-connected component.
- spawning H_s nodes that are linearly connected, and only one of the nodes at the ends of the "line" will get connected to the clique generated in the first step.

As a more complex version of the lollipop graph, we have *Barbell Graph* [35] which does not do anything more but generate two cliques of size K_n that will be linked through the linear hub: one end node links clique one, the other node end link clique two (Figure. 8.2). Barbell graphs specifics are what we will later find in more complex models that

generate small-world networks (high clustering coefficient with clusters of nodes densely connected, but poor connectivity between the clusters).

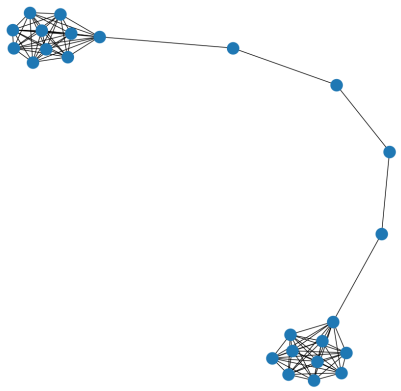


Fig. 8.2 Generated Barbell graph with $K_n = 10$ and $H_s = 4$.

The other two honorable mentions are the null models that generate networks with a star shape, respectively, a circle. Although one may consider them as being generated purely for aesthetics, these two networks can be found as motifs in real-world networks, motifs with powerful functions in the systems they are found (star motifs are widely found in internet networks where a hub node connects multiple hub nodes, like a modem with a series of computers; cycles are popular in the study of natural chains).

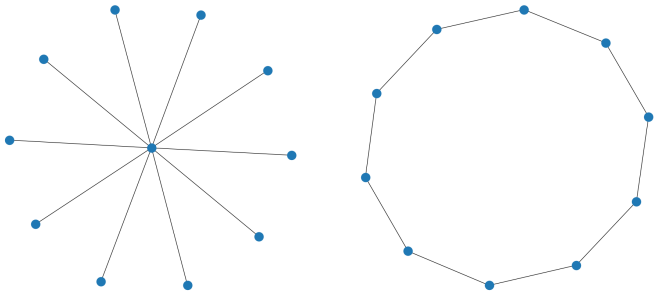


Fig. 8.3 Left: star graph generated with $n = 10$; right: cycle graph generated with $n = 10$.

Although one can consider these null models simple from an algorithmic perspective and what they try to encode, what makes them essential is that they represent just the first iteration of the more complex models we will discuss in the following sections.

8.2.2 Degree sequence model generator

In real-world interactions, one can observe the existence of preferential attachment [73], meaning that the atoms of the systems will tend to interact with other atoms based on specific properties (E.g., in social networks, famous people will look to have relationships with other popular people). This behavior creates the so-called power law distribution characterized by an unbalance between the existence of a phenomenon and its power (E.g., 20% of the world population holds 80% of the economy [101]).

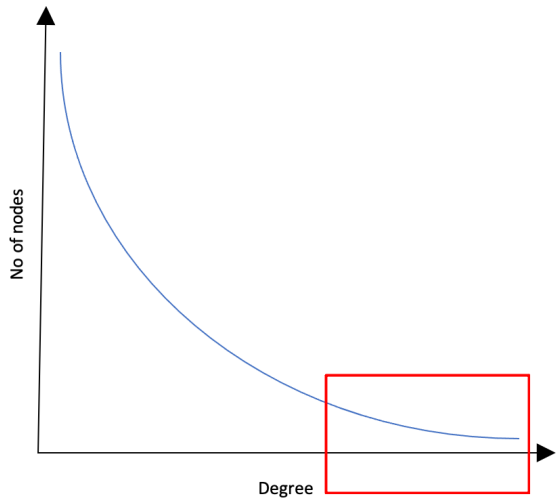


Fig. 8.4 A power-law distribution where the sample of nodes in the red box hold the largest majority of nodes degrees.

In complex networks, this behavior is often found in real-world systems as an unbalance between the number of links and the nodes that hold them (20% of the nodes will have 80% of the links [12]). The class of null models that model this behavior are named degree sequence model generators, mimicking it as creating networks with specific degree sequences by linking nodes randomly until a given degree sequence is met.

One of the most representative such models was proposed by Barabási and Albert in their paper [12], in which they introduce the concept of scale-free networks in the literature. Their model proposal is one that iterates over a sequence of steps until a network of size n is reached. The model starts from a network with $m0$ nodes in a ring topology and generates new nodes that are attached probabilistically to the existing nodes considering their degrees (a high degree means a higher chance of attaching to an existing node). In Figure 8.5, a scale-free network is graphically represented with hub nodes situated at the core and nodes with smaller degree located at the geometric periphery.

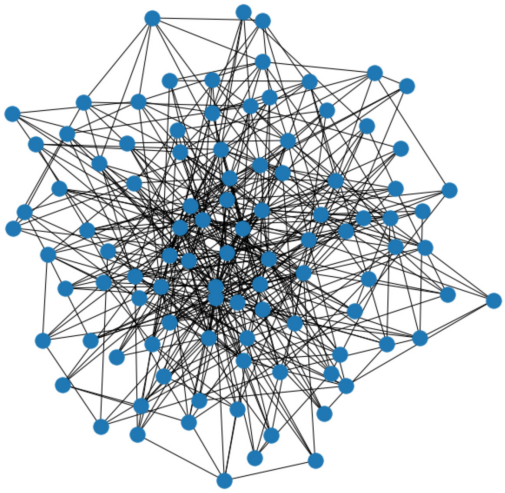


Fig. 8.5 Generated scale-free network using Barabási and Albert model with $n = 100$, $m0 = 15$.

This model and its other variations [2, 91] have proven to be a vital tool in the study of social networks and modeling the evolution of social structures [72], biological networks and the evolution of proteins or genes connections [13], technological networks and internet speed optimization [36] through world nodes and many other examples. Using these null-models, one can gain insights about universal law in the underlying dynamic of the systems studied and come up with predictions about the behavior of the network and how hub nodes evolve.

8.2.3 Small-world model generators

From a complexity perspective, these generators are considered the null-models encoding closest to reality, the dynamic of most social networks. Whenever we think about a small-world network, we will consider a combination between a high clustering coefficient (CC_l) and short average path length, meaning networks' nodes tend to connect to other nodes closer to them, creating well-connected clusters [100]. At the same time, the clusters will be connected through a few short paths between any node part of a cluster. As a general property, we can say that small-world networks are performant in terms of information exchange as the nodes inside a cluster will be quickly reachable through the density of the inter-cluster connections, but also with nodes from other clusters through the direct connections existing between specific nodes part of them.

Firstly introduced by Duncan J. Watts and Steven H. Strogatz in their study [134], and later extended by M. E. J. Newman [102], one of the most known generators capable to

generate synthetic small-world networks is the Newman-Watts-Strogatz model. This model is considered a state-of-the-art generator whenever one needs complex networks holding the topological structure of a small-world network.

From a graphical perspective (Figure 8.6), a small-world network will display a prominent cluster of nodes with a high density of links between the nodes part of the clusters and occasional links between the groups of nodes that reflect the existence of short average path length.

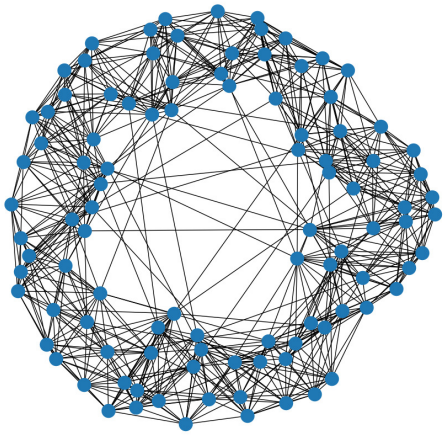


Fig. 8.6 Generated small-world network using Newman-Watts-Strogatz model with $n = 100$, $k = 15$ and $p = 0.1$.

Algorithmically, the model proposed by Newman-Watts-Strogatz is one that starts from a group of nodes n that are grouped in multiple ring (cycle) topologies of k nodes, group of nodes that will be the starting point for building the clusters using a logic of adding randomly links between nodes outside their clusters with a probability p.

The parameter p controls the balance between randomness and regularity, where a high p will generate networks with random topologies as the isolated initial clusters get more and more linked to other clusters until they merge as a single entity. On the other hand, a low p will generate more regular networks as their specific structure of containing isolated clusters that are occasionally linked with other clusters through a few edges will persist.

Newman-Watts-Strogatz model represents a state-of-the-art null model for generating small-world networks due to its capabilities of juggling between randomness and regularity specter that fits and generalizes most real-world systems from social, biological, or technological sciences [3, 90].

8.3 Network motifs null-model proposal

In the previous section, a comprehensive review of the state-of-the-art network generators and their efforts to imitate real-world systems was presented. In this chapter we aim to introduce a new null-model that evolves network motifs, which requires an understanding of the limitations of existing null-models and the reasons why our proposed approach will provide a new category of network generators.

Degree sequence model generators are based on the principle of preferential attachment, which creates specific connections between network nodes that are encoded as patterns of interconnections, or network motifs. However, the mechanism of preferential attachment is rather general and can be defined in multiple ways. One specific type of preferential attachment that drives the evolution of motifs is measured through assortativity degree (ρ). The results of the experiment discussed in Section 5.2 demonstrate that networks with a high ρ are more likely to evolve network motifs due to the tendency of nodes to link with higher probability to nodes with the same degree. The distinction between the degree sequence model generator and the dynamics behind assortativity degree is that state-of-the-art models add new nodes and connect them to existing nodes with a probability proportional to their degree, leading to a few high-degree nodes that are not interconnected and therefore have a low assortativity degree. The objective of the research described in this chapter is to present a novel null-model that builds upon the principle of preferential attachment. Unlike existing models in the literature, this null-model will utilize a mechanism based on assortativity degree which we consider to be the innovative factor, and one that will create a dynamic prone to evolve network motifs with a higher probability than the current model generators.

The model incorporates a second component aimed at increasing the CC_l value. This component differs from small-world network models that randomly create cliques in a ring topology. Instead, the model will use the principle behind assortativity degree to form clusters of nodes that are linked with higher probability to nodes with similar degrees. This approach will allow the model to evolve the two properties most strongly correlated with the presence of network motifs, as described in Chapter 7. The resulting network topologies will mimic the seemingly chaotic behavior of real-world systems, thus generating the patterns that are observed as network motifs.

Classical model generators do not aim to simulate real-world behavior, but instead generate artificial topological structures that can serve as prototypes for applications in areas such as optimization and information flow. Therefore, this type of model will not be included in the experiment proposed and discussed in subsequent sections. Instead, the focus

will be on degree sequence models and small-world network models, which better align with real-world behaviors and they can be compared with the model we propose.

The next sections will present the algorithmic approach for the network motifs null-model, which aims to evolve the network through multiple iterations using a mechanism from the field of artificial intelligence. The formation of motifs is not a purely random or static process, it requires multiple iterations and a continuous dynamic that will eventually result in the emergence of motifs. As such, evolutionary algorithms are considered a suitable framework for implementing the null-model. Despite the limited applications in the literature, the use of artificial intelligence as a framework for generating null-models is a novel approach, with successful results demonstrating its potential to advance the state of the art in model generation [8, 63].

8.3.1 Model arhitecture

There is no better framework to generate networks mimicking real-world dynamics than a real-world process at the base of the nature that surrounds us, a process that optimized the evolution of biological organisms using natural selection process. Evolutionary algorithms (EAs) [42, 61] are a class of algorithms from the field of artificial intelligence that optimize solutions by utilizing a population of candidate solutions. These candidates are evaluated using a fitness function [21, 43] and propagated through multiple generations through a process of selection. The selection process involves selecting candidate solutions that maximize the success rate of the fitness function after the application of operations such as mutation, recombination, and selection on the population. The process stops when an optimal solution was found or any stopping criteria was met. Figure 8.7 aims to depict the typical flow of a generic evolutionary algorithm using the components previously described.

In the initial stage, a population of p chromosomes, serving as candidate solutions, is established. These chromosomes are then combined through the application of a crossover function, resulting in a new offspring chromosome that inherits characteristics from both parents. Subsequently, the offspring chromosomes undergo random mutation to introduce new, potentially advantageous traits in addition to inheriting traits from the previous generation. The process of elitism, which involves the preservation of the best chromosomes from the previous generation, is also employed to ensure the retention of optimal genes.

The proposed null-model (Figure. 8.8) starts with a population of p complex networks represented as chromosomes, each consisting of an adjacency matrix with n nodes. The fitness function is used to evaluate each network, trying to maximize the values of CC_l and

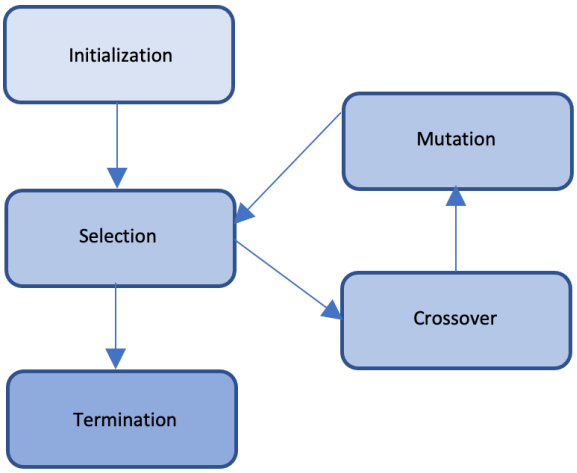


Fig. 8.7 Evolutionary algorithms steps

ρ.

The population t evolves into a new population $t+1$ of the same size p through a combination of mating and elitism-based selection. Mating involves selection of two random chromosomes that will be considered parents, followed by applying a single-point cutting operation to each row of the adjacency matrices of the parents creating a new offset that is considered to be the child chromosome. On child's offset it is applied stochastic mutation that randomly drops or adds a new edge between any two nodes by switching values between 0 and 1 at a position (i, j) in the adjacency matrix. The resulted child is considered to be a new chromosome added to the population $t+1$. The selection by elitism involves choosing the chromosome with the highest score from population t according to the fitness function. The populations evolve for a total of s steps, which serves as the termination condition, at the end of which it will be selected the chromosome with the highest score as being the result of the null-model.

The proposed model's algorithm implemented as a module requires input of parameters to define the generated network topology and parameters for the Evolutionary Algorithm (EA) used to generate the topology (described in Table 8.1).

Table 8.1 Proposed null-model with motifs parameters description.

Property	Annotation	Property	Annotation
Number of nodes	n	EA generations	Gen_{no}
Fitness score threshold	f_s	EA mutation probability	M_p
EA population	p	Early stop threshold	ε

The algorithm described in Algorithm 2 implements a function for generating random

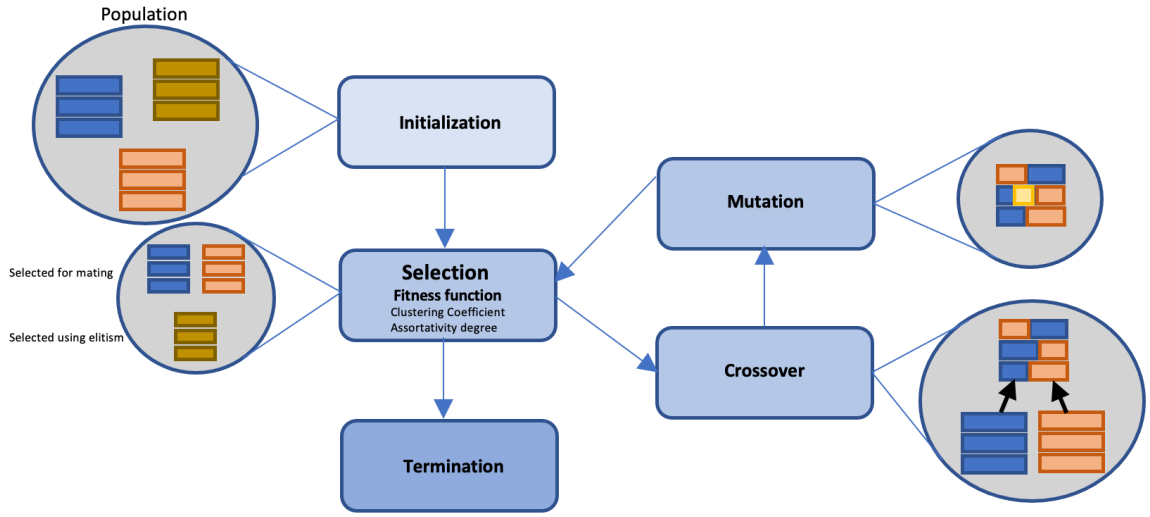


Fig. 8.8 Null-model architecture proposal

graphs with a specified number of nodes (n) and fitness score (f_s) threshold. The function takes as inputs n, f_s, p, Gen_{no}, and M_p, representing the number of nodes, the desired fitness score, the size of the population, the number of generations, and the mutation probability, respectively. We incorporate an early stopping criterion, represented by a pre-defined threshold ε, to terminate the evolutionary process once the fitness score threshold has been achieved within a certain margin of error.

The algorithm starts by generating a random population of p chromosomes, each chromosome representing a pure random graph with n nodes represented as an adjacency matrix. The evolution process runs for Gen_{no} generations, where at each generation a new population is evolved from the previous population by taking the best chromosome and creating new chromosomes through crossover of two randomly selected parents and applying a stochastic mutation with a probability of M_p, until the size of the new population equals p members. The fitness of each chromosome is assessed using the *FitnessScore* function, which calculates the ρ and CC_l values of the chromosome and generates two bi-dimensional points, $p1$ with coordinates (f_s, f_s), respectively $p2$ with coordinates (ρ, CC_l).

These points are located in a bi-dimensional space, enabling the calculation of their Euclidean distance from the desired threshold point $p1$ to evaluate their proximity to the expected f_s value. This approach promotes the evolution of chromosomes with balanced ρ and CC_l values (Figure 8.9). Chromosomes with lower fitness scores, which correspond to smaller distances to the threshold point $p1$, are preferred candidates for the algorithm's evolution.

In the next Section, the proposed null-model will be evaluated for its accuracy by com-

Algorithm 2 NullModelWithNetworkMotifs(n, f_s, p, Gen_{no}, M_p)

Input: n, f_s, p, Gen_{no}, M_p, where $1 \leq n \leq \infty$, $0 \leq f_s \leq 1$, $1 \leq Gen_{no} \leq \infty$, $0 \leq M_p \leq 1$
Output: Graph G =(V,E), where V is the list of vertices and E is the list of nodes.
$G \leftarrow$ EvolutionaryAlgorithm(n, f_s, p, Gen_{no}, M_p)
return G

FitnessScore(*chromosome*, f_s)
$rho \leftarrow$ AssortativityDegree(*chromosome*)
$CC_l \leftarrow$ ClusteringCoefficient(*chromosome*)
$p1 \leftarrow [f_s, f_s]$
$p2 \leftarrow [rho, CC_l]$
return $\sqrt{(p1[0]-p2[0])^2+(p1[1]-p2[1])^2}$

BestChromosome(*population*, f_s)
$best \leftarrow popuation[0]$
for c in *population* **do**
    if FitnessScore(c, f_s) $\leq$ FitnessScore($best$, f_s) **then**
        $best \leftarrow$ c
    end if
end for
return c

EvolutionaryAlgorithm(n, f_s, ε, p, Gen_{no}, M_p)
$population \leftarrow$ list of chromosomes encoding as adjacency matrices, graphs with random links and n nodes
for $g \in [1, Gen_{no}]$ **do**
    $bestChromosome \leftarrow$ BestChromosome($population$, f_s)
    if FitnessScore($bestChromosome$, f_s) $\leq \varepsilon$ **then**
        return $bestChromosome$
    end if
    $population \leftarrow$ [$bestChromosome$]
    for $i \in [1, size(population)-1]$) **do**
        $p1 \leftarrow$ select random chromosome from $population$
        $p2 \leftarrow$ select random chromosome from $population$
        $off \leftarrow$ apply crossover operation with one cut between $p1$, $p2$
        $prob \leftarrow$ random value in interval [0,1]
        if $prob > 1 - M_p$ **then**
            off suffers mutation on random positions i, j
        end if
        $population.insert(off)$
    end for
end for
return BestChromosome($population$)

paring it against state-of-the-art null-models from the degree sequence and small-world categories. The comparison will be performed by generating x networks using the proposed EA null-model, the Newman-Watts-Strogatz model, and the Barabási-Albert model. The existence of network motifs in the underlying topologies will be validated and the frequency of networks with motifs in each category will be computed. In order to determined if a network contains motifs or not, it will be used ESU algorithm discussed in Chapter. 3.

This comparison will provide insight into the reliability of the newly introduced null-model as a generator of synthetic networks that are prone to evolve network motifs. Moreover, we are willing to obtain a sense of determination about the degree of robustness of its

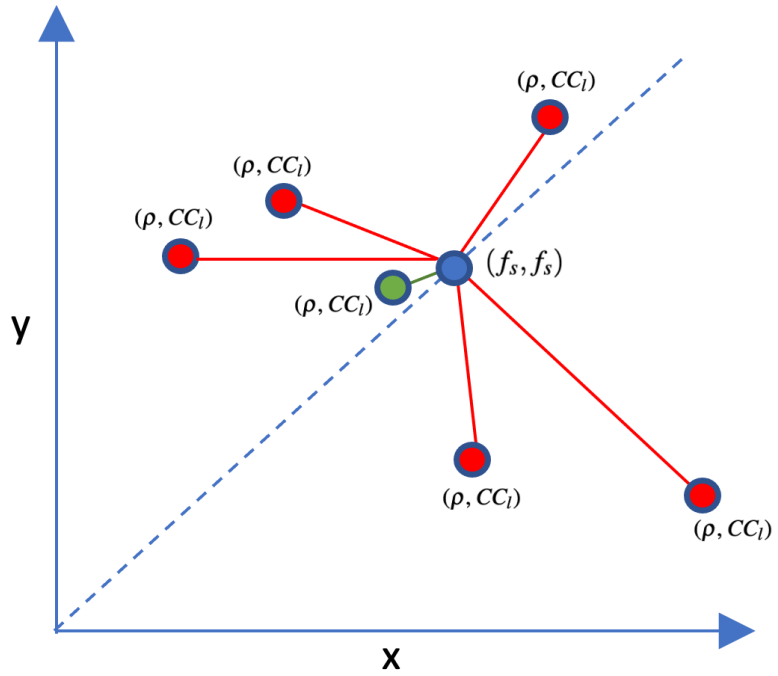


Fig. 8.9 The chromomoses representation in the Euclidean space established according to the values of ρ and CC_l. The chromosome population is plotted in the space, and the fittest chromosome is identified as the one with the minimum distance to the threshold point fs, represented by a green point.

specialization in evolving motifs compared with the others.

8.3.2 Experiment

In this study, a comprehensive evaluation of the proposed null model is conducted using both qualitative and quantitative analysis techniques. We consider an experimental sample size large enough to to validate the accuracy of the model and compare it with existing null models in the literature. The Newman-Watts-Strogatz (NWS) model is selected as the representative null model for small-world networks, and the Barabási-Albert (BA) model for degree sequence networks. A sample of 100 networks is generated for each null model, using the following set of parameters:

- NWS: $n = 100$, $k = 15$, $p = 0.1$
- BA: $n = 100$, $m0 = 15$

For the Newman-Watts-Strogatz (NWS) model, a k and p value is selected that would result in a relative sparse network, allowing for the formation of cliques with a more qual-

itative arrangement of connections (not purely random). This enhances the likelihood of creating patterns that can be detected as network motifs and compete with the proposed model. The same principle of creating qualitative connections instead of a dense network, where the strength of connections is proportional to their rarity, is applied in the selection of $m0$ for the Barabási-Albert (BA) model. This results in a well-distributed number of hub nodes in comparison to the number of nodes in the network, as described by the scale-free distribution shown in Figure 8.4.

To assess the accuracy of the proposed model in comparison with the generated null models mentioned earlier, three test cases are generated, considering low, medium, and high fitness scores. This will determine the values of ρ and CC_l for the final network evolved. This approach will validate the hypothesis that networks with higher values of these parameters have a higher probability of evolving network motifs. The parameters for the three test cases are as follows:

- Low probability: $f_s = 0.2$.
- Medium probability: $f_s = 0.4$.
- High probability: $f_s = 0.7$.

In each of the three cases, we set the parameters for the evolutionary algorithm component, namely the population size $n = 100$, parent population size $p = 100$, number of generations $Gen_{no} = 5000$, early stop $\varepsilon = 0.01$, and mutation probability $M_p = 0.2$ respectively.

The convergence rate of the generated networks with low, medium, and high fitness scores is depicted in Figure 8.10, which confirms the efficacy of the EA parameter configuration in achieving a rapid and consistent convergence towards optimal chromosomes in each generation. On a network with a f_s of 0.2, an average of 90 generations were required to satisfy the stop condition of $\varepsilon = 0.01$, whereas for $f_s = 0.4$ and $f_s = 0.7$, 270 and 1950 generations were needed, respectively. Our implementation, executed on a machine with an i7 CPU and 32GB RAM, required an average time of 35 seconds for running 100 generations. Although our solution's time requirement is significantly higher than NWS and BA models, which are almost instantaneous, we believe that this is an acceptable trade-off considering the evolutionary nature of our approach and our goal of replicating real-world dynamics, which we consider to be the true reason for the existence of motifs. We experimented with various parameter sets, including smaller or larger populations and higher mutation probabilities, but they did not significantly improve the convergence performance, warranting further investigation or optimization.

Furthermore, the evaluation of chromosomes using the *FitnessScore* function (described in Section 8.3.1) demonstrates that the desired equilibrium between the ρ and CC_l network properties is achieved. This results in the evolution of high-quality networks that display an equal improvement in both properties.

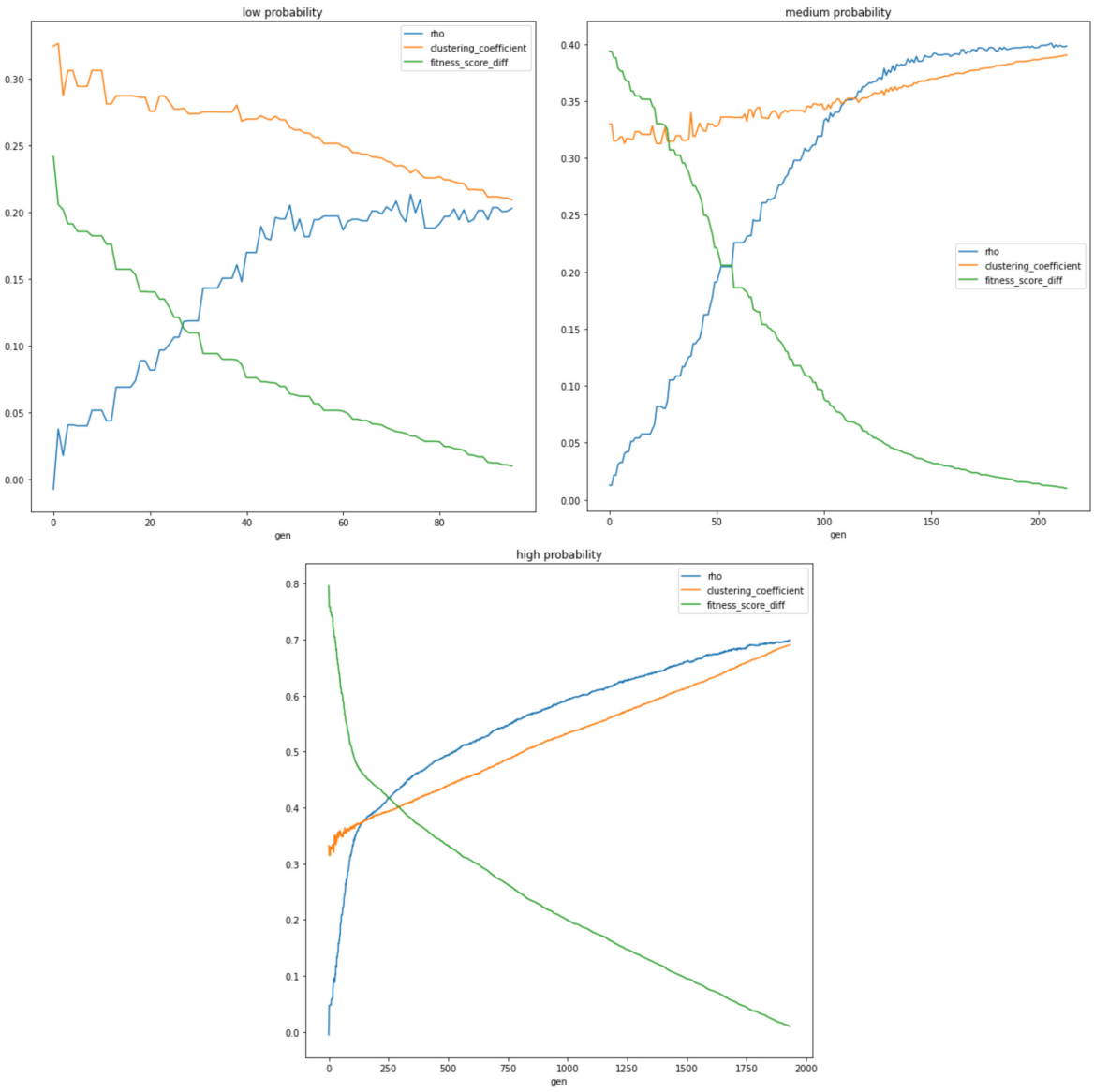


Fig. 8.10 Convergence of f_s, *rho* and CC_l to the optimal solution for each category, low probability (left), medium probability (right), high probability (center).

In the next section we will examine the properties and characteristics of these generated networks, with the aim of comparing the robustness and accuracy of the proposed null-model that aims to evolve network motifs with probability controller through f_s parameter.

8.3.3 Results discussion

For each of the studied model NSW, BA and our proposed null-model (P_m) there were generated 100 distinct undirected networks as described in the methodology of the experiment (Section 8.3.2).

Table 8.2 presents an overview of the topological characteristics of the studied networks, focusing on ρ and CC_l. In addition, part of the same exercise we computed Density (d), which is a measure of how close a network is to achieving the maximum number of edges, as discussed in Subsection 2.5.3. We introduced d as an indicator of the quality of the generated networks, aiming to generate networks that optimize the values of ρ and CC_l while minimizing the number of edges if possible.

Network type	ρ median	ρ 95% CI	CC_l median	CC_l 95% CI	d median	d 95% CI
NSW	0.0380	0.0300 *to* 0.0400	0.5900	0.5900 *to* 0.6000	0.1600	0.1551 *to* 0.1558
BA	-0.0800	-0.0880 *to* -0.0800	0.3600	0.3570 *to* 0.3610	0.2575	0.2575 *to* 0.2575
P_m ∼0.2	0.1994	0.1990 *to* 0.1999	0.2089	0.2087 *to* 0.2090	0.2364	0.2357 *to* 0.2371
P_m ∼0.4	0.3989	0.3986 *to* 0.3991	0.3903	0.3902 *to* 0.3904	0.3493	0.3485 *to* 0.3501
P_m ∼0.7	0.6750	0.6753 *to* 0.6648	0.7263	0.7219 *to* 0.7306	0.2365	0.2359 *to* 0.2371

Table 8.2 Overview of the generated networks along with their corresponding median values and 95% Confidence intervals for ρ, CC_l and d. NSW and BA networks generated based on the setup described in Section 8.3.2.

The networks generated by the NSW model conform well to the small-world network principle, which suggests that nodes tend to form clusters or communities with a higher number of intra-community edges and fewer edges outside the community they are part of. This accounts for the high value of CC_l. However, due to the stochastic nature of intra-community edge creation, there is no preferential attachment specific to assortativity. This explains the nearly neutral value of ρ, as the nodes in the networks generated by this model do not replicate the real-world behavior of preferential attachment, where links are created based on the dynamics of the network.

The BA model, which utilizes preferential attachment to create a highly disproportionate distribution of edges among nodes, results in a negative value for ρ, indicating a tendency for nodes to connect with those possessing a dissimilar degree. This behavior can be attributed to the philosophy of preferential attachment employed by the model, whereby low degree nodes seek to establish connections with nodes of higher degree, thus in contradiction with assortativity's principles. The presence of popular nodes, or hub nodes, in the networks generated by the BA model leads to the formation of weak communities around them, weak as the hubs serving as their centers or single point of failure. This observation explains the

value of CC_l which is not necessarily low, but also not particularly high.

The proposed null-model P_m was successful in evolving balanced values for both ρ and CC_l by utilizing a fitness function that computes the euclidean distance between a threshold point (f_s, f_s) and a bi-dimensional point (ρ, CC_l) (described in Subsection 8.3.2), resulting in the most optimal versions of the networks. Additionally, our objective to maximize ρ and CC_l while maintaining a reasonable d was met, indicating that the EA was able to evolve high-quality networks with the minimum number of links necessary to achieve the goal of maximizing the parameters.

Network type	With motifs	Without motifs	Percentage
NSW	24	76	24%
BA	1	99	1%
P_m ∼0.2	55	45	55%
P_m ∼0.4	90	10	90%
P_m ∼0.7	95	5	95%

Table 8.3 Results of the motifs extraction algorithm applied to the generated networks. The algorithm used was ESU (Z-score = 5) and the motifs extracted were of size k = 3.

Table 8.3 presents the results of the motifs extraction using the ESU algorithm, which was extensively discussed in Chapter 3. As the size of the networks was relatively small ($n = 100$), we employed the classic version proposed by Grochow in paper [52] rather than the parallelized version we proposed (Chapter 3). We focused on extracting motifs of size $k = 3$, which we deemed sufficient for the binary labeling of each processed network as either containing motifs or not. To ensure the reliability of the extracted motifs, we set a high Z-score threshold = 5 for the ESU algorithm. The Z-score reflects the statistical significance of the extracted motif, with higher Z-scores indicating more powerful motifs.

The obtained results support our previous findings presented in Chapter 7, where we identified ρ and CC_l as properties that promote the evolution of network motifs. Despite their different generation mechanisms, neither NSW nor BA models are designed to enhance both properties simultaneously, with only BA exhibiting some preference for high CC_l. The statistical analysis of the results highlights a significant difference between the performance of NSW and BA, and each of the three configurations of the parameter P_m, further demonstrating the superior ability of our proposed null-model in generating networks with not only existing but also strong and qualitative motifs.

A comparison between the different variations of the P_m with f_s values of 0.2, 0.4, and 0.7 was performed, revealing interesting insights. Despite the low probability associated intuitively with f_s = 0.2, this configuration generates a relatively large number of networks

with motifs (44%), compared to $f_s = 0.4$ and $f_s = 0.7$, which exhibit only a 5% difference in the number of generated networks with motifs. These observations raise practical considerations regarding the algorithm's performance, indicating that it is not necessary to set a high f_s value (i.e., greater than 0.7) to achieve comparable results. Rather, an $f_s = 0.4$ can achieve similar results in significantly less time (270 generations compared to 1950), with a small trade-off for accuracy. Therefore, we can conclude that an f_s value of 0.4 strikes a balance between accuracy and performance, making it the optimal choice for our proposed algorithm.

8.4 Conclusions

This chapter describes the result of an iterative experiment that sought to develop an analytical framework capable of elucidating the causal relationship between network motifs and their corresponding topologies. Initially, a micro-level analysis was conducted to establish that certain network properties predispose them to evolve motifs. Based on this observation, the proposed framework identified assortativity degree (ρ) and local clustering coefficient (CC_l) as reliable network properties that could be incorporated into a null-model designed to generate networks containing motifs.

The main challenge of this experiment was to break the paradigms of existing null models that used a stochastic approach to add edges between nodes by maximizing only a given micro-level property at a time. For example, the Newman-Watts-Strogatz model (NWS) is a state-of-the-art in the generation of networks following the principle of small-world networks, characterized by a high local clustering coefficient (CC_l) due to their topologies prone to evolve tight communities. In contrast, the Barabási-Albert model (BA) aims to create networks with the property of preferential attachment, but it is applied differently than the preferential attachment existing in networks with high assortativity degree (ρ). In networks with a high ρ, nodes aim to link with other nodes having the same degree without any preference for the magnitude of the degree value. On the other hand, in BA networks, nodes with low degrees aim to link to hub nodes (nodes with high degree). This difference means that networks generated using the BA model are not ones having a high ρ.

The aim of the proposed null model was not only to maximize two properties simultaneously but also to create a more realistic mechanism for adding edges to the network - something we consider crucial in the reasoning of motifs existence. To achieve this, an evolutionary algorithm (EA) structure was designed to mimic the real-world dynamics of network evolution. This allowed the generated network to evolve gradually over time, sim-

ilar to a dynamical system, until it reached a state modeled as a complex network. The null model was controlled by a set of parameters, which were based on an innovative fitness function we introduced for the first time, that aimed to maximize both ρ and CC_l at the same time.

In this study, we tested the proposed null-model against the BA and NWS models, considering three versions with different values of the probability to contain motifs (f_s): 0.2, 0.4, and 0.7. The results confirmed those presented in Chapter 7, demonstrating that networks with high values of assortativity degree (ρ) and local clustering coefficient (CC_l) are more likely to contain motifs. In comparison to BA and NWS models, our proposed null-model generated a significantly higher number of networks with motifs for each variation of f_s, indicating its effectiveness in generating networks with desired topological features.

The analysis of the variations of the proposed null-model with different values of f_s revealed a proportional relationship between f_s and the number of networks with motifs generated. However, the relationship was not linear, as there was no proportional difference between the number of networks with motifs generated when f_s was set to 0.4 and 0.7. Based on this finding, we can conclude that a value of f_s = 0.4 is sufficient and efficient when using the proposed null-model.

In conclusion, our proposed null-model presents an innovative approach to generating complex networks by utilizing AI-based algorithms, which is a departure from the traditional stochastic-based methods commonly used in the literature. Moreover, our model's impact on the field of network motif analysis is significant, as it allows for a deeper understanding of the emergence of patterns in complex systems. By providing a mechanism for generating networks with high values of assortativity degree and local clustering coefficient, our model enables researchers to investigate the causal effects of network motifs and their potential implications for real-world systems.

To further improve the proposed architecture, the fitness function could be extended to maximize the power of the motifs as well. This would involve merging the fitness function with the Z-score present in the Exact Subgraph Enumeration (ESU) algorithm. Such an improvement would eliminate the current probabilistic nature of evolution, which cannot guarantee the existence of motifs but only increase the probability through the f_s parameter. This could lead to the generation of more reliable networks containing motifs, providing a better understanding of the causal effects of motifs in network topologies. In addition to extending the fitness function to maximize the power of motifs, another potential improvement to our proposed architecture could be the inclusion of other topological properties in the fitness function. Although we found that assortativity degree (ρ) and local clustering

coefficient (CC_l) were highly correlated with the evolution of motifs, there are other properties that could be added to further refine the fitness function. For example, incorporating measures of community structure, relative edge distribution entropy, or power law exponent may lead to a more rapid convergence to a high fitness score. Future work could explore the inclusion of these or other topological properties in the fitness function to improve the accuracy and efficiency of our proposed null model.

Chapter 9

Conclusions and future work

The pursuit of finding solutions to challenges in life requires curiosity and the willingness to explore uncharted territories. Such an endeavor can be both beautiful and demanding, and discovering solutions can bring a sense of unparalleled satisfaction. The roadmap of this thesis reflects such an attitude and the author considers himself fortunate to have been able to engage in the pursuit of new knowledge - a dream he had cherished since childhood. Although the primary objective was well-defined, unexpected findings along the way contributed to the author's personal and professional growth. As the saying goes, "the wiser you become, the more questions you have," and staying focused on the roadmap presented a logistical challenge that was overcome with the guidance of the author's supervisor and mentors - to whom he is grateful.

In conclusion, this thesis marks the beginning of a new phase of research. It has been a fulfilling journey, and we are delighted to present the results in the form of a captivating story. As a network enthusiast, we firmly believe that every network has a story to tell, and this thesis is a testament to that belief. Through the review of thousands of networks, this work validated research questions, supported hypotheses, and demonstrated experiments. The findings of this thesis have contributed to the understanding of complex networks, and we are eager to continue exploring new questions and discovering more stories.

The objective of our research was to investigate the evolution of network motifs and explore the feasibility of controlling their formation. Network motifs are infrequent in real-world systems due to their specific dynamics, making them valuable indicators for understanding complex networks. However, their rarity has impeded the development of universal laws governing their existence, thereby presenting significant challenges to their study. Our research therefore required us to explore unknown territory, acknowledging that there would be times when we would not have access to established solutions or answers.

The initial step of this research project involved an investigation of tools for identifying network motifs. Specifically, we explored various algorithms commonly used for determining if a group of nodes and edges can be classified as a motif. Our study revealed that the algorithms had poor performance due to the NP-hard nature of the motif extraction problem. This posed a significant challenge for our research project, as we would need to run the algorithms numerous times, resulting in unacceptably long waiting times for the results needed for our analysis. To address the performance issue of motif extraction algorithms, a reliable and efficient process was developed prior to conducting the desired analysis. A parallelized version of the ESU algorithm was introduced and presented as a substantially improved solution in comparison to other approaches (Chapter 3). The development of this algorithm was a significant accomplishment that not only served the goal of the research project but also benefited the larger research community in the field of complex networks.

The research continued with the collection of a diverse dataset of networks from various domains and sizes to ensure a balanced dataset for analysis. Alongside the extraction of motifs from these networks, we also collected their topological properties (Chapter 2) to investigate potential correlations with the presence of motifs. The parallelized ESU algorithm proved to be a significant time-saver, allowing us to efficiently gather a comprehensive dataset in a surprisingly short period.

Following the principles of dynamical convergence from the Stability Theory (Section 2.3), we developed a set of research questions to guide our experiments. These questions were centered around the atoms or nodes of the system, as they encode the dynamics that converge to them and eventually settle into a new state. We used this approach to formulate a hypothesis explaining why and how nodes and edges appear or disappear in complex networks, providing an overarching explanation for the evolution of motifs. According to our hypothesis, when a specific event occurs in a system, the atoms behave in a way that creates structures guided by the event's dynamics. From the perspective of a modeled system as a network, this is encoded as nodes having specific edges, which are detected as network motifs.

Building upon the theoretical insights obtained from our analysis, we conducted a series of statistical tests to investigate the relationship between the extracted topological properties of the networks and the presence of motifs. The resulting complex statistical analysis provided answers to all of the research questions we had formulated, confirming our convergence hypothesis. Our findings revealed that a set of topological properties exhibit a strong correlation with the existence of motifs in the network.

We started by investigating the correlation between the presence of Articulation Points

(*AP*) in a network and the existence of network motifs. *AP* nodes are critical bridges between two or more communities and play a significant role in facilitating information flow between them. Through a series of experiments presented in Chapter 4, we demonstrated that *AP*s have a controllability property, which enables them to propagate specific dynamics to the communities they control while isolating them from others. This leads to a rapid evolution of motifs in those networks. Our findings have important implications, particularly in social networks, where *AP*s are often influential people who can induce ideologies in the communities they lead. This has a dangerous application, especially in social networks, where *AP*s are usually influential people and can induce ideologies in the communities that they lead.

The other two topological properties highly associated with the presence of network motifs were assortativity degree (ρ) and local clustering coefficient (CC_l), owing to their ability to encode specific network behaviors. ρ is a measure that indicates whether nodes in a network are more likely to be connected to other nodes with the same degree, which is comparable to people in a community being more likely to have friends who are equally popular. This particular behavior is highly specific and explains the emergence of certain motifs during the process of forming links between nodes of the same degree. Local clustering coefficient (CC_l) is another topological property that is highly correlated with the existence of network motifs. It indicates the tendency of nodes in a network to form tightly interconnected groups or clusters. This behavior provides a favorable environment for the evolution of motifs among the nodes within the same cluster, particularly if they tend to link with nodes of similar degrees following the ρ principles of preferential attachment.

After obtaining the results of the analysis, we proposed a comprehensive framework that confirmed the convergence properties from Stability Theory in complex networks, thereby achieving our initial objective of comprehending the evolution of network motifs in such networks (Chapter 7). Furthermore, based on these outcomes, we suggested a new tool that simplifies the extraction of motifs as a preprocessing step. The task of extracting topological properties from complex networks is significantly faster than extracting motifs, even with the use of parallelization. Since specific topological properties and their values have been shown to indicate the presence of motifs in networks, we propose training a series of AI-based models that can predict the presence or absence of motifs in a network based on its topological properties. In this way, it is possible to determine from an early stage whether the network is worth processing using the trained AI-based model as a recommender system. Of note, one of the trained models achieved outstanding performance, detecting the presence or absence of motifs with over 95% accuracy (Chapter 5).

Despite the promising results obtained from our previous analyses, gave birth to a new more supreme goal. We identified a research gap in the current state-of-the-art literature, specifically the absence of a null-model that can generate networks containing motifs. While existing null-models aim to generate specific network types using stochastic methods to add links between nodes until a desired property is met, none address the challenge of creating networks that contain motifs. We believed that the algorithmic approach used by most null-models in the literature, which aims to simulate a synthetic behavior, contradicts the natural environment required for motifs to exist. As a result, we developed a new null-model architecture based on an Evolutionary Algorithm (EA) that can evolve networks to replicate the continuous dynamics of a real-world system and able to generate complex networks containing motifs. The EA component aimed to optimize ρ and CC_l properties simultaneously and to devise a more realistic mechanism to add edges to the network, a crucial factor for motif existence. We controlled the proposed null-model using a set of parameters based on a novel fitness function, which we introduced for the first time in the literature. The experiment described in Chapter 8 showed that the proposed architecture was reliable, as it generated networks with strong motifs. This contribution represents an important addition to the literature, with potential implications for future research in this field.

In future work, we consider the study of coexistence dynamics between network motifs and other meso-scale topological structures, such as communities or chains. This exploration holds the potential to unveil nuanced behaviors in the evolution of network motifs' controllability. Leveraging these structures could enhance the precision of the existing null-model introduced in Chapter 8, enabling the design of a mechanism for regulating motif size and not only their existence. Additionally, we plan to explore diverse null-model architectures, such as stochastic block models or deep generative models, and conduct comparative analyses. This approach promises to unlock further avenues for controlling motif evolution in generated networks.

In conclusion, while we acknowledge that no work can ever be considered completely finished, we are confident that our research has contributed significantly to the field of complex network analysis. Throughout this journey, we have proposed a series of results innovative to the state-of-the-art and opened up new research tracks and considered suggestions for improvement at the end of each chapter, with the hope of inspiring young researchers to continue pushing the boundaries of knowledge. Looking back on our achievements, we take pride in the completeness of the thesis and we are excited to see how the result will impact future research in this area.

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